August

Date: 20260801
Title: Miniprep (liquid culture 0731)
Purpose: to extract plasmid from liquid culture
Procedure:
A1, A2, A3: pSTK-1A - PtagC+RBS - Ho1:pcyA - L3S2P21
C6, C9: pSTK-1B - PlepA - RBS0 - ccaSm3 - B0015
I3: pSTK-1D - PcpcG2-veg - EEBD-RBS0 - Csn:ACCD - B0015

1. Transfer 1000 µL of liquid culture to 1.5 mL microtube.
2. Put in a centrifuge and centrifuge at 16000 g for 1 minute.
3. Discard the supernatant.
4. Repeat steps 1–3 until all mediums are collected.
5. Add 200 µL of MX1 buffer.
6. Resuspend by pipetting and vortexing.
7. Add 250 µL of MX2 buffer.
8. Invert slowly for 8–10 times, incubate at RT for 2 mins.
9. Add 350 µL of MX3 buffer and invert slowly for 4–5 times.
10. Centrifuge at 23444 g for 5 minutes.
11. Transfer 880 µL of supernatant to column.
12. Centrifuge at 7000 g for 30 seconds.
13. Discard the flow through.
14. Add 500 µL of WN buffer to the column.
15. Centrifuge at 7000 g for 30 seconds.
16. Discard the flow through.
17. Add 700 µL of WS buffer to the column.
18. Centrifuge at 7000 g for 30 seconds.
19. Discard the flow through.
20. Centrifuge the empty column for 8 mins. at 10000 g.
21. Place column into 1.5 mL tube.
22. Add 50 µL EB buffer to the center of the column membrane.
23. Incubate at RT for 3 mins.
24. Centrifuge at 10000 g for 3 mins.
25. Measure the concentration by using Biodrop

Done by: Naomi, Sarah, Sophie H.

BioDrop Results:

Sample	Concentration (ng/µL)	A260/A280	A260/A230
A1	107	2.019	1.163
A2	89	1.854	0.671
A3	94	1.808	0.890

C6	98	1.607	0.782
C9	78	1.560	0.395
I3	165	1.897	0.487

Date: 20260801
Title: Collect B. sub supernatant and pellet of J
Purpose: To prepare supernatant and pellet for analyses
Procedure:
J: Pveg-EEBD-RBS0-csn:ACCD-B0015

1. Transfer culture of J1, J2, and J3 to 1.5 mL tube.
2. Centrifuge at 16000 g for 1 minute.
3. Collect supernatants in 1.4 mL tubes.
4. Repeat steps 1–3 until all cultures are collected.

Done by: Sarah, Naomi, Sophie H.

Date: 20260801
Title: Make acrylamide gel
Purpose: create a fine-pored matrix that separates proteins or small nucleic acids by size during electrophoresis.
Procedure:

1. Assemble holder, glass plates, comb.
2. Prepare resolver solution:
    a. Resolver A — 3 mL
    b. Resolver B — 3 mL
    c. 10% APS — 30 µL
    d. TEMED — 3 µL
3. Fill to 1.5 - 1 cm below the bottom of comb teeth with a resolver solution (~ 6mL).
4. Prepare stacker solution:
    a. Resolver A — 1 mL
    b. Resolver B — 1 mL
    c. 10% APS — 10 µL
    d. TEMED — 2 µL
5. Fill the top with a stacker solution (~2 mL).
6. Insert comb & allow gel to polymerize.

Done by: Sophie C., Chloe

Date: 20260801
Title: Make 0.8% agarose gel (x3)
Purpose: To prepare gel for enzyme digestion
Procedure:

1. Weigh 0.48 g of agarose powder and 60 mL 1x TAE buffer.
2. Add the components into a conical flask.

3. Assemble mold, tray, and comb.
4. Microwave for 30 sec after plastic wrap & holes poking. Agitate. Repeat until there are no sediments.
5. Wait until no more smoke, add 3 µL of safety dye.
6. Pour the melted gel into the three assembled molds.
7. Put the comb and wait for it to solidify.

Done by: Sophie H.

Date: 20260801
Title: Enzyme Digestion
Purpose: To confirm plasmid size for each samples before moving on to golden gate assembly
Procedure:

Samples:

- A1, A2, A3: pSTK-1A - PtagC + RBS - Ho1:pcyA - L3S2P21
- C6, C9: pSTK-1B - PlepA - RBS0 - ccaSm3 - B0015
- I3: pSTK-1D - PcpcG2-veg - EEBD-RBS0 – csn:ACCD - B0015

1. Prepare following reaction:
    a. A1 (base pair 3754)
        i. 10× Cutsmart buffer: 2 µL
        ii. DNA: 4.67 µL
        iii. ddH_2O: 12.83 µL
        iv. EcoRI (1 site): 0.5 µL
        v. Total volume: 20 µL
    b. A2 (base pair 3754)
        i. 10× Cutsmart buffer: 2 µL
        ii. DNA: 5.62 µL
        iii. ddH_2O: 11.88 µL
        iv. EcoRI (1 site): 0.5 µL
        v. Total volume: 20 µL
    c. A3 (base pair 3754)
        i. 10× Cutsmart buffer: 2 µL
        ii. DNA: 5.32 µL
        iii. ddH_2O: 12.18 µL
        iv. EcoRI (1 site): 0.5 µL
        v. Total volume: 20 µL
    d. C6 (base pair 3508)
        i. 10× Cutsmart buffer: 2 µL
        ii. DNA: 5.1 µL
        iii. ddH_2O: 12.4 µL
        iv. EcoRI (1 site): 0.5 µL
        v. Total volume: 20 µL
    e. C9 (base pair 3828)

i. 10× Cutsmart buffer: 2 μL
ii. DNA: 6.41 μL
iii. ddH_2O: 11.09 μL
iv. EcoRI (1 site): 0.5 μL
v. Total volume: 20 μL

f. I3 (base pair 3757)
i. 10× Cutsmart buffer: 2 μL
ii. DNA: 3.03 μL
iii. ddH_2O: 14.47 μL
iv. EcoRI (1 site): 0.5 μL
v. Total volume: 20 μL

2. For digested samples, add 0.5 μL of restriction enzyme EcoRI.
3. For undigested samples, add 2 μL more of ddH_2O.
4. Incubate the mixture at 37°C for 1 hour.
5. After finishing incubation, run gel electrophoresis.

Done by: Sarah, Naomi, Sophie H.

Date: 20260801
Title: Electrophoresis for Enzyme Digestion
Purpose: To separate fragments and ensure plasmid size for each samples from 20260801 enzyme digestion
Procedure:

1. Load the gel in the following order:

M	A1	A1*	A2	A2*	A3	A3*	C6	C6*	C9	C9*	I3	I3*	M
5	5	5	5	5	5	5	5	5	5	5	5	5	5

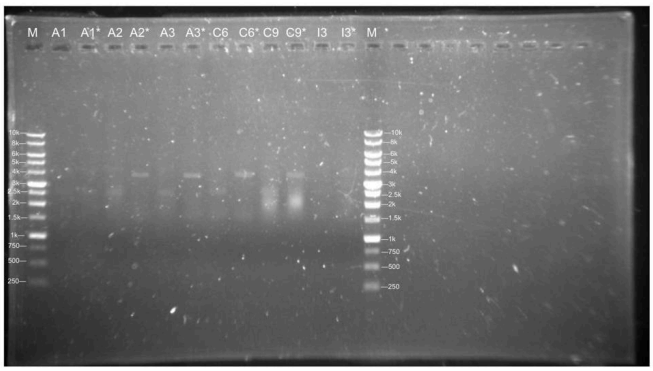
- M: 1kb DNA ladder
- A1~3: PtagC+RBS-Ho1:pcyA-L3S2P21 (base pair: 3754 bp)
- C6,9: PlepA-RSS0-ccaSm3-B0015 (base pair: 3828 bp)
- I3: PcpcG2-Veg-EEBD-RBSO-csn:ACCD-B0015 (base pair: 3857 bp)

- *: digested
- Enzyme: EcoRI

2. Resolve the gel at 100V for 30 minutes

Done by: Sarah, Naomi, Sophie H.

Results



M: 1kb DNA ladder

A1~3: PtagC+RBS-Ho1:pcyA-L3S2P21
(base pair: 3754 bp)

C6,9:
PlepA-RSS0-ccaSm3-B0015
(base pair: 3828 bp)

I3:
PcpcG2-Veg-EEBD-RBSO-csn:ACCD-B0015
(base pair: 3857 bp)

*: digested

Date: 20260801
Title: Colony PCR
Purpose: screen for successful cloning after transformation
Procedure:
Samples:

1. Prepare the following reaction (3 master mix) and dispense solutions into PCR tubes

G: PcpcG2-ver-EEBD-RBS0-Csn:LEA-bkdBcoldD
bp: 1375

	1 reaction	13 reactions
Forward primer (CsnF)	1μL	13μL
CDS primer (LEA R)	1μL	13μL
Q-amp master mix	10μL	130μL
ddH_2O	8μL	104 μL

M: PcpcG2-veg-EEBD-RBS0-GanA-Gala-L3S2P21
bp: 2837

	1 reaction	13 reactions
Forward primer (Biobrick Prefix)	1μL	13μL
CDS Reverse primer (STK seq rv)	1μL	13μL
Q-amp master mix	10μL	130μL
ddH_2O	8μL	104μL

L: 3P-EEBD-RBS0-GanA-Gala-L3S2P21
bp: 2411

	1 reaction	17 reactions
Forward primer (pSTK-Exp-seq-rv)	1μL	17μL
CDS reverse primer (Exp-seq4-rv)	1μL	17μL
Q-amp master mix	10μL	170μL
ddH_2O	8μL	136 μL

2. Circle 12 colonies from plate G and M, 16 colonies from L
3. Pick up colonies from each agar plate with tip and streak the labelled grid
4. Dispense the tips with colonies into the PCR strips and wait 5 minutes
5. Take out the tips and close the lid
6. Place into PCR machine for thermal cycle

95°C	2 min	1 cycle
95°C	30 sec	30 cycles
65°C	40 sec	
72°C	3 min	
72°C	1 min	1 cycle

Done by Ryan Y, Olivia L, Abby T

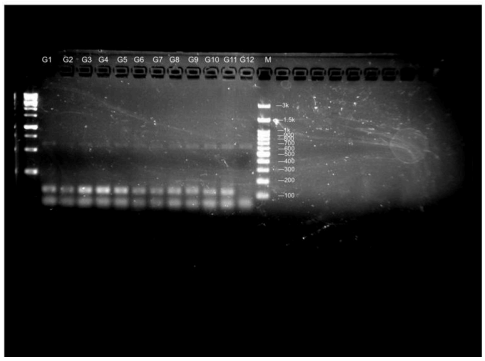
Date: 20260801
Title: Gel electrophoresis
Purpose: validate the bps of pcr results
Procedure:

1. Load gel in the following format
    - M: PcpcG2-veg-EEBD-RBS0-GanA-Gala-L3S2P21
    - G: PcpcG2-veg-eEBD-RBS0-csn:LEA-bkdb-cold (G10~12 picked for liquid culture)
    - L: 3P-EEBd-RBS0-ganA-galA-L3S2P21

Gel1
M M12 M11 M10 M9 M8 M7 M6 M5 M4 M3 M2 M1
5 5 5 5 5 5 5 5 5 5 5 5 5
Gel2
M G1 G2 G3 G4 G5 G6 G7 G8 G9 G10 G11 G12
5 5 5 5 5 5 5 5 5 5 5 5 5
Gel3
M L1 L2 L3 L4 L5 L6 L7 L8 L9 L10 L11 L12 L13 L14 L15 L16 A1u A2D I1 I3
5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5

2. Resolve the gel at 100V for 30min

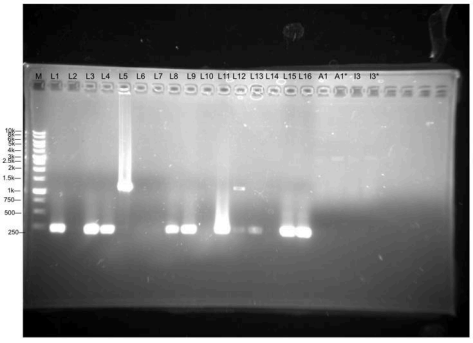
20260801 Colony PCR (G)



L: 1kb DNA ladder

G: PcpcG2-veg-EEBD-RBS0-Csn-LEA-bkdB Cold (12)
Primers:
Csn F
LEA R
bp: 585

20260801 Colony PCR (L) + Enzyme Digestion of A1 I3



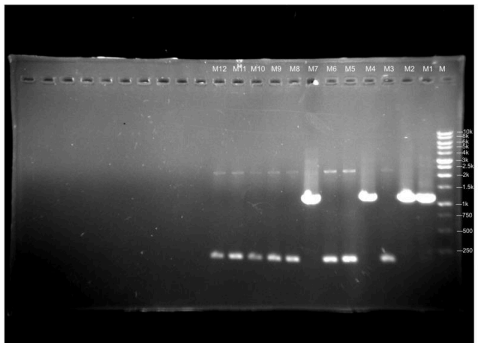
M: 1kb DNA ladder

L: PtagC+RBS-GanA-Gala-L3S2P21 (16)
Primers:
EXP-seq fw
EXP-seq4 rv
bp: 2411

A1, A3: PtagC+RBS-Ho1:pcyA-L3S2P21
(base pair: 3754 bp)

*: Digested

20260801 Colony PCR (M)



M:PcpcG2-veg-EEBD-RBS0-GanA-Gala-L3S2P21 (12)
Primer
Biobrick prefix
STK rv
bp:2837

Done by Olivia Lin, Abby Tsai,Ryan Y

Date: 20260801
Title: Liquid culture
Purpose: bacteria culture from successful colony PCR product
Procedure:
Samples from 0801 colony PCR

- PcpcG2-veg-EEBD-RBS0-csn:LEA-bkdb-cold(G10)

- PcpcG2-veg-EEBD-RBS0-csn:LEA-bkdb-cold(G11)
- PcpcG2-veg-EEBD-RBS0-csn:LEA-bkdb-cold(G12)
- Pveg-EEBD-RBS0-csn:AccD-B0015(J1)
- Exp1-Pveg-SamyQ-TAT-HA-AmilCP(#8)

1. Label 5 liquid culture tube
2. Turn on bunsen burner
3. Add 3mL of TB medium and 3μL of ampicillin to each of the G10,G11,G12 tube
4. Add 1mL of TB medium and 5μL of erythromycin to J1 and #8 tubes
5. Dip a pipette tip into the colonies
6. Dispense tips in their respective tubes
7. Place the tubes in the incubator, slant around 45%

Done by Olivia Lin, Ryan Y
Supervised by Gabriel

Date: 20260801
Title: Level 2 Golden Gate Assembly
Purpose: assemble level 1 constructs from different modules
Procedure:

1. Prepare the reaction as follows

(A)pSTK-1A-PtagC+RBS-Ho1:pcyA-L3S2P21
(B)pSTK-1B-PliaG-RBS0-ccaSm3-B0015
(C)pSTK-1B-PlepA-RBS0-ccaSm3-B0015
(D)pSTK-1C-Pveg-RBS0-ccaR-B0015
(E)pSTK-1D-PcpcG2-veg-EEBD-RBS0-csn:AmilCP-B0015
(I)pSTK-1D-PcpcG2-veg-EEBD-MF001-csn:ACCD-B0015
A, I, C-> liquid culture
A+B+D+I

Part	Original (μL)	Dilution
Vector: pSTK-EXP-2-sfGFP - Concentration: 163 - Bp: 5104	0.2035337423	1.01766871(1/5 dilution)
(A)pSTK-1A-PtagC+RBS-Ho1:pcyA-L3S2P21 - concentration:107 - Bp: 3754 1953	0.4560934579	1.36828037(⅓ dilution)
(B)pSTK-1B-PliaG-RBS0-ccaSm3-B0015 - concentration: 83ng/ul - Bp: 3792	0.5939277108	1.78178313 (1/3 dilution)

(D)pSTK-1C-Pveg-RBS0-ccaR-B0015 - Concentration: 134ng/μL - Bp: 3243 bp 1442	0.314619403	1.25847761 (1/4 dilution)
(I)pSTK-1D-PcpcG2-veg-EEBD-MF001-csn:ACCD-B0015 - Concentration: 165 ng/μL - Bp: 3757	0.2960060606	1.4800303 (1/5 dilution)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water	μL	9.09375988 μL
T4 DNA ligase	1 μL	1 μL
BsmB1	1 μL	1 μL
Total	20 μL	20 μL

A+C+D+E

Part	Original (μL)	Dilution
Vector: pSTK-EXP-2-sfGFP - Concentration: 163 - Bp: 5104	0.2035337423	1.01766871(1/5 dilution)
(A)pSTK-1A-PtagC+RBS-Ho1:pcyA-L3S2P21 - Concentration: 107 ng/μL - Bp: 3754	0.4560934579	1.36828037 (1/3 dilution)
(C)pSTK-1B-PlepA-RBS0-ccaSm3-B0015 - Concentration: 98 ng/μL	0.5077959184	1.52338776(⅓ dilution)

- Bp: 3828 bp		
(D)pSTK-1C-Pveg-RBS0-ccaR-B0015 - Concentration: 134ng/μL - Bp: 3243 bp	0.314619403	1.25847761 (1/4 dilution)
(E)pSTK-1D-PcpcG2-veg-EEBD-RBS0-csn:AmilCP-B0015 - Concentration:168ng/μL - Bp: 3485 bp	0.269672619	1.3483631(⅛ dilution)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water	μL	9.48382245μL
T4 DNA ligase	1 μL	1 μL
BsmB1	1 μL	1 μL
Total	20 μL	20 μL

A+B+D+E

Part	Original (μL)	Dilution
Vector: pSTK-EXP-2-sfGFP - Concentration: 163 - Bp: 5104	0.2035337423	1.01766871(1/5 dilution)
(A)pSTK-1A-PtagC+RBS-Ho1:pcyA-L3S2P21 - Concentration: 107 ng/μL - Bp: 3754 **1953**	0.4560934579	1.36828037(⅓ dilution)
(B)pSTK-1B-PliaG-RBS0-ccaSm3-B0015	0.5939277108	1.78178313 (1/3 dilution)

- concentration: 83ng/μL - Bp: 3792 **1991**		
(D)pSTK-1C-Pveg-RBS0-ccaR-B0015 - Concentration: 134ng/μL - Bp: 3243 bp **1442**	0.314619403	1.25847761 (1/4 dilution)
(E)pSTK-1D-PcpcG2-veg-EEBD-RBS0-csn:AmilCP-B0015 - Concentration:168ng/μL - Bp: 3485 bp **1684**	0.269672619	1.3483631(⅛ dilution)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water	μL	9.22542708μL
T4 DNA ligase	1 μL	1 μL
BsmB1	1 μL	1 μL
Total	20 μL	20 μL

A+C+D+I

Part	Original (μL)	Dilution
Vector: pSTK-EXP-2-sfGFP - Concentration: 163 - Bp: 5104 \	0.2035337423	1.01766871(1/5 dilution)
(A)pSTK-1A-PtagC+RBS-Ho1:pcyA-L3S2P21 **1953** - Concentration: 107 ng/μL - Bp: 3754	0.4560934579	1.36828037(⅓ dilution)

(C)pSTK-1B-PlepA-RBS0-ccaSm3-B0015 - Concentration: 98 ng/μL - Bp: 3828 bp **2027**	0.5077959184	1.52338776(⅓ dilution)
(D)pSTK-1C-Pveg-RBS0-ccaR-B0015 - Concentration: 134ng/μL - Bp: 3243 bp **1442**	0.314619403	1.25847761 (1/4 dilution)
(I)pSTK-1D-PcpcG2-veg-EEBD-MF001-csn:ACCD-B0015 - Concentration: 165 ng/μL - Bp: 3757 **1956**	0.2960060606	1.4800303 (1/5 dilution)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water	μL	9.35215525μL
T4 DNA ligase	1 μL	1 μL
BsmB1	1 μL	1 μL
Total	20 μL	20 μL

2. Run thermal cycle

55°C	5 min	45 cycles
16°C	5 min	
55°C	10 min	1 cycle
80°C	20 min	

Done by Abby T, Audrey H

Date: 20260801
Title: Level 1 Golden Gate Assembly
Purpose: assemble parts
Procedure:

1. Prepare the reaction as follows

L: 3P-EEBD-RBS0-GanA-Gala-L3S2P21

Part	Original (μL)	Dilution
Vector (shuttle vector): pSTK-EXP-1-sfGFP concentration: 162 ng/ μl 6929 bp	0.2780154321	1.4 μL (dilute 1/5)
Insert promoter(3P) concentration: 106 ng/ μl 2909bp	0.3567641509 μL	1.43 μL (dilute 1/4)
Insert RBS (EEBD-RBS0) concentration: 151 2392 bp	0.2059337748	1.03 (dilute 1/5)
Insert CDS(GanA-Gala) concentration: 177 4148 bp	0.3046553672	1.22 (dilute 1/4)
Insert Terminator (L3S2P21) concentration: 170 2145 bp	0.1640294118 μL	1.64 μL (dilute 1/10)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water	9.28 μL	
T4 DNA ligase	1 μL	1 μL
BsaI-HFv2	1 μL	1 μL
Total	20 μL	20 μL

Done by Sophie C, Abigail

Date: 20260801
Title: Western blot

Purpose: detect specific protein within a complex biological sample
Procedure:

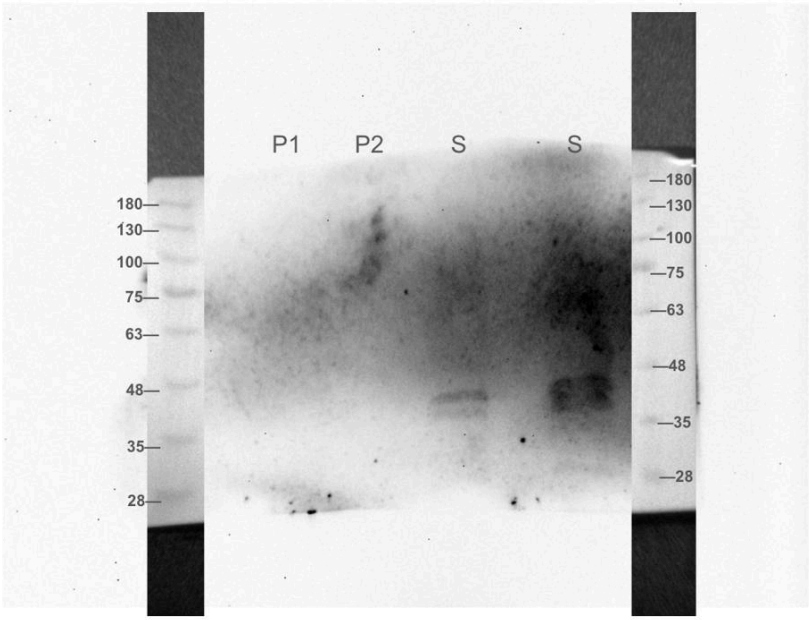
1. Collecting *Bacillus subtilis* by centrifuge at 16000g 30sec
2. Transfer supernatant to the new 15ml tube. Add 1mL cellytic B lysis buffer and resuspend
3. Add 10ng/mL lysozyme 20 mL. Incubate at RT for 15 min
4. Transfer 125mL lysate to the new microtube. Add 25mL dumpling buffer and incubate at 95°C for 5min as P1
5. Transfer 125mL medium to new microtube. Add 25mL sampling buffer and incubate at 95°C for 5 min as S.
6. Centrifuge the lysate at 16000 g for 30sec. Transfer 125mL supernatant to new microtube. Add 25mL sampling buffer and incubate at 95°C for 5min as P2
7. Resolve samples in SDS-PAGE(10%) by the following layout

ladder	P1	P2	S	S	ladder
5mL	50mL	50mL	50mL	50mL	2mL

8. Transfer to PVDF membrane
9. Blocking in 5% milk in TBST for 2 hr
10. Replace blocking solution by anti His-tag mouse m/tb(1:1000) in 5% Milk. Incubate on the orbital shaker at 4°C for overnight

Done by Chloe, Sophie C, Chars

Results



P1: Bacillus total pellet
P2: Bacillus soluble pellet
S: Supernatant

Exposure: 6 min

Date: 20260802
Title: Miniprep
Purpose: isolate plasmid DNA
Procedure:
G10, G11, G12: PcpcG2-veg-EEBD-RBS0-Csn:LEA-bleb-cold double GanA-GalA Level 0

1. Transfer 1000μL liquid culture to 1.5ml microtube
2. Centrifuge at 16,000g for 1min, discard supernatant
3. Repeat step 1&2 until all medium are collected
4. Add 200μL of MX1 buffer
5. Resuspend by pipetting or vortexting
6. Add 250μL of MX2 buffer
7. Invert slowly 8-10 times (don't pipette or vortex!), and incubate at RT for 2min
8. Add 350μL of MX3 buffer
9. Centrifuge at 23444g for 5min
10. Transfer 800μL of supernatant to column
11. Centrifuge at 7000g for 30sec and discard the flow through
12. Add 500μL of WN buffer, centrifuge at 7000g for 30sec
13. Centrifuge at 7000g for 30sec and discard the flow through
14. Centrifuge empty column for 8min at 10000g and place column into 1.5mL tube
15. Add 50μL EB buffer at the center of the column
16. Centrifuge at 23444g for 5min

	Concentration (ng/μL)	A260/230	A260/280
G10	48.0	1.655	1.920
G11	57.0	1.676	1.966
G12	62.0	1.879	1.958
Gana-GalA level 0	278	2.122	1.9021

Done by Olivia L, Ryan Y

Date: 20260802
Title: PCR
Purpose: amplify specific segment of DNA
Procedure:

A:MF001
B:MF002
C:L3S1P32-L3S2P56

1. Prepare the reaction as follow:

	MF001 5.2rxn	MF002 5.2rxn	L3S1P32-L3S2P56 5.2rxn
1.0xEx taq buffer	13μL	13μL	13μL
2.5uM dNTP Mix	10.4μL	10.4μL	10.4μL
Forward Reverse MF001(each)	2.6μL	x	x
Forward Reverse MF002(each)	x	2.6μL	x
Forward Reverse L3S1P32-L3S2P56	x	x	2.6μL
Sterile water	98.15μL	98.15μL	98.15μL

2. Distribute 24.875μL of the 5rxn mixture into each tube for the corresponding part
3. Add 0.125μL of Ex Tax polymerase to each tube
4. Place in PCR machine applying the thermal cycle below:

98°C	30sec	1cycle
98°C	10sec	25cycle
55°C	30sec	
72°C	15sec	
72°C	5min	1cycle
25°C	∞	

Done by Ryan Y, Sophia Y
Redone by Ryan Y

Date: 20260802
Title: Prepare 2% gel for PCR
Purpose: validate PCR results
Procedure:

1. Weight 1.6g of agarose powder and measure 80mL of 1X TAE buffer
2. Add the ingredients to a flask and cover the flask with a plastic wrap punctured with holes
3. Microwave for 30 secs and agitate it. Repeat until there are no sediments.
4. Assemble the trays and the combs
5. Wait until no smoke , add 4μL of safety dye and agitate it
6. Pour melted gel to the molds
7. Wait until the gel solidifies

Done by Naomi

Date: 20260802
Title: Prepare 1% gel for PCR
Purpose: validate PCR results
Procedure:

1. Weigh 0.2g of agarose powder and measure 20mL of 1X TAE buffer
2. Add the ingredients into the flask, cover the flask with plastic wrap with punctured holes.
3. Microwave for 30 seconds and agitate it. Repeat until no particles are present
4. Assemble the tray and the comb
5. Add 1μL of safety dye to the gel
6. Pour the gel into the mold
7. Wait until the gel solidifies
   Done by Naomi

Date: 20260802
Title: Enzyme Digestion
Purpose: Check plasmid size before DNA sequencing
Procedure:

1. Make sample for each vector to be digested and undigested using:
   G10: PcpcG2-veg-EEBD-RBS0-csn:LEA-bkdb-cold double

10x restriction buffer	2μL
DNA plasmid	10.916μL
Nuclear free water	7.984μL

G11: PcpcG2-veg-EEBD-RBS0-csn:LEA-bkdb-cold double

10x restriction buffer	2μL
DNA plasmid	8.7719μL
Nuclear free water	9.2281μL

G12: PcpcG2-veg-EEBD-RBS0-csn:LEA-bkdb-cold double

10x restriction buffer	2μL
DNA plasmid	8.0645μL
Nuclear free water	9.9355μL

GanA-GalA level 0

10x restriction buffer	2μL
DNA plasmid	1.7985μL
Nuclear free water	10.2015μL

2. For digested sample , add 0.5 μL of EcoR1
3. For undigested sample , add 2μL of ddH_2O
4. Incubate mixture at 37℃
   Done by Ryan Y, Olivia Lin

Date: 20260802
Title: Transformation
Purpose: Transform and cultivate engineered bacteria on agar plate
Procedure:
ABDI:(PtagC+RBS-Ho1:PcyA-L3S2P21)A+(PliaG-RBS0-CcaSm3-B0015)B+(Pveg-RBS0-ccaR-B0015)D+(PcpcG2-veg-EEBD–RBS0-csn:AccD-B0015)I
ACDE:(PtagC+RBS-Ho1:PcyA-L3S2P21)A+(PlepA-RBS0-CcaSm3-B0015)C+(Pveg-RBS0-ccaR-B0015)D+(PcpcG2-veg-EEBD-RBS0-csn:AmilcP-B0015)E
ABDE:(PtagC+RBS-Ho1:PcyA-L3S2P21)A+(PliaG-RBS0-CcaSm3-B0015)B+(Pveg-RBS0-ccaR-B0015)D+(PcpcG2-veg-EEBD-RBS0-csn:AmilcP-B0015)E
ACDI:(PtagC+RBS-Ho1:PcyA-L3S2P21)A+(PlepA-RBS0-CcaSm3-B0015)C+(Pveg-RBS0-ccaR-B0015)D+(PcpcG2-veg-EEBD–RBS0-csn:AccD-B0015)I
L:3P-EEBD-RBS0-GanaA-Gala-L3S2P21

1. Set the water bath to 42°C and prepare ice
2. Get 1 LBA plate and 4 LBC plates
3. Thaw competent cells until they’re 50% thawed and add 5μL of plasmid
4. Incubate on ice for 20min
5. Heat shock at 42°C for 45 seconds
6. Incubate on ice for 3 min
7. Add 1mL SOC medium
8. Incubate for 1 hr at 37°C
9. Centrifuge at 5000g for 5 min
10. Remove 1 mL of supernatant then vertex
11. Spread on their respective plates
12. Incubate at 37°C for 16hr
    Done by Alex
    Supervised by Gina

Date: 20260803
Title: Gel extraction of Mf001, Mf002, L3S2P56
Purpose: Extract targeted DNA
Procedure:

1. Place the tubes in a dry bath on 60℃ to dissolve the gel slices from 20260802
2. Load no more than 700μL of melted gel into the collection tube
3. Centrifuge for 30sec at 7000g, discard the flow through. Repeat step 2 and 3 until all the for rest of the mixture
4. Wash the column once with 0.5mL of WN buffer by centrifuging for 30sec.Discaid FT
5. Wash the column once with 0.5mL of WS buffer by centrifuging for 30sec. Discard FT
6. Centrifuge the column at full speed 23444g for 3min. To remove residual ethanol
7. Place column into new 1.5mL microtube. Add 15μL of EB buffer onto the center of the membrane
8. Stand column for 2 min. Centrifuge at 23444g for 2 min to elute DNA

Done by Sophie C

Biodrop Results:

part	Concentration(ng/μL)	A230/260	A260/180
MF001	15	1.071	1.875
MF002	19	1.583	1.900
L3S1PS2-L3S2P56	14	0.875	1.750

Date: 20260803
Title: Colony PCR
Purpose: Amplify the number of targeted gene
Procedure:

1. Prepare the reaction as followed:

	1rxn	26 rxn
Fr premier (Exp)	1μL	26μL
Rv promier (Exp)	1μL	26μL
Q-amp master mix	10μL	260μL
dd H_2O	8μL	208μL

ABDE: ACDI: 6486 bp

2. Separate 20μL of the master mix into each PCr tubes (24 PCR tubes in total)
3. Pick the total of 24 colonies from the three plates under UV light (

4. Pick up colonies using tip after turning on the bunsen burner
5. Point the tip(with colony) on another plate (labeled) and slightly point it for 3 times
6. Place the tip in PCR tubes and let it incubate for 5min, gently mix by shaking horizontally
7. Put it in the PCR machine

Thermal cycle:

98°C	2 min	1 cycle
98°C	30 secs	30 cycles
55°C	4 secs	
72°C	2 min	
25°C	1 min	1 cycle

Done by Sophie H, Sophia Y.

Date: 20260803
Title: Golden gate assembly
Purpose: Assemble level 0 parts
Procedure:

A. MF001

Component	Amount	Dilution
Vector STK001_pSTK-0-sfGFP ● concentration(ng/μL): 346 ● bp: 3090	0.05804913295 μL	1.16 μL (dilute 1/20)
Insert MF001 ● concentration(ng/μL): 15 ● bp: 82	0.07106666667 μL	1.14 μL (dilute 1/16)
T4 DNA ligase buffer	2 μL	
Sterile water	13.7μL	
T4 DNA ligase	1 μL	
BbsI-HF	1 μL	

Total	20 µL	

B. MF002

Component	Amount	Dilution
Vector STK001_pSTK-0-sfGFP ● concentration(ng/µL): 346 ● bp: 3090	0.05804913295 µL	1.16 µL (dilute 1/20) 0.5µL vector 9.5µL ddH_2O
Insert MF002 ● concentration(ng/µL): 19 ● bp: 82	0.056 µL	1.12 µL (dilute 1/20)
T4 DNA ligase buffer	2 µL	
Sterile water	13.72 µL	
T4 DNA ligase	1 µL	
BbsI-HF	1 µL	
Total	20 µL	

C. L3S1P52:L3S2P56

Component	Amount	Dilution
Vector STK001_pSTK-0-sfGFP ● concentration(ng/µL): 346 ● bp: 3090	0.05804913295 µL	1.16 µL (dilute 1/20)
Insert L3S1P52:L3S2P56 ● concentration(ng/µL): 14 ● bp: 175	0.1625 µL	1.14 (dilute 1/7)
T4 DNA ligase buffer	2 µL	

Sterile water	13.7 µL	
T4 DNA ligase	1 µL	
BbsI-HF	1 µL	
Total	20 µL	

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	1.5 minutes	25-45 cycles
16 °C (Ligation)	3 minutes	
37 °C (Restriction enzyme digestion)	5 minutes	1 cycle
60 °C (Heat inactivation of BsaI restriction enzyme)	10 minutes	1 cycle

Done by Sophie C.

Date: 20260803
Title: Make 0.8% gel
Purpose: To prepare gel for gel electrophoresis
Procedure:

1. Assemble comb, tray mold
2. Add 0.64g of agarose powder and 80mL 1XTAE buffer into conical flask
3. Cover flask with plastic wrap & puncture holes
4. Microwave for 30sec. Agitate. Repeat fill no sediments
5. When no smoke, add 4µL safety dye.Agitate
6. Pour melted gel into the mold
7. Wait until gel solidifies

Done by Sophie C/H,Naomi

Date: 20260803
Title:Gel electrophoresis
Purpose: ensure plasmid size for each samples

Procedure:

1. Load gel in the following format

   Gel 1

   M L1 L2 L3 L4 L5 L6 L7 L8 L9 L10 L11 L12 L13 L14 L15 L16 W1 X1 X2 X3 X4 X5 X6 X7

   5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5

   M:1kb ladder

   L:3P-EEBD-RBS0-GanaA-Gala-L3S2P21

   W:ABDE

   X:ACDI

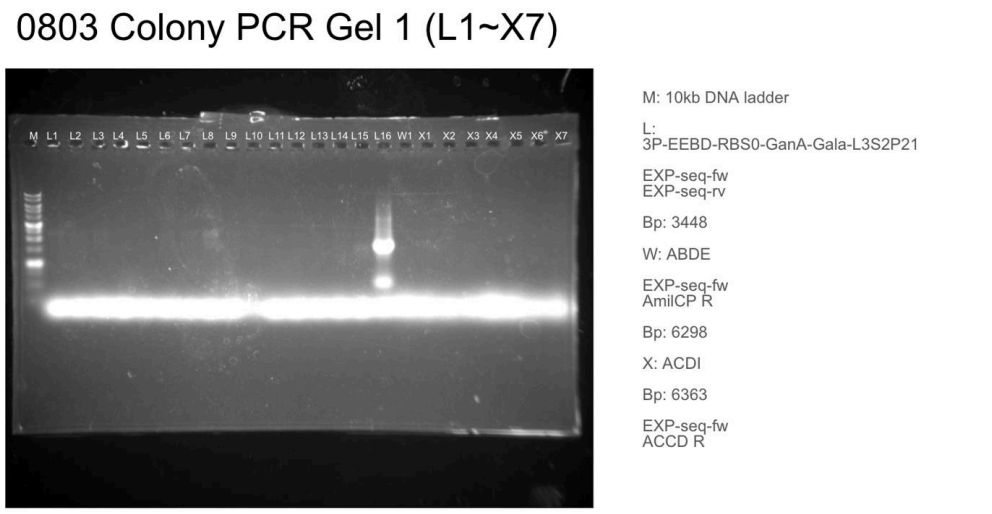
   Gel2

   M A1 A2 A3 A4 A5 A6 A7 A8 H1 H2 H3 H4 H5 H6 H7 H8

   5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5

   A:PlagC-RBS-H01:PcyA-L3S2P21

   H:Pveg-EEBD-RBS0-csn:LEA:lkdB coldD





Done by Ethan C, Sophie C

Date: 20260803
Title: Liquid Culture
Purpose: bacteria culture
Procedure:
J: Pveg-EEBD-RBS0-csn:AccD-B0015

1. Label culture tube
2. Turn on bunsen burner
3. Add 5mL LB medium and 25μL of erythromycin to the tube
4. Dip a pipette in J1
5. Dispense tip into tube
6. Place tube in the incubator(shape)

Done by Sophie C

Date: 20260804
Title: 250mL TB medium preparation
Purpose: Prepare TB medium for Arabidopsis planting on plates
Procedure:

1. Weight 11.9g of broth powder, 250mL ddH_2O, and 2mL trehalose
2. Add the ingredient above in order (broth powder→ddH_2O→trehalose) in 250mL flask and add a stirring bar
3. Open stirring machine and let it stir until all the broth powder was fully dissolved
4. Cover the flask with aluminum foil and stick an autoclave tape on top
5. Autoclave for 20 minutes at 127°C.

Done by Sophie H, Naomi

Date:20260804
Title: Miniprep of A8, H4, and H5
Purpose: Isolate plasmid DNA
Procedure:
A8: Pveg-Ho1:PcyA (did to step 5 because of too obvious blue color through observation)
H4: Pveg-Csn:LEA
H5: Pveg-Csn:LEA

1. Transfer 1.5mL Tb medium to a 2mL microtube
2. Centrifuge at 16000g for 1min
3. Discard the supernatant
4. Repeat step 2 and 3 until all the liquid culture were collected
5. Add 200μL of MX1 buffer
6. Resuspend the pellet by vortexing or pipetting
7. Add 250ml of MX2 buffer
8. Gently mix the buffer well by inverting the microtube 8-10 times
9. Incubate at Rt for 2min
10. Add 350mL of MX3 buffer
11. Gently mix via inversion 4~5 times
12. Centrifuge at 23444g for 5 min
13. Assemble the column and the tube while waiting for centrifuge
14. Transfer the supernatant (approximately 800μL) to the column
15. Centrifuge the column at 7000g for 30sec
16. Discard the flow through
17. Add 500μL WN buffer
18. Centrifuge the column at 7000g for 30sec
19. Discard the flowthrough
20. Add 700μL of WS buffer
21. Centrifuge the column at 7000g for 30sec
22. Discard the flow through
23. Centrifuge the empty column 10,000g for 3min
24. Place the column into new 1.5mL microtube
25. Add 50μL of EB buffer to the center of the membrane

26. Incubate at RT for 3 min
27. Centrifuge at 10,000g for 3min

Done by Sarah, Naomi

Biodrop Result:

	Concentration(ng/μL)	A260/230	A260/280
H5	131.0	1.899	1.048
H4	132.0	1.913	1.048

Date: 20260804
Title: Colony PCR
Purpose: Amplify DNA plasmid sequence G,L
Procedure:

G: PcpcG2-veg-EEBD-RBS0-csn:LEA-bkdb cold double
L:3P-EEBD-RBS0-GanaA-Gala-L3S2P21

1. Prepare the following reaction:

G:

	1rxn	26rxn
Csnfw	1μL	26μL
pSTK-seq rv	1μL	26μL
Q-amp master mix	10μL	260μL
ddH_2O	8μL	208μL

L:

	1rxn	26rxn
pStK-Exp-seq fw	1μL	26μL
pSTK-Exp-seq rv	1μL	26μL
Q-amp master mix	10μL	260μL
ddH_2O	8μL	208μL

2. Pick up colonies with tip and streak on grid
3. Wait for 5 mins and pick up tips
4. Put in thermal cycle

95°C	5min	1cycle
95°C	20sec	35cycle
65°C	20sec	

72°C	3min40sec	
72°C	5min	1cycle

Done by Alex, Sophie C, Chloe

Date:20260804
Title: Colony PCR
Purpose: Amplify DNA plasmid sequence W,ACDI,ACDE
Procedure:

1. Prepare the following reaction
   W:

	1rxn	3rxn
CcaR Fw	1μL	3μL
PcpcG2-veg rv	1μL	3μL
Q-amp master mix	10μL	30μL
ddH_2O	8μL	24μL

   X:

	1rxn	12rxn
CcaR Fw	1μL	12μL
PcpcG2-veg rv	1μL	12μL
Q-amp master mix	10μL	120μL
ddH_2O	8μL	96μL

   V:

	1rxn	4rxn
CcaR Fw	1μL	4μL
PcpcG2-veg rv	1μL	4μL
Q-amp master mix	10μL	40μL
ddH_2O	8μL	32μL

2. Pick up colonies with tip & streak on grid
3. Wait 5 min and take out tips
4. Put in thermal cycle(same as previous page)
   Done by Chloe & Sophie C

Date:20260804

Title: make 0.8% gel
Purpose:prepare gel for electrophoresis
Procedure:

1. Add 0.64g agarose powder and 80mL of 1XTAE buffer into a conical flask
2. Cover it with a plastic wrap and puncture holes
3. Assemble tray, mold, comb
4. Microwave flask for 30sec. Agitate. Repeat until no sediments
5. When no smoke, add 4μL safety dye. Agitate
6. Pour melted gel into mold
7. Let it solidify at RT
   Done by Sophie C, Chloe

Date:20260804
Title: make 1% gel
Purpose:prepare gel for electrophoresis
Procedure:

1. Add 0.4g agarose powder and 40mL 1XTAE buffer into conical flask
2. Cover it with a plastic wrap and puncture holes
3. Microwave flask for 30sec Agitate. Repeat until no sediments
4. Assemble mold, fray, comb
5. When there is no smoke, add 2μL safety dye. Agitate
6. Pour melted gel into mold
7. Let it solidify at RT
   Done by Sophie C, Chloe

Date:20260804
Title: Transformation
Purpose: Amplification MF001, MF002,L3S1PS2-L3S2P56, V:ABDI, X:ACDI
Procedure:

1. Set the water bath to 42°C and prepare ice
2. Get 1 LBA and 5 LBC plate
3. Thaw competent cells until they're 50% thawed and add 10μL plasmid to 100μL competent cell
4. Incubate on ice for 20min
5. Heat shock at 42°C for 45 sec
6. Incubate on ice for 3 min
7. Add 1mL SOC medium
8. Incubator for 1hr at 37°C
9. Centrifuge at 5000g for 5 min
10. Remove 1mL of supernatant & vortex
11. Spread on their respective plates
12. Incubate at 37°C for 16hr
    Done by Sophie C, Chloe

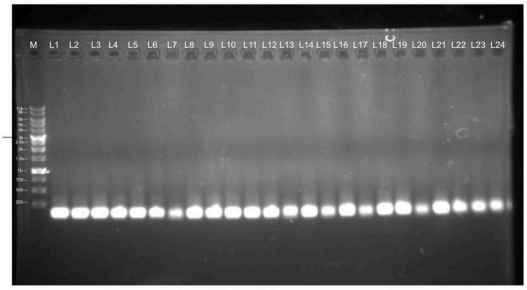
Date:20260804
Title: Liquid culture
Purpose: bacteria culture from successful colony PCR product
Procedure:
A:PtagC+RBS-Ho1-PcyA-L3S2P21
G7~9:pcpcG2-veg-eEBD-RBS0-csn:LEA-bkbd cold

1. Label 4 liquid culture tubes
2. Add 3mL TB medium and 3 μL ampicillin in each tube
3. Dip a pipette into colonies (each)
4. Dispense tips in tubes(each)
5. Place tubes in incubator(shake)
   Done by Sophie C

Date:20260804
Title: Gel electrophoresis
Purpose: validate the bps of pcr results
Procedure:
W:ABDE
X:ACDI
V:ACDE

1. Load the samples in the following order:
   Gel 1
   M W2 W3 X8 X9 X11 X12 X13 X14 X15 X16 X17 X18 V1 V2 V3
   5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5
   Gel2
   M L1 L2 L3 L4 L5 L6 L7 L8 L9 L10 L11 L12 L13 L14 L15 L16 L17 L18 L19 L20 L21 L22
   5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5
   L23 L24
   5 5
2. Resolve the gel at 100V for 30min

0804 Colony PCR Gel 1



M: 10kb DNA ladder

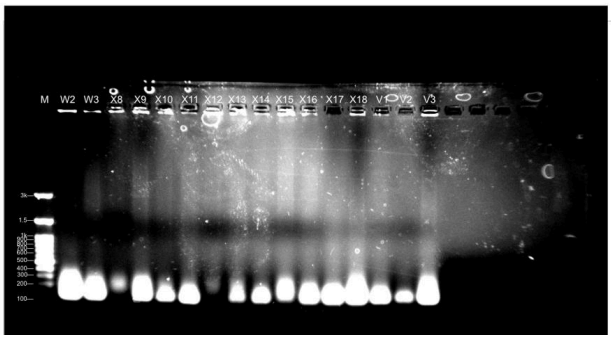
L:
3P-EEBD-RBS0-GanA-Gala-L3S2P21

Bp: 3516

EXP-seq fw primer
EXP-seq rv primer

All fail -> ReGolden Gate

0804 Colony PCR Gel 2



M: 3kb DNA ladder

W: ABDE
X: ACDI
V: ACDE

CcaR fw primer
PcpcG2-veg rv primer
Bp:1045

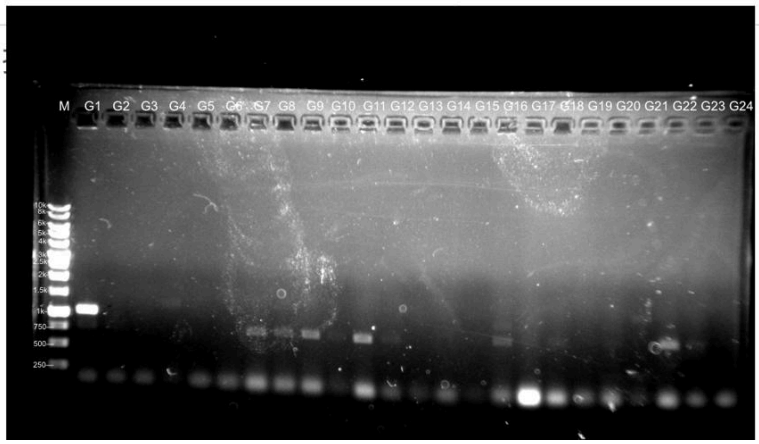
All fail -> Retransform 0804

Done by Sophie C, Chloe

Date:20260804
Title: Gel electrophoresis
Purpose: validate the bps of pcr results
Procedure:
G:PcpcG2-veg-EEBD-RBS0-csn:LEA-blckB-cold

1. Load the samples in the following order:
   M G1 G2 G3 G4 G5 G6 G7 G8 G9 G10 G11 G12 G13 G14 G15 G16 G17 G18 G19
   5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5
   G20 G21 G22 G23 G24
   5 5 5 5 5
2. Resolve at 100V for 30min

0804 Colony PCR Gel 3



M: 10kb DNA ladder

G:
PcpcG2-veg-EEBD-RBS0-Csn:Lea-bkdB-Cold

Csn fw primer
STK-seq rv primer

- Bp: 787

G7, G8, G9 -> liquid culture 0804

Done by Sophie C

Date:20260804
Title: Add 正瀚protectant to exp7. Circular plates (pretreatment group: 0, 正瀚/ 100 NaCl, 正瀚)
Purpose: to test when to add the protectant is the best

Procedure:

1. Dilute 正瀚protectant by 1/500 (add 6986 μL water + 14μL protectant) into the 15mL tube in the BSC
2. Add 1mL of the diluted protectant into the 2 agar plates (0, 正瀚/100mM NaCl, 正瀚)
3. Evenly distribute the liquid protectant by filling the plate sideways until all the plants are covered with a thin layer protectant
4. Seal the plates with tape and label the pretreatment date

Done by Sophia Y.
Supervised by Audrey H, Abby T., Abigail

Date: 20260804
Title: Agar exp set 6 fresh weight
Purpose: compare different groups' growth results
Procedure:

1. Using a tweezer, lift and pick all 9 arabidopsis simultaneously from the agar plate in a single motion
2. Careful not to hurt the plants by squeezing the tweezer
3. Lift from below the leaf area
4. Put the 9 plants on a tissue paper, lightly press to remove the agar sticking to the plants
5. Careful not to press the leaf areas too hard
6. Place all 9 plants in a weighing plate, weight the fresh weight of arabidopsis (a plate at a time)
7. Pick them up and put them(plants) in a 2mL microtube, take care that the entire plant and the roots need to go in too
8. Label the group concentration as well as the respective fresh weight on the microtube
9. Repeat these steps for as many times as the sets of arabidopsis (8 sets)

Done by Audrey H
Supervised by Abby T

Result:

Group	Weight(g)
150mM NaCl	0.0175
Control	0.01756
100mM NaCl 5mM Trehalose	0.0517
75mM NaCl 5mM Trehalose	0.0655
75mM NaCl	0.0655
5mM Trehalose	0.1755

100mM NaCl	0.0298
150mM NaCl 5mM Trehalose	0.0187

Date: 20260805
Title: Miniprep
Purpose: Isolate plasmid DNA
Procedure:
G7, G8, G9: PcpcG2-veg-EEBD-RBS0-Csn:LEA-bdlB-cold
A1: PtagC+RBS-Ho1-pcyA-L3S2P2

1. Transfer 1.8mL of TB medium to a 2mL microtube
2. Centrifuge for 30sec 16000g, then discard the supernatant
3. Repeat set 1&2 until all the liquid culture were collected
4. Add 200μL MX1 buffer into each tube
5. Resuspend the pellet by vortexing or pipetting
6. Add 250μL of MX2 buffer
7. Let it incubate in RT for 2 min after inverting for 8~10times
8. Add 350μL of MX3 buffer, then slowly invert it 4~5times
9. Centrifuge at 23444g for 5min
10. Assemble the column and the tube
11. Transfer about 400μL of supernatant into the column
12. Centrifuge the column at 7000g for 3 min
13. Discard the flow through, then add 500μL WN buffer
14. Centrifuge the column at 7000g for 3 min
15. Discard the flow through, then add 500μL of WS buffer
16. Centrifuge the column at 7000g for 3min
17. Discard the flow through
18. Centrifuge the empty column at 1000g for 3 min and place it into new 1.5mL microtube
19. Add 50μL EB buffer into the column (right in the center)
20. Incubate at RT for 3 min
21. Centrifuge at 10000g for 3 min
22. Using biodrop testing the concentration

	Concentration(ng/μL)	A260/280	A260/230
A1	109.0	1.912	0.940
G7	115.0	1.917	0.966
G8	145.9	2.014	1.280
G9	147.0	1.934	1.114

Done by Naomi, Sophie H

Date: 20260805
Title: Colony PCR
Purpose: Amplify DNA plasmid sequence
Procedure:

MF: MF002
LL:L3S1PS2-L3S2P5G
X:XACDI

1. Prepare the following reactions and distribute into PCR tube

LLx12rxn

Fw primer L3S1PS2 fw	12µL
Rv primer L3S1PS2 rv	12µL
Q-amp master mix	120µL
ddH_2O	96µL

MFx13rxn

Fw primer MF002 fw	13µL
Rv primer MF002 rv	13µL
Q-amp master mix	130µL
ddH_2O	104µL

Lx26rxns

Fw primer Exp-seq-fw	26µL
Rv primer Exp-seq4-rv	26µL
Q-amp master mix	260µL
ddH_2O	208µL

Xx2rxn

Fw primer CcaR fw	2µL
Rv primer PcpcG2-veg rv	2µL
Q-amp master mix	20µL

ddH_2O	16μL

2. Pick up colonies with tip&shake on grid
3. Wait 5min&take out tips
4. Put in thermal cycle:

95℃	5min	x1
95℃	20sec	x35
60℃	20sec	
72℃	4min	
72℃	5min	x1

Done by Sophie C, Abby T

Date: 20260805
Title: Make 1% & 0.8% gel
Purpose: make gel for electrophoresis of colony PCR
Procedure:

1. Add the ingredient as follow in to two 250ml flask
2. 1%-> 0.4g agarose powder / 40ml TAE buffer
   0.8%->0.36g agarose powder / 40ml TAE buffer
3. Put the flask into microwave and heat until no powder in the flask
4. Wait for it to stop smoking and add 2μL safety dye into each flask
5. Pour the gel into two assembled tray, use tip to push the air out and pop bubbles
6. Assemble the comb
7. Wait for the two tray to solidify
   Done by Sophie H

Date: 20260805
Title: Protein Purification & extraction
Purpose: Isolating a specific protein based on binding affinity(his-tag)
Procedure:

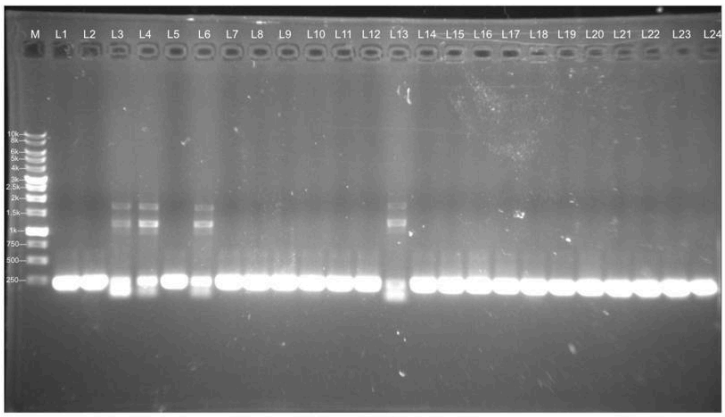
1. Take 24 2mL centrifuge tubes and add 1800μL of overnight liquid culture(*Bacillus subtilis*)
2. Centrifuge 16000g for 30seconds
3. Take tubes out of microcentrifuge and pour the supernatant into a 50mL centrifuge tube
4. Repeat the steps 1-3 until the 50mL centrifuge tube is filled
5. Pour the supernatant into a syringe connected to 0.22um filter and filter the supernatant into a 50mL centrifuge tube
6. Repeat the previous steps until all culture is filtered
7. Centrifuge the Ni-NTA column at 175g for 2 min to remove the storage buffer

8. Add 6mL of equilibrate buffer into the column by dispensing the buffer along the walls of the tube
9. Add 237.5μL of imidazole into supernatant
10. Add all supernatant from 1 tube into the column
11. Add 6mL of wash buffer and remove the flow-through repeat this step 3 times
12. Centrifuge at 175g for 2 min
13. Collect the flow through from the last wash buffer step
14. Add 3mL of elute buffer
15. Centrifuge at 175g for 2min
16. Collect the flow-through as the purified protein
    Note: only did one tube, others filtered store in refrigerator for 0806 purification
    Done by Abigail
    Supervised by Gabriel

Date: 20260805
Title: Gel electrophoresis
Procedure:
MF: MF002
LL:L3S1PS2-L3S2P5G
L:3P-GanaA-Gala
X:XACDI

1. Load the gels in the following order
   Gel1:M L1 L2 L3 L4 L5 L6 L7 L8 L9 L10 L11 L12 L13 L14 L15 L16 L17 L18 L19 L20
   5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5
   L17 L18 L19 L20 L21 L22 L23 L24
   5 5 5 5 5 5 5 5
   Gel2: M MF1 MF2 MF3 MF4 MF5 MF6 MF7 MF8 MF9 MF10 MF11 MF12 LL1 LL2 LL3
   5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5
   LL4 LL5 LL6 LL7 LL8 LL9 LL10 LL11 X1
   5 5 5 5 5 5 5 5 5
2. Resolve the gel at 100V for 30sec

0805 Colony PCR Gel 1



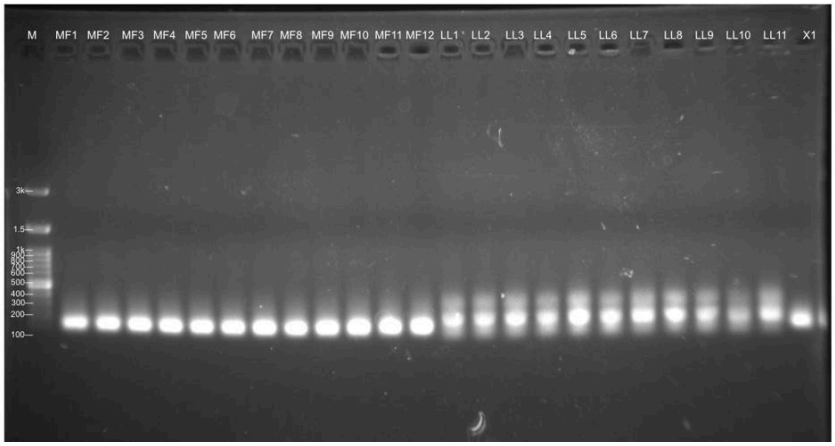
M: 10kb DNA ladder

L: 3P-EEBD-RBS0-GanA-Gala-B0015

Primers : EXP-seq-fw & EXP-seq4-rv

Bp: 3626 bp

0805 Colony PCR Gel 2



M: 3k DNA ladder

MF: MF002

Primers : MF002 fw & MF002 rv

Bp:46 bp

LL: L3S1PS2-L3S2P56

Primers: L3S1PS2-L3S2P56 fw & L3S1PS2-L3S2P56 rv

Bp: 139 bp

Done by Sophie C

Date: 20260805
Title: Golden Gate Assembly
Purpose: assemble level 1 parts to level 2
Procedure:
A: pSTK-1A-Ptag+RBS-Ho1:pcayA-L3S2P21
B:psTK-1B-Pllaq-RBS0-CcaSm3-B0015
C:pSTK-1B-PcepA-RBS0-CcaSM3-B0015
D:pSTK-1C-Pveg-RBS0-CcaR-B0015
I:pSTK-1D-PcpcG2-veg-EEBD-RBS0-csn:AccD-B0015
C:pStK-Exp1-3P-EEBD-RBS0-GanA-Gala-L3S2P21

1. ✓ABDI

Part	Original	Dilution
pSTK-Exp2-SfGfP (163ng/μL 5104bp)	0.2μL	1.02μL(⅕)
A (109ng/μL 3754bp)	0.46μL	1.37μL(⅓)
B (83ng/μL 3793bp)	0.59μL	1.78μL(⅓)
D (134ng/μL 3243bp)	0.31μL	0.26μL(⅓)
I (165ng/μL 3757)	0.29μL	1.48μL(⅕)
T4 DNA ligase buffer	2μL	
T4 DNA ligase	1μL	
BsmBI-r2	1μL	
ddH_2O	9.05μL	

x:ACD1

Part	Original	dilution
pSTK-Exp2-SfGfP (163ng/μL 5104bp)	0.2μL	1.02μL(⅕)
A	0.46μL	1.37μL(⅓)
C (98ng/μL 3744bp)	0.31μL	1.26μL(¼)
I	0.29μL	1.48μL(⅕)
T4 DNA liq buff	2μL	
T4 DNA liq	1μL	
BsmB1-v2	1μL	
ddH_2O	9.35μL	

L

Part	Original	dilution

pSTK-Exp1	0.28μL	1.4μL(⅛)
3P	0.36μL	1.43μL(¼)
EEBD-RBS0	0.21μL	1.05μL(⅛)
GanaGala	0.3μL	1.22μL(¼)
L3S2P21	0.16μL	1.64μL(1/10)
pSTK-ocx-spacer	0.11μL	1.07μL(1/10)
pSTK-DC2-nissx	0.12μL	1.16μL(1/10)
T4 DNA liquid buffer	2μL	
T4 DNA liq	1μL	
nsa-Hf-r2	1μL	
ddH_2O	7.05μL	

Done by Chloe, Sophic C

Date: 20260805
Title: Make acrylamide gel
Purpose:separates proteins or small nucleic acids by size during electrophoresis.
Procedure:

1. Assemble holder, plates, comb
2. Prepare resolves solution

Resoever A	3mL
Resoever B	3mL
10% APS	30μL
TMLd	3μL

3. Full to 1.5-1cm below bottom of comb teeth w resolver solution(~6mL)
4. Prepare stacker solution

Stacker A	1ml
Stacker B	1mL
10% APS	10μL
Tmed	2μL

5. Fill to top with stacker solution(~2mc)

6. Insert comb and allow the gel to polymerize
   Done by Sophie C

Date: 20260805
Title: agar (exp set #6) chlorophyll extraction
Purpose: compare different groups' growth results
Procedure:

1. Take the microtubes from 0804
2. Dilute acetone to 80%, add 4mL to each tube
3. Add 10 metal bits into each tube
4. Wait for the chlorophyll to extract, and use a vortex to crush the plants to speed up the process (done when the plants are no longer green)
5. Add the liquid to the 96 well, 200μL in each well, 3 wells for each sample
6. ELISA Reader, insert the 96 well, run the results through the equation
7. Make sure to select the correct wavelength
    a. A633 -> used the wrong wavelength this time, it should be 663

Done by Audrey H, Phoebe, Sarah C

Date: 20260805
Title: Seed planting on agar
Purpose: grow the seeds on agar
Procedure:

1. Open the BSC
2. Use the seed template (circular plates) to plant 25 seeds on each agar plate
3. Use breathable tape to seal each of the agar plates
4. Label the vernalization date, incubator date, and seed type
5. Wrap the agar plates with aluminium foil
6. Vernalize in the 4°C refrigerator for 4 days and avoid light

Done by Audrey H, Naomi, Sophie H

Date: 20260805
Title: Add CH protectant and NaCl on horizontal agar plates(exp set #7)
Purpose: to test when to add protectant is the best (Co-treatment)
Procedure:

1. Use the following amount to apply for each plate

3M NaCl	1667μL
CH biopeptide	1000μL (dilution rate 1/500 -> 6986 μL sterile water + 14 μL peptide)

2. Apply NaCl for the following plates

Co-treatment	100mM NaCl, CH biopeptide (Day 0)
Pre-treatment	100mM NaCl, CH biopeptide (Day -1)
After-treatment	100mM NaCl, CH biopeptide (Day 1)

3. Apply CH biopeptide to the following plates

Co-treatment	100mM NaCl, CH biopeptide	(Day 0)
	0mM NaCl, CH biopeptide	

Done by Audrey H

Date: 20260806
Title: Colony PCR of G, H
Purpose: Amplify desired DNA sequence
Procedure:

1. Prepare the master mix as below:

	1 rxn	34rxn
Fr Primer (biobrick prefix)	1μL	34μL
Rv primer (LEA reverse)	1μL	34μL
ddH_2O	8μL	272μL
Qamp master mix	10μL	340μL

	1 rxn	30rxn
Fr Primer (pSTK Exp1 Fw)	1μL	30μL
Rv Primer (pSTK Exp1 Rv)	1μL	30μL
ddH_2O	8μL	240μL
Qamp master mix	10μL	300μL

2. Mix the master mix by vortexing
3. Distribute the master mix into PCR tubes(remember to label the PCr tubes first, each PCR tube should have 20μL master mix）
4. Centrifuge PCR tubes before starting to pick up colonies
5. Open bunsen burner, and use pipette tip to pick up labelled single colony
6. Dip/draw a few lines with the tip on another labeled LBA plate
7. Dip the tip into respective PCR tubes and dispense it inside
8. After all the labeled colonies were collected, let the PCR tubes incubate at RT for 5min, gently shake the tip at the same time
9. Remove pipette tips and put PCr tubes in the PCR machine
10. Set up thermal cycle as below:

95°C	2 min	1 cycle
95°C	30 sec	30 cycles
65°C	40 sec	
72°C	2.5 min	
72°C	1 min	1 cycle

Done by Sophie H, Sarah

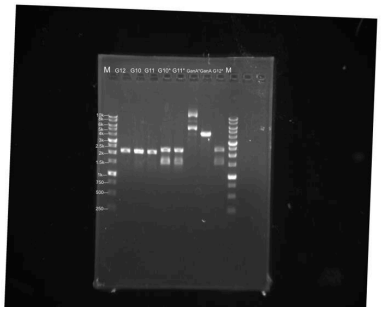
Date: 20260806
Title: Make 0.8% gel for enzyme digestion in 8/2
Purpose: Make gel for electrophoresis
Procedure:

1. Weight 0.32g of Agarose powder and 40mL TEB buffer
2. Add the ingredients above in 250mL flask and cover the top with double layer plastic wrap
3. Poke some holes on plastic wrap with pipette tip
4. Microwave the gel until the gel is clear
5. Remove the plastic warp and wait until the gel stops smoking but is still hot
6. Add 2μL of diluted safety dye into liquid gel and mix it by shaking the flask gently
7. Pour the gel into assembled tray to the black line in the tray
8. Use pipette tip to squeeze the air out and pop bubbles
9. Wait for gel to solidify about 30min)
   Done by Naomi

Date: 20260806
Title: Gel electrophoresis of 0802 enzyme digestion
Purpose: check the result of 0802 enzyme digestion

1. Load gel in the following format
   M G12 G10 G11 G10* G11* GanA* GanaA G12*
   5 5 5 5 5 5 5 5 5
   M:1kb DNA ladder
   G10~12:PcpcG2-veg-EEBD-RBS0-csn:LEA-bkdb-cold double
   GanA:GanA-Gala level0
   *undigested
2. Resolve in 100V for 30 min

0806 Enzyme digestion(8/3)



M: 10kb DNA ladder

G10~12
PcpcG2-veg-EEBD-RBS0-csn:LEA-bkdb-cold double
- Enzyme: EcoR1
- Bp:
  - 3234 (undigest)

GanA-GalA level 0
- Enzyme: EcoR1
- Bp:
  - 4148 (undigest)

*= undigested

Done by Sarah , Naomi

Date: 20260806

Title: Make 1% gel for colony PCR(25 well*3)
Purpose: Make gel for electrophoresis
Procedure:

1. Measure 1.2g of Agarose powder and 120mL of TEB buffer
2. Add the ingredient above into 250mL flask and cover the flask with double layer
3. Pop a few holes in plastic wrap with pipette tip
4. Microwave liquid gel until no powder can be seen
5. Take out plastic wrap and wait until no smoke
6. Add 6μL of safety dye into liquid gel
7. Pour the gel into 3 assembled tray
8. Wait for them to solidify
   Done by Sarah Chou, Sophie H

Date: 20260806
Title: Load gel for colony PCR
Purpose: Load gel for colony PCR product of G and H
Procedure:

1. Load the gel as following order:
   Gel1:

M	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12	G13	G14	G15	G16	G17
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5

G18	G19	G20	G21	G22	G23	M
5	5	5	5	5	5	5

   Gel2:

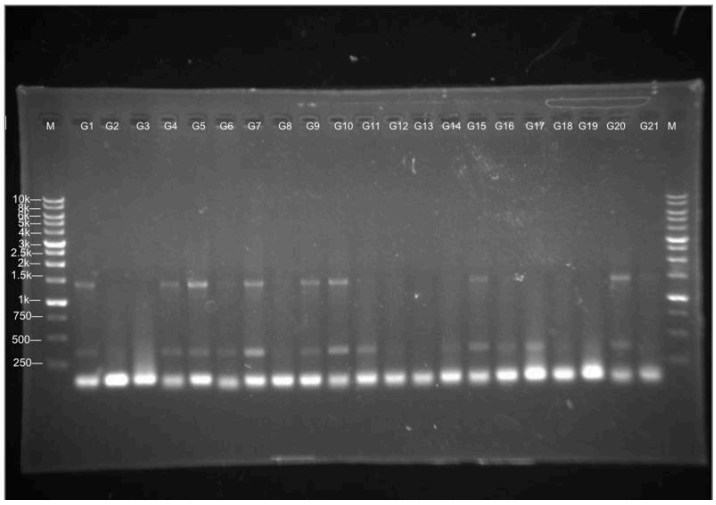
M	G24	G25	G26	G27	G28	G29	G30	G31	G32	H1	H2	H3	H4	H5	H6	H7	H8
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5

H9	H10	H11	H12	H13	H14	M
5	5	5	5	5	5	5

   Gel3:

M	H15	H16	H17	H18	H19	H20	H21	H22	H23	H24	H25	H26	H27	H28	M
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5

0806 Colony PCR G1~21



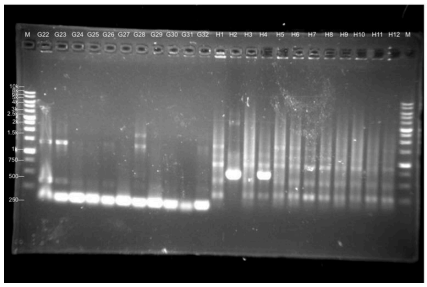
M: 10kb DNA ladder

G1~21
PcpcG2-veg-EEBD-RBS0-csn:LEA-bkdb-col d double

- FW primer: biobrick prefix
- RV primer: LEA reverse

Bp: 3.1K

0806 Colony PCR(G22~H12)



M: 10kb DNA ladder

G22~28:
PcpcG2-veg-EEBD-RBS0-csn:LEA-bkdb-col d double

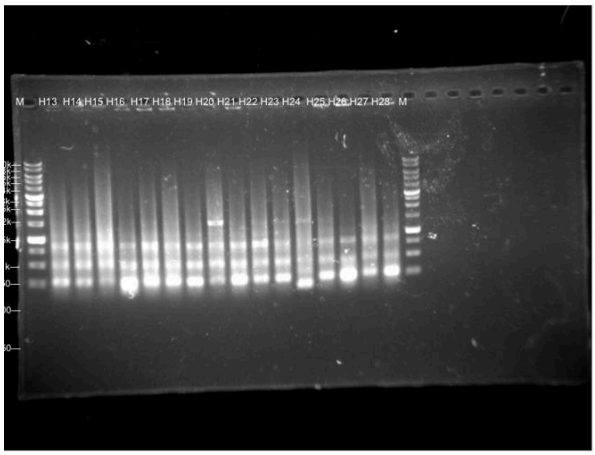
- FW primer: biobrick prefix
- RV primer: LEA reverse

H1~H12:
3P-EEBD-RBS0-Csn:LEA-B0015

- FW primer: pSTK Exp1 Fw
- RV primer: pSTK Exp1 Rv

Bp: 3.1K

0806 Colony PCR(H13~H28)



M: 10kb DNA ladder

H13~H28:
3P-EEBD-RBS0-Csn:LEA-B0015

- FW primer: pSTK Exp1 Fw
- RV primer: pSTK Exp1 Rv

Bp of H: 3.1K

Date: 20260806
Title: make LBA plate with glycerol
Purpose: prepare LBA plate
Procedure:

1. Weight 250mL ddH_2O and 6.25 agar powder*2
2. Add 1mL glycerol and put a stirring bar in
3. Stir the liquid mixture until no powder on glycerol can be seen
4. Cover the flask with double-layer aluminum foil
5. Autoclave for 20 minutes at 127°C.
6. Add 250μL ampicillin in the BSC while it's still a bit hot
7. Pour 20μL into each plate and wait for it to solidify
   Done by Sophi H

Date: 20260806
Title: Protein purification(take from 0805 filtered tubes)
Purpose: isolated ACCD protein
Procedure:

1. Use the following buffer for the procedures:(formula for 50mL and buffers are prepared/stocked beforehand)

Buffer	Imidazole final concentration(mM)	10xPBS(mL)	2M imidazole(pM)	water(mL)
Equilibrate buffer	10	5	250	44.75
Wash buffer	25	5	625	44.375
Elute buffer	150	5	3750	41.25

2. Centrifuge the Ni-NTA column at 175g for 2 min to remove storage buffer
3. Add 6mL of equilibrate buffer
4. Before adding supernatant, add and pipette 237.5 μl of imidazole to the tube of supernatant with 47.5mL volume and add 62.5 μl to the tube of supernatant with 12.5mL volume
5. Adding all supernatant into the column by washing the walls(sides) of the column
6. Add 6mL of wash buffer and remove the flow-through. Repeat this step 3 times. *At the 3rd time, collect the flowthrough to load gel
7. Centrifuge at 175g for 2 min
8. Collect the flow-through from the last wash buffer step
9. Add 3mL of elute buffer
10. Centrifuge at 175g for 2 min

11. Collect flow-through as purified protein

Note 1: did 2 tubes with volumes of 47.5mL and 12.5mL and store others in 4°C
Note 2: method to store Ni-NTA column after use

1. Add 30mL of Ni-NTA regenerate buffer(10x of resin's volume(3mL)) to the column to wash
2. Add 30mL ddH_2O(10x) of resin's volume) to the column to wash
3. Add 20% of ethanol in the column
4. Store at 4°C

Done by Abigail and Sophie C
Supervised by Gabriel

Date: 20260806
Title: SDS-PAGE electrophoresis
Purpose: separate proteins by size
Procedure:

1. Assemble casting frames with two 1.5mm glass plates
2. Prepare solutions for the stacking(top) and resolving(bottom) gels

components	stacker(top gel)	resolver(bottom gel)
Resolver A	x	4.5mL
Resolver B	x	4.5mL
Stacker A	1.5mL	x
Stacker B	1.5mL	x
Total Volume	3mL	9mL
TEMED	3 µl	4.5 µl
10% APS	15 µl	45 µl

3. Transfer resolving gel between glass plates using a 1000 µl pipette
4. Add the stacking gel to the top of the green part(line) of the assembled rack ~6mL
5. Insert comb to form wells and allow it to set for ~10-15min
6. Remove the comb and mount the gel into the electrophoresis tank
7. Mix 50 µl of protein sample (25mM and 150mM) with 10 µl of 6x Laemml: Sample Buffer(final volume's ⅙)
8. Heat samples at 95°C for 5 minutes
9. Briefly spin down
10. Fill the inner and outer chambers of the gel tank with 1x running buffer
11. Load the protein ladder in the first and last lane
12. Load prepared protein samples into remaining wells

Gel one(purpose: western blot):

	(0805)	(0805)	(0805)	(0805)	(0806)	(0806)	
Ladder	25mM	10mM	150mM	250mM	25mM	150mM	Ladder
10 µl	25 µl	25 µl	25 µl	25 µl	25 µl	25 µl	3 µl

Gel two(purpose: coomassie staining):

	(0805)	(0805)	(0805)	(0805)	(0806)	(0806)	
Ladder	25mM	10mM	150mM	250mM	25mM	150mM	Ladder
3 µl	25 µl	25 µl	25 µl	25 µl	25 µl	25 µl	10 µl

13. Close the lid and connect the electrodes
14. Run at constant voltage of 200V for 10 min until the dye reaches the bottom of the stacking gel then continue electrophoresis at a constant voltage(100-150V) for 40 min until the dye reaches near the bottom of the gel
15. Carefully disassemble the glass plates and collect the gel for subsequent analysis

Done by Abigail and Sophie C
Supervised by Gabriel

Date: 20260806
Title: Coomassie Blue Staining
Purpose: Visualize proteins in polyacrylamide gels after electrophoresis
Procedure:

1. Make the solutions and buffer according to the recipe below:

Coomassie blue solution recipe(40mL)	
methanol	50%
Acetic acid	10%

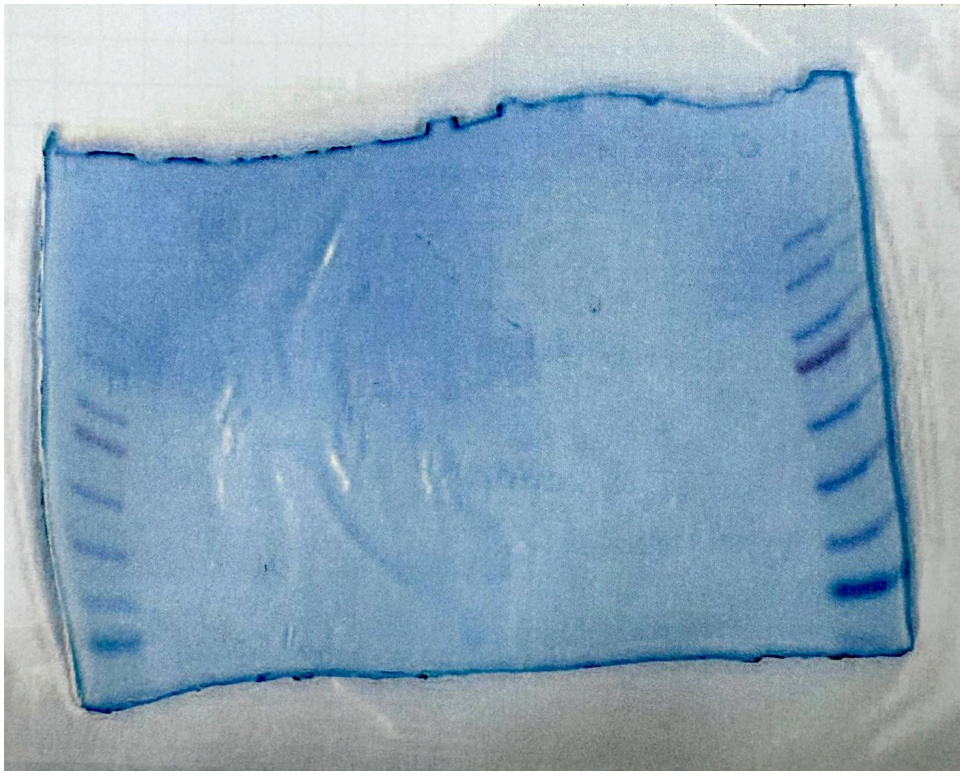
Coomassie blue destain buffer(40mL)	
methanol	30%
Acetic acid	10%

2. Transfer gel 2 to a staining tray
3. Wash gel with ddH_2O for 5 minutes, repeat 3 times

4. Stain the gel with 0.05% Coomassie Blue R250 for 1 hour on the shaker at 50 rpm
5. Destain the gel by soaking in Coomassie Blue destain buffer for at least 1 hour until protein bonds are clearly visible against a transparent background
6. Document the gel image and compare the protein bands to the ladder

Done by Abigail and Sophie C
Supervised by Gabriel

Results



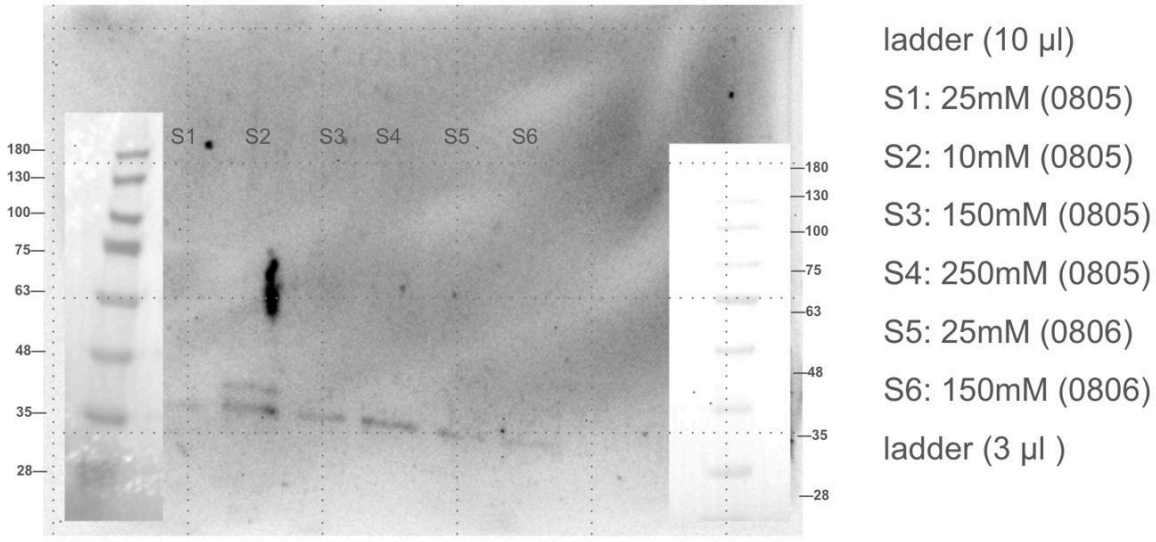
Date: 20260806
Title: Western Blotting
Purpose: to identify specific protein within a complex sample mixture using gel separation and antibody binding
Procedure:

1. After SDS-PAGE, carefully collect gel 1 for transfer
2. Soak the PVDF membrane in methanol for 1 minute.
2. Equilibrate the membrane in 1X transfer buffer for 3 minutes
3. Equilibrate the gel in 1X transfer buffer for 5 minutes
4. Wet 3M papers in 1X transfer buffer
5. Assemble the transfer sandwich in the cassette in the following order: sponge pad → filter paper → gel → PVDF membrane → filter paper → sponge pad. Ensure there are no air bubbles between the gel and membrane
6. Place the cassette into the tank filled with cold 1X transfer buffer and an ice block
7. Transfer at 100 V for 90 minutes

8. Block in 5% non-fat milk in TBST for 2 h at room temperature
9. Replace blocking solution by anti- His-tag mouse mAb (1:1000) In 5% milk
10. Incubate on the orbital shaker at 4°C overnight

Done by Sophie C and Abigail
Supervised by Gabriel

Results



Date: 20260806
Title: Protein Dialysis
Purpose: Remove imidazole which may interfere with future experiments
Procedure:

1. Dilute 10x PBS: Mix 160mL of 10xPBS and 1.6L of deionized water
2. Adjust pH to 8
3. Top up H_2O to 1.6L and wrap opening of beaker with plastic wrap
4. Prepare three beakers of 1.6L PBS
5. Open the dialysis cassette by gently twisting the cap counterclockwise
6. Add the protein lysate to the cassette, slowly withdrawing the pipette while dispensing. Don't overload the cassette
7. Remove the excess air in the cassette by simultaneously pressing the membrane gently on both sides using gloved thumb and forefinger and insert the cap
8. Insert cap and lock by gently twisting it clockwise until the arrows align and a firm stop is felt
9. Place dialysis cassette in 2 L beaker with 1.6 L 1X PBS and stir bar. Stir on ice for 2 hours
10. Replace 1X PBS and stir on ice for another 2 hours
11. Replace 1X PBS and stir on ice overnight
12. Transfer lysate containing recombinant protein from dialysis cassette to 1.5 mL microtubes

13. The recombinant protein is stored at -80°C

Done by Sophie C and Abigail
Supervised by Gabriel

Date: 20260806
Title: Transformation
Purpose: to amplify DNA part
Procedure:
L: 3P-EEBD-RBS0-GanA-Gala-L3S2P21
X: ACDI
V: ABDI

1. Set the water bath to 42°C and prepare ice
2. Get 1 LBA and 2 LBC plates
3. Thaw competent cells until they are 50% thawed and add 10 µl of plasmid to 100 µl of competent cell
4. Incubate on ice for 20 min
5. Heat shock 42°C for 45 seconds
6. Incubate on ice for 3 min
7. Add 1 mL SOC medium
8. Incubate for 1 hr at 37°C
9. Centrifuge at 5000g for 5 min
10. Remove 1 mL supernatant and vortex
11. Spread on plates
12. Incubate at 37°C for 16hours

Done by Sophie C

Date: 20260806
Title: Add CH protectant on horizontal agar plates (exp set #7)
Purpose: test CH protectant
Procedure:

1. Use the following amount to apply for each plates CH biopepitde 1000µL (dillution 1/500-> 6986µL of sterile water +14µL of peptide)
2. Apply the protectant for the following plants
    - Aftertreatment
        - 100mM NaCL , CH biopeptide (Day 1)
        - 0,CH biopeptide (Day1)

Done by Sophic C and Naomi
Supervised by Abigail

Date:20260807
Title: Colony PCR
Purpose: Amplify desired DNA sequence
Procedure:
L: 3P-EEBD-RBS0-GanA-Gala-L3S2P21:

1. Prepare the following reaction

L *40

	40 reactions	1 reaction
Fw primer Exp1	40 µl	1 µl
RV primer Exp1	40 µl	1 µl
Q amp master mix	400 µl	10 µl
ddH_2O	320 µl	8 µl

2. Picked up colonies with tips and streak on grid
3. Wait 5 minutes and take out tip
4. Put in thermal cycle

95°C	5 min	1 cycle
95°C	20 sec	35 cycles
65°C	20 sec	
72°C	3 min 30 sec	
72°C	5 min	1 cycle

Done by Sarah, Sophie L, Sophia Y, and Sophie H

Date:20260807
Title: Golden Gate Assembly of G, H, L, X, V
Purpose: Assemble level 2 construct
Procedure:

G: PcpcG2-veg-EEBD-RBS0-Csn:LEA-B0015
H: 3P-EEBD-RBS0-Csn:LEA-B0015
L: 3P-EEBD-RBS0-GanA-Gala-L3S2P21
X: A+C+D+I (1:1)
V: A+B+D+I (1:1)

1.

G	
Part	**Amount**

Vector 1D (132ng/μL, 3120bp)	1.5μL (1/10)
PcpcG2-veg (168ng/μL, 235bp)	1.175μL (1/50)
EEBD-RBS0 (151ng/μL, 2392bp)	1.04μL (1/5)
Csn:LEA (212ng/μL, 634bp)	0.76μL (1/20)
B0015 (107ng/μL, 2213bp)	1.34μL (1/5)
T4 DNA ligase buffer	2μL
Sterile ddH_2O	10.4μL
T4 DNA ligase	1μL
Bsa1-HFv2	1μL

H	
Part	**Amount**
Vector Exp1 (162ng/μL, 6929bp)	1.4μL (1/5)
3P (106ng/μL, 2909bp)	1.43μL (1/9)
EEBD-RBS0 (150ng/μL,2392bp)	1.04μL (1/5)
Csn:LEA (212ng/μL, 634bp)	0.76μL (1/20)
B0015 (107ng/μL, 2213bp)	1.34μL (1/5)
T4 DNA ligase buffer	2μL
Sterile ddH_2O	10.3μL
T4 DNA ligase	1μL
Bsa1-HFv2	1μL

L	
Part	**Amount**
Vector Exp1 (162ng/μL, 6929bp)	1.4μL (1/5)
3P (106ng/μL, 2909bp)	1.43μL (1/9)
EEBD-RBS0 (150ng/μL,2392bp)	1.04μL (1/5)

GanA-Gala (177ng/μL, 4198bp)	1.22μL (1/9)
L3S2P21 (170ng/μL, 2195bp)	1.64μL (1/5)
T4 DNA ligase buffer	2μL
Sterile ddH_2O	10.3μL
T4 DNA ligase	1μL
Bsa1-HFv2	1μL

X	
Part	**Amount**
Vector Exp2 (163ng/μL, 5104bp)	1.017μL
A (109ng/μL, 3754bp)	1.119μL
C(98ng/μL,3828bp)	1.27μL
D (134ng/μL, 3243bp)	0.786μL
I (165ng/μL, 3757bp)	0.74μL
T4 DNA ligase buffer	2μL
Sterile ddH_2O	11.06μL
T4 DNA ligase	1μL
BsmBI	1μL

V	
Part	**Amount**
Vector Exp2 (163ng/μL, 5104bp)	1.017μL
A (109ng/μL, 3754bp)	1.119μL
C(98ng/μL,3828bp)	1.484μL
D (134ng/μL, 3243bp)	0.786μL
I (165ng/μL, 3757bp)	0.74μL
T4 DNA ligase buffer	2μL

Sterile ddH_2O	10.85μL
T4 DNA ligase	1μL
BsmBI	1μL

Done by Sophie C, Chloe

Date: 20260807
Title: Liquid culture of ABDI
Purpose: bacteria culture from successful colony PCR product
Procedure:

1. Label liquid culture tube
2. Add 5mL LB medium and 25μL chloramphenicol to the tube
3. Dip a pipette on a colony
4. Dispense the tip nto tube
5. Place the tube into the incubator at an angle
   Done by Sarah, Sophie L

Date: 20260807
Title: Liquid culture of B-sub WBS00
Purpose: bacteria culture from successful colony PCR product
Procedure:

1. Label liquid culture tube
2. Add 5mL LB medium to the tube
3. Dip a pipette on a colony
4. Dispense the tip into tube
5. Place the tube into the incubator at an angle
   Done by Sophie L, Sarah

Date: 20260807
Title: Make 250mL LBC with glycerol agar
Purpose: Make LBC agar plate
Procedure:

1. Weight 250mL dd H_2O and 9.25g LB agar powder
2. Add the ingredient above in 500mL flask and pu ta stirring bar
3. Stir the mixture until all the powder were fully dissolved
4. Cover the flask with double layer aluminum foil and put a autoclave sticker on it as well
5. Autoclave for 20 minutes at 127°C.
6. Put the flask into BSC after incubating (or put in 55℃ water bath if not pour shortly after incubating)
7. Add 250μL chloramphenicol into the liquid gel and gently mix it
8. Pour 20mL of liquid agar into each plate
9. Wait for it to solidify (about 40min later)
   Done by Sarah, Sophie L, Sophie H.

Date: 20260807
Title: Make 0.8% Agarose gel for two 25well big tray
Purpose: Make gel for electrophoresis
Procedure:

1. Weight 0.64g agarose powder and 80mL TAE buffer
2. Add the ingredient above in 250mL flask and cover the flask with double layer plastic wrap
3. Poke some holes on plastic wrap with pipette tip
4. Microwave the flask until all the powder were fully dissolved
5. Take the plastic wrap off and wait till it stop smoking
6. Add 4μL of diluted safety dye and mix it
7. Pour the gel into 2 assembled tray evenly
8. Use pipette tip to press the air out and pop bubbles
9. Wait for it to solidify
   Done by Sarah, Sophie H, Sophie L.

Date: 20260807
Title: Electrophoresis of L
Purpose: Check the bp of colony PCR product
Procedure:

1. **Load the gel in the following order:**

   Gel1:

M	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	L11	L12	L13	L14	L15	L16	L17	L18	L19	L20
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5

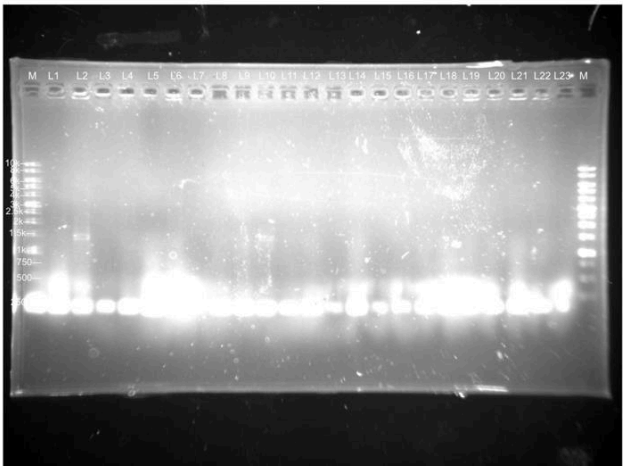
L17	L18	L19	L20	L21	L22	L23	M
5	5	5	5	5	5	5	5

   Gel2:

M	L24	L25	L26	L27	L28	L29	L30	L31	L32	L33	L34	L35	L36	L37	M
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5

2. Resolve 100V for 30min

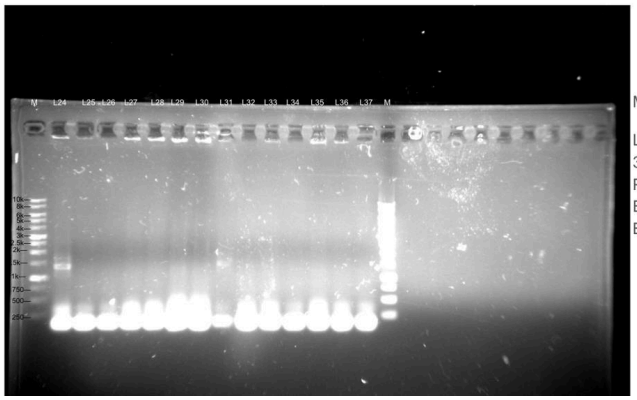
0807 Colony PCR (L1-L23)



M: 1Kb DNA ladder

L:
3P-EEBD-RBS0-GanA-Gala-L3S2P21
Primer: Exp-sq Fw
Exp-sq Rv
Bp: 3516

0807 Colony PCR (L24-L37)



M: 1Kb DNA ladder

L:
3P-EEBD-RBS0-GanA-Gala-L3S2P21
Primier: Exp-sq Fw
Exp-sq Rv
Bp: 3516

Done by Sarah, SophieL, Sophie H

Date:20260808
Title: Miniprep
Purpose: Isolate and purify Level 2 GGA product
Procedure:

ABDI:
PtagC-RBS-Ho1-pcyA-L3S2P21+Pliag-RBS0-ccaSm3-B0015+Pveg-RBS0-ccaR-B0015
+PcpcG2-veg-EEBD-RBS0-Csn:ACCD-B0015

1. Transfer 1000μL liquid culture to 1.5ml microtube
2. Centrifuge at 16,000g for 1min, discard supernatant
3. Repeat step 1&2 until all medium are collected
4. Add 200μL of MX1 buffer
5. Resuspend by pipetting or vortexting
6. Add 250μL of MX2 buffer
7. Invert slowly 8-10 times (don't pipette or vortex!), and incubate at RT for 2min
8. Add 350μL of MX3 buffer
9. Centrifuge at 23444g for 5min
10. Transfer 800μL of supernatant to column
11. Centrifuge at 7000g for 30sec and discard the flow through
12. Add 500μL of WN buffer, centrifuge at 7000g for 30sec
13. Centrifuge at 7000g for 30sec and discard the flow through
14. Centrifuge empty column for 8min at 10000g and place column into 1.5mL tube
15. Add 50μL EB buffer at the center of the column
16. Centrifuge at 23444 for 5min

	Concentration (ng/μL)	A260/230	A260/280
ABDI	66.00	1.222	1.886

Done by Sarah, Sophie H.

Date:20260808
Title: B Sub electrocompetent cells prep(WN800)
Purpose: electroporation of foreign DNA into a protease-deficient strain to safely express and secrete recombinant proteins.
Procedure:

1. Dilute overnight culture of B Sub WN800 (312.5 μl) in 16 fold in growth medium(5mL) and grow at 37°C to an OD600 of
2. Cool the cells on ice cold water and put on shaker for 10 minutes
3. Centrifuge at 4°C at 5000g for 5 minutes to harvest cells
4. Add 1 mL ice cold electroporation medium to resuspend cells in 1.5mL microtube centrifuge 5000g for 1 minute and discard supernatant. Repeat 4 times to wash cells
5. Suspend the cells in 1/40 of the culture volume of the electroporation solution (0.125mL) with a cell concentration of 1-1.3x10^{10}cfu/mL

6. Store competent cell at -80°C
   Done by Chloe, Abigail, Sophie C

Date:20260808
Title: PCR
Purpose: rapidly copy and amplify a specific DNA sequence
Procedure:

1. Prepare the following reactions
   Qx2

reagent	1rxn	2rxn
10x ExTAQ buffer	5μL	10μL
2.5uM DNTP mix	4μL	8μL
Fw primer(CcaSm3)	1μL	2μL
Rv primer(CcaR)	1μL	2μL
DNA temp(V)	1μL	2μL
PCR-grade H_2O	37.75μL	75.5μL
Ex-taq poly	0.25μL	0.5μL

R

reagent	1rxn
10x ExTAQ buffer	5μL
2.5uM DNTP mix	4μL
Fw primer(Csn)	1μL
Rv primer(AccD)	1μL
DNA temp(V)	1μL
PCR-grade H_2O	37.75μL
Ex-taq poly	0.25μL

S

reagent	1rxn
10x ExTAQ buffer	5μL
2.5uM DNTP mix	4μL
Fw primer(Ho1)	1μL

Rv primer(pcyA)	1μL
DNA temp(V)	1μL
PCR-grade H_2O	37.75μL
Ex-taq poly	0.25μL

2. Cycle

95°C	2 min	1 cycle
95°C	30 sec	30 cycles
65°C	40 sec	
72°C	2.5 min	
72°C	1 min	1 cycle

Done by Sophie C, Chloe

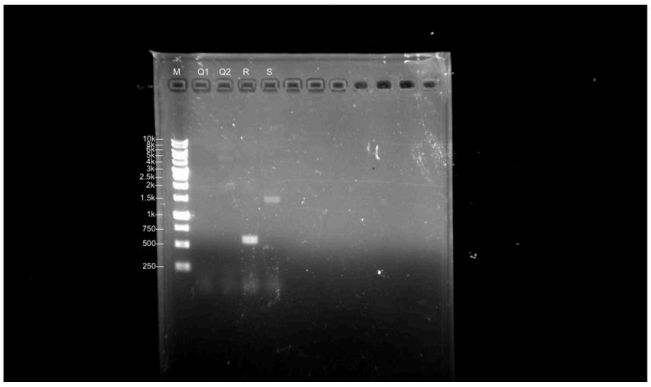
Date:20260808
Title: make 1% agarose gel
Purpose: prepare gel for electrophoresis
Procedure:

1. Assemble tray, mold, comb
2. Weigh 0.2g agarose powder and measure 20mL 1xTAE buffer
3. Add all to a conical flask
4. Cover with plastic wrap&puncture holes
5. Microwave for 30 sec Agitate. Repeat till no sediments
6. When no smoke, add 1μL safety dye
7. Pour into mole
8. Let solidify at RT
   Done by Sophie C

Date:20260808
Title: Gel electrophoresis of PCR
Purpose: check the PCR result
Procedure:

1. Load samples in the following order
   M Q1 Q2 R S
   5 5 5 5 5
2. Resolve the gel at 100V for 30min

0808 Level 2 V PCR



M: 1kb DNA ladder

Q1-2: ABDI (CcaSm3 F, CcaR R)
- bp: 2626

R: ABDI (Csn F, AccD R)
- bp: 1195

S: ABDI (Ho1 F, pcyA R)
- bp: 1549

Done by Sophie C

Date:20260808
Title: Transformation (G,H,L,X,V)
Purpose: amplification DNA part
Procedure:

1. Set up water bath at 42°C and prepare ice
2. Get 3 LBA and 2 LBC plates
3. Thaw competent cells till they're 50% thawed and add 10μL plasmid to 100μL competent cell
4. incubate on ice for 20min
5. Heat shock at 42°C for 45 sec
6. Incubate on ice for 3 min
7. Add 1mL SOC medium
8. Incubate for 1 hr at 37°C
9. Centrifuge at 5000g for 5min
10. Remove 1mL of supernatant and vortex
11. Spread on their respective plates
12. Incubate 37°C for 16hr
    Done by Sophie C, Ryan W, Chloe, Abby T, Ryan Y

Date:20260808
Title: Colony PCR
Purpose: Amplify the number of desired plasmid
Procedure:
pSTK-Exp1-P3p:RBS0-EEBD-GanA

1. Prepare the Master mix

	1rxn	25rxn
pSTK-Exp-seq fw	1μL	25μL
pSTK-Exp-seq rv	1μL	25μL
ddH_2O	8μL	200μL
Q-amp master mix	10μL	250μL

2. Thermal cycle

95°C	5min	1 cycle
95°C	30 sec	30 cycles
65°C	40 sec	
72°C	3min 30sec	
72°C	5 min	1 cycle

Done by Chars

Date: 20260809
Title: Colony PCR of G,H,X,V
Purpose: Amplify the number of desired plasmid
Procedure:
G:PcpcG2-veg-EEBD-RBS0-csn:LEA-B0015
H:3P-EEBD-RBS0-csn:LEA-B0015
X:ACDI
v:ABDI

1. Prepare the master mix

X:ACDI

	1rxn	14rxn
CcaR fw	1μL	14μL
PcpcG2-veg rv	1μL	14μL
Q-amp master mix	10μL	140μL
ddH_2O	8μL	112μL

v:ABDI

	1rxn	14rxn
CcaR fw	1μL	14μL
PcpcG2-veg rv	1μL	14μL
Q-amp master mix	10μL	140μL
ddH_2O	8μL	112μL

G

	1rxn	8rxn
Csnfw	1μL	8μL
LEA rv	1μL	8μL
Q-amp master mix	10μL	80μL
ddH_2O	8μL	64μL

H

	1rxn	2rxn
Csnfw	1μL	2μL
LEA rv	1μL	2μL

Q-amp master mix	10μL	20μL
ddH_2O	8μL	160μL

2. Thermal cycle

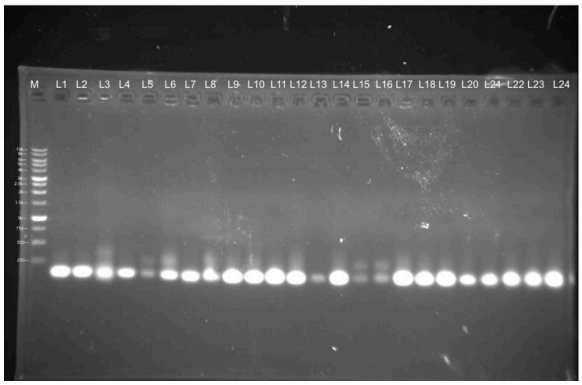
95°C	5min	1cycle
95°C	20sec	35cycle
65°C	20sec	
72°C	1min 30 sec	
72°C	5min	1cycle

Done by Abigail, Chloe W
Supervised by Chars

Date: 20260809
Title: Gel electrophoresis of Colony PCR
Purpose: check the PCR result
Procedure:

1. Load the sample in the following order
   M L1 L2 L3 L4 L5 L6 L7 L8 L9 L10 L11 L12 L13 L14 L15 L16 L17 L18 L19 L20 L21 L22
   5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5
   L23 L24
   5 5
   M:1kb ladder
   L:pSTK-Exp1-P3p-RBS0-EEBD-GanaA-Gala
2. Resolve gel in 100v for 30min

0809 colony pcr L



M: 1kb DNA ladder
L1-L24:
pSTK-EXP1-P3p:RBS0-EEBD-GanA-Gala
- bp: 3516 bp
- forward primer: pSTK-EXP-seq fw
- reverse primer: pSTK-EXP-seq rv

Date: 20260809

Title: Measure Arabidopsis fresh weight

Purpose: To determine which experimental set is best for Arabidopsis growing

Procedure:

1. Prepare seven 1.5mL tube and label information on the agar plate (vernalized, incubated, planted, and the date of measuring fresh weight)
2. Open the electronic scale and tare the scale after putting a labeled 1.5mL tube inside.
3. Take out the tube
4. Carefully take out all the Arabidopsis from the corresponding agar plate (careful not to break the root) and place them inside the tube
5. Put the tube back onto the scale and measure fresh weight
6. Wait until the weighting value stops changing
7. Took the tube out and label the fresh weight on the tube
8. Repeat step 2~7 until all agar plate sets were finished measuring
9. Store the tubes in -80°C and wait for extraction

Done by Naomi, Sophia L, Sophie H

Results:

Group	Freshweight (in grams)	Number of arabidopsis
Control(0,0)	0.1537	23
Pretreatment(w/ 100mM NaCl)	0.0203	24
Pretreatment(no salinity)	0.1013	23
Cotreatment(w/ 100mM NaCl)	0.0168	25
Cotreatment(no salinity)	0.0406	25
Aftertreatment(w/ 100mM NaCl)	0.0234	25
AFtertreatment(no salinity)	0.1830	24

Date: 20260809

Title: Measure OD of hay infusion

Purpose: To observe if there is bacillus subtilis growth in hay

Procedure:

1. Take out the incubated centrifuge tubes(incubated overnight)
2. Disinfect all equipment and move it into the BSC for aseptic operation

3. Transfer 200 µl from each centrifuge tube into a separate well of a 96-well plate
4. Transfer 200 µl of LB(blank) and filtered hay infusion(blank) into another two wells
5. Place the 96-well plate into the OD microplate reader(Elisa) and record the OD600 result
6. Repeat steps 1-5 for each hour for a total of 7 hours

Hour	LB	Hay infusion
1	0.083	0.080
2	0.090	0.081
3	0.163	0.079
4	0.366	0.081
5	0.353	0.081
6	0.469	0.079
7	0.355	0.081
Blank	0.038	0.065

Done by Sophia L and Abigail

Date: 20260809
Title: Vernalize exp#7 trial 2 seeds
Purpose: Vernalize new seeds for the second trial.
Procedure:

1. Open the BSC
2. Use the seed template(0.8% agar circular plates) to plant 40 seeds on each agar plates
3. Use breathable tape to seal each of the agar plates
4. Label the vernalization date, incubator date and seed type
5. Wrap the agar plates with aluminum foil
6. Vernalize in the 4°C refrigerator for 4 days

Date: 20260809
Title: agar experiment set 7 fresh weight
Purpose: measure the arabidopsis weight
Procedure:

1. Set the scale by weighing the empty tube
2. Use a tweezers to pick arabidoopsis from each agar plate
3. Slightly press the plants on tissue paper to remove agar
4. Put all arabidopsis into a ube labelled with their experiment group
5. Weigh the tube with collected arabidopsis and record the fresh weight

6. Repeat steps1-5 until all experiment groups are collected
   Result:

group	freshweight(g)	# of arabidopsis
Control (0,0)	0.1587	23
Pretreatment (without 100mM NaCl)	0.0203	24
pretreatment	0.1013	23
Contreatment (without 100mM NaCk)	0.0168	25
cotreatment	0.0406	25
After treatment (without 100mM NaCl)	0.0234	25
After treatment	0.1830	24

   Done by Naomi, Sophia L, Sophie H

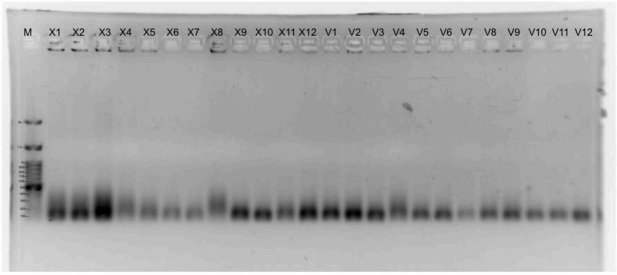
Date: 20260810
Title: Make 1% gel
Purpose: prepare gel for electrophoresis
Procedure:

1. Assemble tray,comb,mold
2. Weigh 0.8g agarose powder&80mL 1XTAE buffer
3. Add all in conical flask
4. Wrap with plastic wrap and puncture holes
5. Microwave for 30sec Agitate.Repeat till no sediments
6. When there is no smoke, add 4μL safety dye. Agitate
7. Pour melted gel into mold
8. Let it solidify at RT
   Done by Sophie C

Date: 20260810
Title: Electrophoresis
Purpose:Check PCR result
Procedure:
X: ACDI
V:ABDI
G:PcpcG2-veg-EEBD-RBS0-csn:LEA-B0015
H:3P-EEBD-RBS0-csn:LEA-B0015

1. Load the samples in the following order
   M X1 X2 X3 X4 X5 X6 X7 X8 X9 X10 X11 X12 V1 V2 V3 V4 V5 V6 V7 V8 V9 V10 V11
   5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5
   V12
   5
   M H1 G1 G2 G3 G4 G5 G6 G7
   5 5 5 5 5 5 5 5 5
2. Resolve the gel in 100v for 30min

0809 colony pcr x v



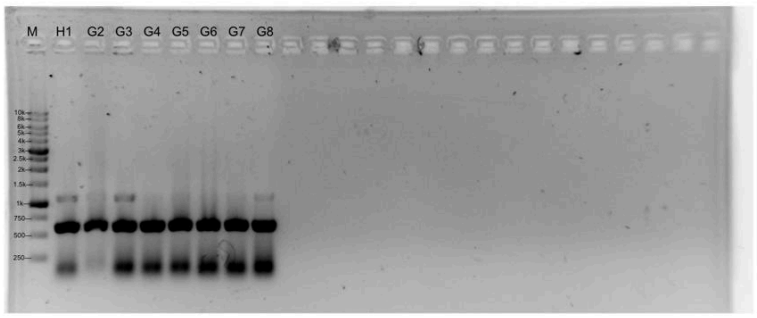
M: 3k DNA ladder

X: ACDI
- CcaR fw + PcpcG2-veg rv
- bp: 1032 bp

V: ABDI
- CcaR fw + PcpcG2-veg rv
- bp: 1032 bp

All fail -> RE-GGA

0809 colony pcr g h



M: 10k DNA ladder

G: PcpcG2-veg-EEBD-RBS0-Csn:LEA-B0015
- Csn fw + Lea rv
- bp: 584

H: 3P-EEBD-RBS0-Csn:Lea-B0015
- Csn fw + Lea rv
- bp: 584

Liquid culture 8/10: H1, G4, G6

Done by Sophie C

Date: 20260810
Title: Colony PCR of L
Purpose:Amplify the number of desired plasmid
Procedure:

1. Prepare the following master mix
   Lx25

Exp-seq fw	25μL
Exp-seq rv	25μL
Q-amp master mix	250μL
ddH_2O	200μL

2. Pick up colonies with tips& streak on grids
3. Wait 5 min , take out tips
4. Thermal cycle

95°C	5min	1cycle
95°C	20sec	35cycle
60°C	20sec	
72°C	3min 30 sec	
72°C	5min	1cycle

Done by Sophie C, Chloe

Date: 20260810
Title: electroporation
Purpose:allowing foreign molecules like DNA to enter the cells
Procedure:
J:pSTK-Exp1-Pveg-EEBD-RBS0-csn:AccD-B0015
X:pSTK-Exp2-Ptag+RBS-H01:PcyA-L3S2P21-PrepA-RBS0-CcaSm3-B0015-Pveg-RBS0-CcaR-B0015-PcpcG2-veg-EEBD-RBS0-csn:AccD-B0015

1. Prewarm the sOC medium at 37°C and pre-cool a 0.2cm electroporation curette
2. Mix 15.2μL plasmid in 100μL electrocompetent cell in a 1.5mL microtube, and mix 2.4μL plasmid (400ng/μL)J in 100μL electrocompetent cell
3. Set up electroporator and voltage resistance
    - V:2.3Kv
    - Resistance :300Ω
    - Capitance: 35F
4. Transfer mixture of cells &DNA to electroporation curette&tap suspension to bottom
5. Place in shock pad, close lid&wait until beep
6. Immediately remove form shock pad and add 1mL SOC medium
7. Transfer to liquid culture tube and incubate at 37°C for 3 hours, shaking at 200rpm
8. check&record parameters
9. Centrifuge at 5000g for 5min
10. Spread LB plate with erithmycin

J

TC	6.3
V	2284

X

TC	5.5
V	2282

Done by Chloe, Sophie C

Date: 20260810
Title: Colony PCR
Purpose:Amplify the number of desired plasmid
Procedure:
pSTK-Exp1-3P-EEBD-RBS0-GanA-Gala-L3S2P21

1. Prepare the following mixture

	1rxn	24rxn

pSTK-Exp-seq fw	1μL	24μL
pSTK-Exp-seq rv	1μL	24μL
Q-amp master mix	10μL	240μL
ddH_2O	8μL	192μL

2. Apply the following thermal cycle

95°C	5min	1cycle
95°C	20sec	35cycle
60°C	20sec	
72°C	3min 30 sec	
72°C	5min	1cycle

Done by Chloe, Sophie C

Date: 20260810
Title: Golden Gate Assembly
Purpose: Assemble level 1 parts to form level 2 parts
Procedure:
X:ACDI
V:ABDI

1. Prepare the following reaction
   X

Part	amount
Exp2 (163ng/μL,5104bp)	1.017μL
A:PtagC-RBS-H01:pcyA-I3S2P21 (109ng/μL,3754bp)	1.119μL
C:PlepA-RBS0-CcaSm3-B0015 (98ng/μL,3828bp)	1.27μL
D:Pveg-RBS0-CcaR-B0015 (134ng/μL,3243bp)	0.786μL
I:PcpcG2-veg-EEBD-RBS0-csn:AccD-B0015 (165ng/μL,3757bp)	0.74μL
T4 DNA ligase buffer	2μL

Sterile water	11.06μL
T4 DNA ligase	1μL
BsmBI	1μL

V

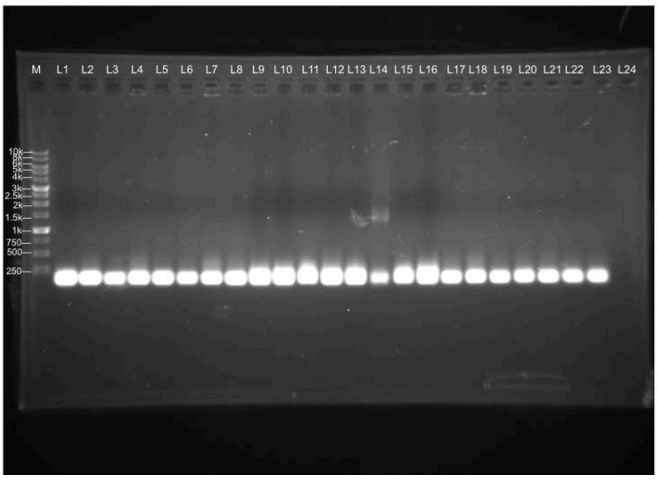
Part	amount
Exp2 (163ng/μL,5104bp)	1.017μL
A:PtagC-RBS-H01:pcyA-I3S2P21 (109ng/μL,3754bp)	1.119μL
B:PliaG-RBS0-CcaSm3-B0015 (83ng/μL,3792bp)	1.484μL
D:Pveg-RBS0-CcaR-B0015 (134ng/μL,3243bp)	0.786μL
I:PcpcG2-veg-EEBD-RBS0-csn:AccD-B0015 (165ng/μL,3757bp)	0.74μL
T4 DNA ligase buffer	2μL
Sterile water	10.85μL
T4 DNA ligase	1μL
BsmBI	1μL

Done by Chloe

Date: 20260810
Title: Gel electrophoresis
Purpose: Checking PCR result
Procedure:

1. Load samples in the following layout:
   M L1 L2 L3 L4 L5 L6 L7 L8 L9 L10 L11 L12 L13 L14 L15 L16 L17 L18 L19 L20 L21 L22
   5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5
   L23
   5
2. Resolve at 100V for 30 minutes

0810 colony PCR (GanA-Gala)



M: 1k ladder

L: 3P-EEBD-RBS0-GanA-Gala-L3P2S21

Primers: pSTK-EXP-seq-fw & pSTK-EXP-seq-rv

Bp: 3516 bp

Done by Chloe

Date: 20260810
Title: Liquid culture
Purpose: bacteria culture form colony PCR product
Procedure:

1. Label 6 liquid culture tubes
2. Add 3mL TB medium and 3μL of ampicillin in culture tubes
3. Dip a pipette on a colony
4. dispense the tip into tube
5. Place the tube into the incubator (slanted angle)

Done by Sophie C

Date: 20260810
Title: 1/10 MS medium preparation (1000mL)
Purpose: bacteria culture form colony PCR product
Procedure:

1. Weight the ingredient below
2. Add the ingredient above in 800mL distilled water
3. Add a stirring bar and stir it
4. Add 200mL distilled water
5. Adjust the pH of the medium until it reach 5.7~5.8 (ori:3.75)
6. Separate the medium into 2 500mL flask
7. Cover flask with double layer aluminum foil and stick autoclave tape
8. Autoclave for 20 minutes at 127°C.
   Done by Sophie H

Date: 20260810

Title: agar (exp set #7 trial 1) chlorophyll extraction

Purpose: to isolate green pigment from plant

Procedure:

1. Take the microtubes form 0809
2. Flute acetone to 80% add 4mL to each tube
3. Add to 10 metal bits into each tube
4. Wait for chlorophyll to extract, and use a vortex to crush the plants to speed up the process
5. Add liquid to 96 wells, 200µL in each well. Three wells for each sample
6. ELISA reader, Run results through equation:

	F4	F5	F6	F8	F9	F10	G5	G6	G7	G8	G9	G10	H5	H6	H7	H8	H9	H10
C	P100	P0	CO	A100	AO	P100	P0	C100	CO	A100	A0	P100	PO	C100	CO	A100	AO	

7. Correct wavelength
   A663
   A645
   Done by Sophi C, Audrey H
   Supervised by :Gabriel

Date: 20260810

Title:Arabidopsis soil planting (tray 5 & tray6)

Purpose: planting arabidopsis

Procedure:

1. Autoclave a bag of soil
2. Seak 8 flower pot in 1/10 bleach and wait for 20 min
3. Rinse the put with water for 5 min(until there is no bleach smell)
4. Fill the put with full soil and slightly press it down
5. Gently shake it vertically to soften the soil
6. Water the soil until the upper part of the soil become fully wet
7. Use tip to pick up a seed from preservation solution
8. Dispense the seed with solution on the soil in fixed position (6 seeds in a pot)
   - Make sure not to cover the seed or move it
9. After all the pots are done, cover the pot with plastic , wrap and place them in incubator.
   Done by Sophie H
   Supervised by Gina

Date: 20260811
Title: Colony PCR (of J in *Bacillus subtilis*)
Purpose: Amplify the number of desired plasmid
Procedure:

1. Prepare master mix as follow X2:

	1 reactions	2 reaction
Fw primer (Exp-seq-fw)	1 µl	2 µl
RV primer Exp1 (Rv primer)	1 µl	2 µl
Qamp master mix	10 µl	20 µl
ddH_2O	8 µl	16 µl

2. Distribute 20µL each into each PCR tube
3. Pick up colonies with tip and streak grid
4. Dispense in PCR tube of master mix
5. Set up thermal cycle and run the machine:

95°C	5 min	1cycle
95°C	20 sec	
55°C	20 sec	35 cycle
72°C	2min 20sec	
72°C	5 min	1 cycle

Done by Sophie C.

Date: 20260811
Title: Miniprep
Purpose: Collect concentrated liquid culture
Procedure:

1. Transfer 1000µL liquid culture in 1.5mL microtube
2. Centrifuge for 30sec at 16000g , discard the supernatant
3. Repeat step 1&2 until all the liquid were collected
4. Add 200µL of MX1 buffer
5. Resuspend the pellet by pipetting
6. Add 250µL of MX2 buffer
7. Invert 8-10times to mix slowly
8. Incubate at RT for 2 min
9. Add 350µL of MX3 buffer
10. Invert 4-5 times to mix gently

11. Centrifuge at 23444g for 5 minute
12. Assemble the column and tube
13. Transfer the supernatant (~800μL) to the column
14. Centrifuge the column at 7000g for 30sec
15. Discard the flow-through
16. Add 500μL of WNfuvver
17. Centrifuge the column at 7000g for 30sec
18. Add 700μL of WS buffer
19. Discard the flow-through
20. Centrifuge empty column at 10000g for 3 min
21. Place column into 1.5mL tube and add 50μL of EB buffer to center of column membrane
22. Incubate at RT for 3mins
23. Centrifuge at 10000 g for 3min

	concentration(ng/μL)	A260/230	A260/180
G4	295	1.724	2.095
G6	187	1.558	2.033
H2	198	1.503	2.010

Done by Sophia Y, Sophie C

Date: 20260811
Title: Gel Electrophoresis
Purpose: Check the approximate length of colony PCR product
Procedure:

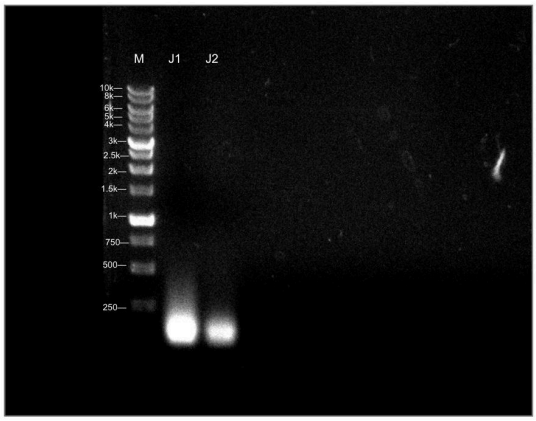
1. Load the gel in the following order:

M	J1	J2
5	5	5

2. Resolve 100V for 30min

Result:

0811 colony PCR J



M: 1kb DNA ladder
J1, J2: Pveg-EEBD-RBS0-Csn:ACCD-B0015

- Bp: 1868
- Fw premier:Exp-sq-Fw
- Rv premier: ACCD Rv

Date: 20260811
Title: Transformation
Purpose: to cultivate and amplify DNA parts
Procedure:

1. Set up water bath at 42℃
2. Thaw competent cells until 50% melted, add 10μL plasmid in 100μL competent cell
3. Ice bath for 20min
4. Heat shock for 45 sec
5. Ice bath for 3 min
6. Add 1ml soc medium
7. Incubate 1hr at 37 ℃
8. Centrifuge 5000g for 5 min
9. Take out 900μL supernatant
10. Resuspend pellet
11. Spread on LBC plate
12. Incubate for 16hr

Done by Sophie C.

Date: 20260811
Title: Electroporation
Purpose: Introduce DNA sequence into competent cells
Procedure:

1. Prewarm the SOC medium at 37℃ and pre-cool a 0.2cm electroporation cuvette

2. Mix 15.2μL plasmid in 100μL electroporation cell in a 1.5mL microtube, and mix 2.4μL plasmid (410ng/μL0 J in 100μL electrocompetent cell
3. Set up electroporator and voltage resistance:
   V: 2.3kv
   Resistance: 300Ω
   Capitance: 35F
4. Transfer mixture of cells & DNA to electroporation cuvette & tap suspension to bottom
5. Place in shock pad, close lid and wait until you hear the beeping sound
6. Immediately remove from shock pad and add 1mL SOC medium
7. Transfer to liquid culture tube and incubate at 37℃ for 3 hr, shaking at 200rpm
8. Check and record parameters
9. Centrifuge at 5000g for 5min
10. Spread LB plate with erithmycin

Done by Sophie C
Recorded by Sophie H.

Date: 20260812
Title: Hay Infusion
Purpose: Observe the growth of *Bacillus subtilis* in different hay infusion lake water
Procedure:

1. Use an electronic pipette controller with a serological pipette to draw 15mL of hay infusion
2. Transfer the 15mL into a syringe fitted with a filter.
3. Filter the hay infusion and collect it in a 15mL tube.
4. Replace the filter with a new one if it becomes clogged and difficult to press down.
5. Repeat step 1 to 3 for the three hay infusions with different lake water.
6. Disinfect all equipment and move it into the BSC for aseptic operation.
7. Use a new serological pipette to transfer 5 mL of the filtered water into a new syringe fitted with a sterile filter.
8. Filter the filtered hay infusion into a centrifuge tube.
9. Use a pipette to add 200 µL of *Bacillus subtilis* into the centrifuge tube
10. Repeat step 6 and 7 for the three filtered hay infusions with different lake water.
11. Place the three centrifuge tubes in the centrifuge overnight.

Done by Sophia, recorded by Sophia.

Results:

Source of hay infusion water	OD of blank(without *Bacillus subtilis*)	OD of *Bacillus subtilis* after one day
淡水	0.054	0.042
新竹	0.043	0.055
大湖	0.064	0.064

Date: 20260812
Title: Colony PCR
Purpose: Amplify engineered DNA plasmid
Procedure:
J: Pveg-EEBD-RBS0-Csn:ACCD-B0015(*Bacillus subtilis*)
V: ABDI
X: ACDI

1. Make the following reactions:

J

	1 reactions	8 reaction
Fw primer (CsnFw)	1 µl	8 µl

RV primer Exp1 (ACCD Rv)	1 µl	8 µl
Qamp master mix	10 µl	80 µl
ddH_2O	8 µl	64 µl

X,V

	1 reactions	27 reaction
Fw primer (CcaR Fw)	1 µl	27 µl
RV primer Exp1 (pcpcG2-veg Rv)	1 µl	27 µl
Qamp master mix	10 µl	270 µl
ddH_2O	8 µl	216 µl

2. Distribute in PCR tubes
3. Dip pipette in colonies
4. Streak grid
5. Dispense tip in PCR tubes
6. Close lid
7. Thermal cycle:

95°C	5 min	1cycle
95°C	20 sec	
55°C	20 sec	35 cycle
72°C	2min	
25°C	5 min	1 cycle

Done by Sophie C, Abby T

Date: 20260812
Title: Make 1% Gel
Purpose: Make Gel for electrophoresis of colony PCR product
Procedure:

1. Assemble tray, comb, and mold
2. Weight 0.8g agarose powder and 80mL 1X TAE buffer
3. Add in 250mL conical flask
4. Wrap the flask with plastic warp and puncture holes on it with tip
5. Microwave until the powder fully dissolved

6. When there's no smoke, add 4μL safety dye and agitate
7. Pour the melted gel in the mold
8. Let it solidify at room temperature

Done by Chloe, Abby T

Date: 20260812
Title: Gel Electrophoresis
Purpose: Check the approximate length of colony PCR product
Procedure:

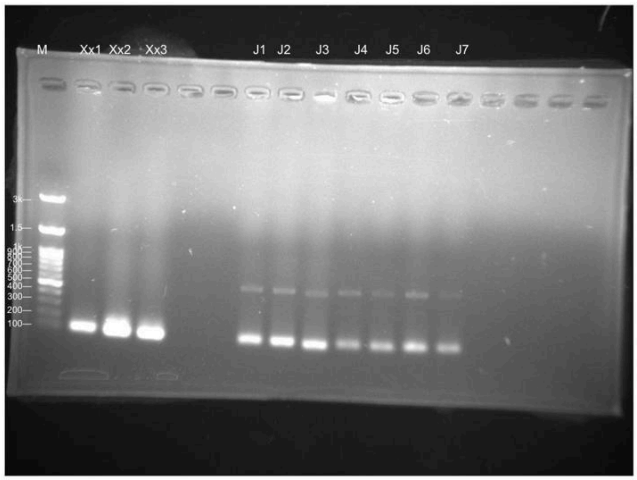
1. Load the gel in the following layout:
   Gel 1:

M	X1	X2	X3	_	_	J3	J4	J6	J7	J1	J2
5	5	5	5			5	5	5	5	5	5

2. Resolve the gel at 100V for 30min.

Result:

8/12 COLONY PCR X, J (GEL 1)



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder

X: ACDI (plepA, accd)(for amplification)

ccar fw primer

pcpcG2-veg rv

bp: 1032

J: Pveg-EEBD-RBS0-csn:ACCD-B0015 (electroporation)

csn fw primer

ACCD reverse primer

bp: around 615

Done by Abby T

Date: 20260812
Title: Gel electrophoresis
Purpose: Check the approximate length of colony PCR product

Procedure:

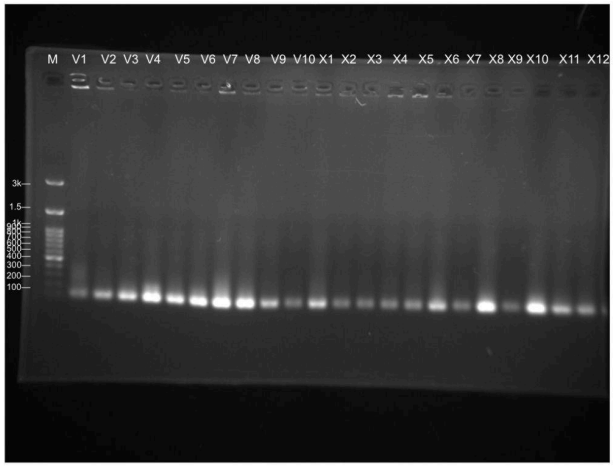
1. Load the gel in the following layout:
   Gel 1:

M	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	X11	X12
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5

2. Resolve the gel at 100V for 30min.

Result:

8/12 COLONY PCR v, x (GEL 2)



M: M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder

X: ACDI (plepA, accd)

V: ABDI (pliag, accd)

bp: 1032

ccar fw primer

pcpcG2-veg rv

Done by Chloe

Date: 20260812
Title: Liquid Culture
Purpose: Transfer colonies on agar to medium
Procedure:

1. Prepare liquid culture by the following table:

Bacteria	Liquid Culture	Antibiotics
WB800	5mL LB	None
pSTK-0 CX-Csnfrom glycerol stock	5mL LB	chloramphenicol 5μL
pSTK-Exp1-Csn:ACCD	5mL LB	Erytheromycin 25μL

#J1 in *Bacillus subtilis* WB800		
pSTK-Exp1-Csn:ACCD #J2	5mL LB	Erytheromycin 25μL
pSTK-Exp1-Csn:ACCD #J6	5mL LB	Erytheromycin 25μL

2. Incubate in 37℃ at 200rpm for 16 hr

Done by Chars

Date: 20260812
Title: L0 Golden Gate Assembly
Purpose: Assemble parts
Procedure:

1. Prepare the following reaction:

1. pSTK-0C-SamyQ-Bopep4-v2t1

Part	amount
STK001-pSTK-0-sfGFP (346ng/μL, 3090bp)	1.16μL (1/20)
SamyQ-Bopep4-v2t1 (10ng/μL, 180bp)	1.17μL (1/5)
T4 DNA ligase buffer	2μL
Sterile water	13.67μL
T4 DNA ligase	1μL
Bbs1-HF	1μL
Total:	20μL

2.pSTK-0C-SamyQ-Bopep4-v2t2

Part	amount
STK001-pSTK-0-sfGFP (346ng/μL, 3090bp)	1.16μL (1/20)
SamyQ-Bopep4-v2t1 (10ng/μL, 180bp)	1.17μL (1/5)
T4 DNA ligase buffer	2μL
Sterile water	13.67μL
T4 DNA ligase	1μL

Bbs1-HF	1μL
Total:	20μL

3.pSTK-0C-SamyQ-Csn-Lea

Part	amount
STK001-pSTK-0-sfGFP (346ng/μL, 3090bp)	1.16μL (1/20)
Csn-Lea (PCR product,212ng/μL, 634bp)	0.97μL (1/50)
T4 DNA ligase buffer	2μL
Sterile water	13.87μL
T4 DNA ligase	1μL
Bbs1-HF	1μL
Total:	20μL

Done by Sophie C.

Date: 20260812

Title: plant new set of seeds for hydroponic system

Purpose: prepare new seed for hydroponic system

Procedure:

1. Prepare 0.8agar 1/10 MS medium & microtube holder
2. Spray the tip holder with ethanol & shine under UV light for 30 min
3. Fill 48 wrack of the tip holder with 2.2 mL of 1/10 MS medium
4. Microwave the agar until it melts, and pour 250μL agar into each 48 eppenthrf lids
5. Wait for them to solidify
6. Move the lids into respective wrack
7. Use autoclaved tooth pick to pick up one seed and add the seed onto the lid
8. Repeat step 7 until all the lid have the seed
9. Wrap the tube which alumium foil
10. Place the tube wrack into 4 ℃ refrigerator
11. Wait for 4 days for seeds to vernalize
    Done by Sophie H

Date: 20260812

Title:Hay infusion

Purpose: observe the growth of *Bacillus subtilis* in different lake water

Procedure:

1. Use an electronic pipette controller with a serological pipette to draw 15mL of hay infusion

2. Transfer the 15 mL into a syringe fitted with a filter
3. Filter the hay infusion and collect it in a 15mL tube
4. Replace the filter with a new one if it become clogged and difficult to press down
5. Repeat step 1 to 3 for the three hay infusions with different lake water
6. Disinfect all equipment and more it into BSC for aseptic operation
7. Use a new serological pipette to transfer 5mL of the filtered water into a new syringe filtered with a sterile filter
8. Filter the filtered hay infusion into a centrifuge tube
9. Use a pipette to add 200μL of B.cub into the centrifuge tube
10. Repeat step 6 and 7 for the tree filtered hay infusions with different lake water
11. Place the three centrifuge tubes in the centrifuge overnight, start from 18:00

Result:

Source of hay infusion water	OD of blank (without *Bacillus subtilis*)	OD of B.sub after one day
淡水	0.054	0.042
新竹	0.043	0.055
大湖	0.064	0.064

Done by Sophia L, Abigail

Date: 20260812

Title:fortilize plants

Purpose: fortilize plants

Procedure:

1. Add 0.5g of fertilizer into 500mL of water(1g/1000mL)
2. Add the solution to the plant trays

Done by Alex

Date: 20260813
Title: Hay Infusion
Purpose: Compare the OD of hay infusion with and without *Bacillus subtilis*
Procedure:

1. Take out the three centrifuge tubes from the centrifuge
2. Transfer 200 μL from each centrifuge tube into a separate well of a 96-well plate.
3. Disinfect all equipment and move it into the BSC for aseptic operation.
4. Transfer 200 μL from the 0812 filtered hay infusion into a separate well of a 96-well plate.
5. Place the 96-well plate into the OD microplate reader and record the result.

Done by Sophia L, Abigail

Results:

Source of hay infusion water	OD of blank(without *Bacillus subtilis*)	OD of *Bacillus subtilis* after one day
淡水	0.054	0.042
新竹	0.043	0.055
大湖	0.064	0.064

Date: 20260813
Title: PCR
Purpose: Amplify plasmid number
Procedure:
H: Pveg-Csn:LEA

1. Prepare the following reaction

	1 reaction
Forward primer (EEBD F)	1μL
Reverse primer (LEA R)	1μL
Q-amp master mix	10μL
ddH_2O	7μL
DNA template	1 ul

2. Put in PCR machine for thermal cycle

95°C	2 min	1 cycle

95°C	20 sec	35 cycles
60°C	20 sec	
72°C	1 min 20 sec	
25°C	5 min	1 cycle

Done by Sophie C.

Date: 20260813
Title: Gel electrophoresis (colony pcr)
Purpose: Check the approximate length of colony PCR product
Procedure:

1. Load the gels in the following order:
   Gel 1:

M	X13	X14	X15	X16	X17	X18	X19	X20	X21	X22	X23	X24	X25	X26	X27	X28	X29	X30	X31	X32	X33	X34	X35	X36
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5

   M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder
   X1-24: ACDI
   - Primers: CcaR F & PcpcG2-veg R
   - Bp: 1032

   Gel 2:

M	V13	V14	V15	V16	V17	V18	V19	V20	V21	V22	V23	V24	V25	V26	V27	V28	V29	V30	V31	V32	V33	V34	V35	V36
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5

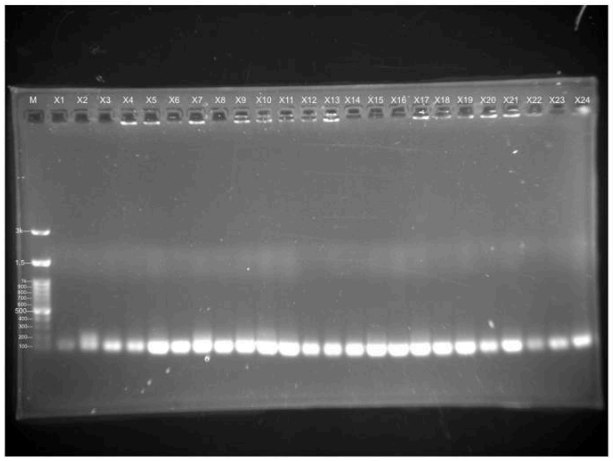
   M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder
   V1-24: ABDI
   - Primers: CcaR F & PcpcG2-veg R
   - Bp: 1032

2. Resolve the gel at 100V for 30min.

Done by Sophie C.

Results:

0813 Colony PCR X(gel 1)



M: M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder

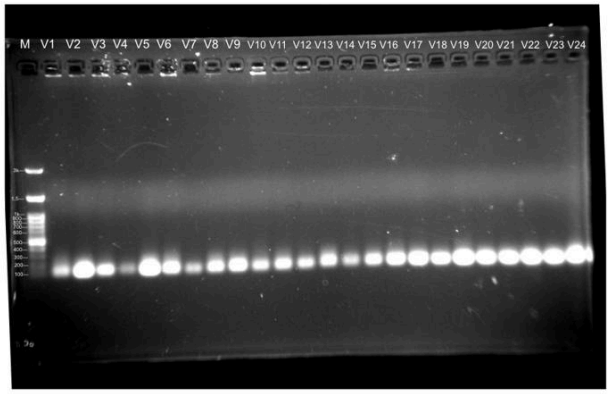
X1-24: ACDI

CcaR F & PcpcG2-veg R

Bp: 1032

-> all failed(re-gga)

0813 Colony PCR V(gel 2)



M: M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder

V1-24: ABDI

CcaR F & PcpcG2-veg R

Bp: 1032

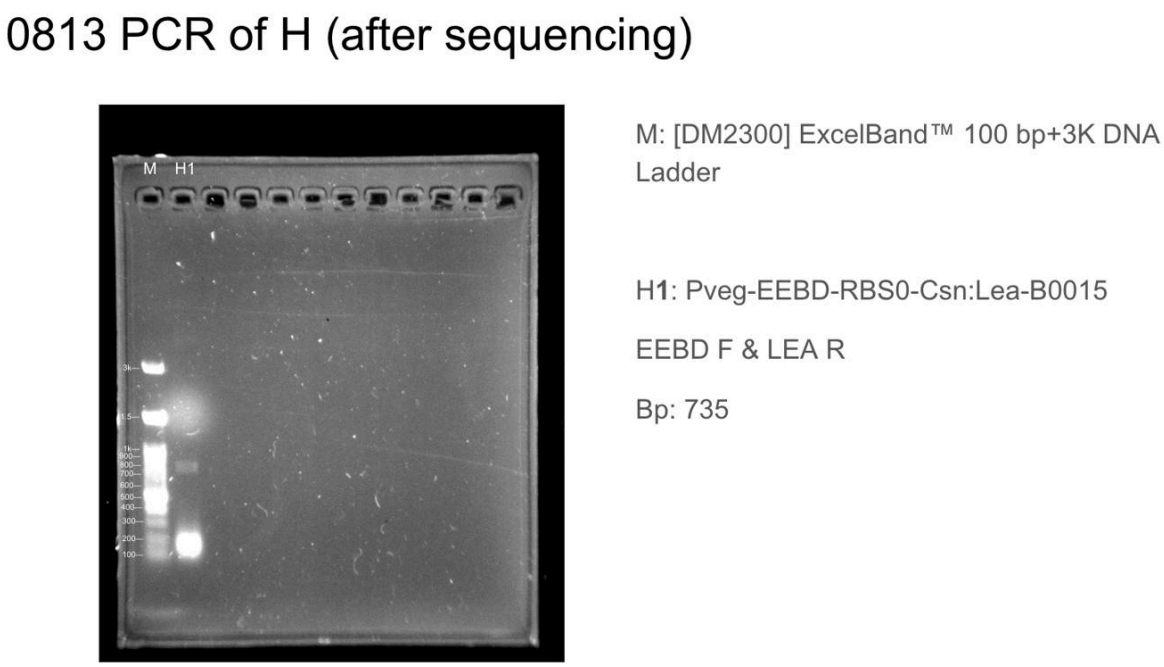
-> all failed(re-gga)

Date: 20260813
Title: Gel electrophoresis (PCR)
Purpose: Check the approximate length of PCR product
Procedure:

1. Load gel in following order:
   M H1
   5 5
   M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder
   H1: Pveg-EEBD-RBS0-Csn:Lea-B0015
   - Primers: EEBD F & LEA R
   - Bp: 735

Done by Sophie C.

Results:



Date: 20260813
Title: Miniprep
Purpose: to extract plasmid from liquid culture
Procedure:
Samples:

- F10: Pveg-EEBD-RBS0-Csn:AmilCP-B0015

1. Transfer 1000 µL of liquid culture to 1.5 mL microtube.
2. Put in a centrifuge and centrifuge at 16000 g for 1 minute.
3. Discard the supernatant.
4. Repeat steps 1–3 until all cultures are collected.
5. Add 200 µL of MX1 buffer.
6. Resuspend by pipetting and vortexing.
7. Add 250 µL of MX2 buffer.
8. Invert slowly for 8–10 times, incubate at RT for 2 mins.
9. Add 350 µL of MX3 buffer and invert slowly for 4–5 times.
10. Centrifuge at 23444 g for 5 minutes.
11. Transfer 880 µL of supernatant to column.
12. Centrifuge at 7000 g for 30 seconds.
13. Discard the flow through.
14. Add 500 µL of WN buffer to the column.

15. Centrifuge at 7000 g for 30 seconds.
16. Discard the flow through.
17. Add 700 µL of WS buffer to the column.
18. Centrifuge at 7000 g for 30 seconds.
19. Discard the flow through.
20. Centrifuge the empty column for 8 mins. at 10000 g.
21. Place column into 1.5 mL tube.
22. Add 50 µL EB buffer to the center of the column membrane.
23. Incubate at RT for 3 mins.
24. Centrifuge at 10000 g for 3 mins.
25. Measure the concentration by using Biodrop

Done by Ryan W.

BioDrop Results:

Sample	Concentration (ng/µL)	A260/A280	A260/A230
F10	81	1.929	1.227

Date: 20260813
Title: Transformation
Purpose: Transform and cultivate engineered bacteria on agar plate
Procedure:

1. Set the water bath to 42°C and prepare ice
2. Thaw 100µL competent cells until they're 50% thawed and add 10µL (L0 GGA 20260812) of plasmid
3. Incubate on ice for 20min
4. Heat shock at 42°C for 45 seconds
5. Incubate on ice for 3 min
6. Add 1mL SOC medium
7. Incubate for 1 hr at 37°C
8. Centrifuge at 5000g for 5 min
9. Remove supernatant, leave around <100µL
10. Resuspend pellet
11. Spread on LBC plate
12. Incubate at 37°C for 16hr

Done by Sophie C.

Date: 20260813
Title: Golden Gate Assembly (L)
Purpose: To assemble the parts for L

Procedure:

Sample: pSTK-EXP1-3P-EEBD-RBS0-Csn:GanA-Gala-L3S2P21

1. Add the following

Part	Original	Dilution
Vector (shuttle vector): pSTK-EXP-1-sfGFP concentration: 162 ng/ µl 6929 bp	0.2780154321	1.4 ul (dilute 1/5)
Insert promoter(3P) concentration: 106 ng/ µl 2909bp	0.3567641509 µL	1.43 µL (dilute 1/4)
Insert RBS (EEBD-RBS0) concentration: 151 2392 bp	0.2059337748	1.03 (dilute 1/5)
Insert CDS(GanA-Gala) concentration: 177 4148 bp	0.3046553672	1.22 (dilute 1/4)
Insert Terminator (L3S2P21) concentration: 170 2145 bp	0.1640294118 µL	1.64 µL (dilute 1/10)
STK331_pSTK-0x-Cansecretion tag 2188 bp 81 ng/µL	0.3511604938 µL	1.07 µL (dilute 1/10)
Histag (pSTK-OCZ-HisGX) ● concentration(ng/µL): 135 ● bp: 2107	0.2028962963 µL	1 µL (dilute 1/5)
T4 DNA ligase buffer	2 µL	2 µL
Sterile water	7.21 µL	
T4 DNA ligase	1 µL	1 µL
BsaI-HFv2	1 µL	1 µL

Total	20 µL	20 µL

Done by Sophie C.

Date: 20260813
Title: Golden Gate Assembly (V)
Purpose: to assemble the parts for V
Procedure:
Sample: V:A+B+D+I

1. Assemble the following

V	
Part	Amount
Vector Exp2 (163ng/µL, 5104bp)	1.017µL
A (109ng/µL, 3754bp)	1.119µL
C(98ng/µL,3828bp)	1.484µL
D (134ng/µL, 3243bp)	0.786µL
I (165ng/µL, 3757bp)	0.74µL
T4 DNA ligase buffer	2µL
Sterile ddH_2O	10.85µL
T4 DNA ligase	1µL
BsmB1	1µL

Done by Sophie C, Sarah, Abigail

Date: 20260813
Title: Add zhenhan protectant on horizontal agar(trial #2)
Purpose: test zhenhan protectan
Procedure:

1. Use the following amount & apply on 4 pretreatment plant
    - [1mL per plate]
    - [1/500 dilution -> 30µL of CH protectant with 14.97 mL ddH_2O]
2. Evenly distribute liquid protectan by fillting the plate sideways until all plants are converted with a thin layer of protectant
3. Seal plates with tpae and label date of pretreatment
   Done by Chloe

Supervised Abigail

Date: 20260813

Title: hay infusion

Purpose: compare the OD of hay infusion with and without *Bacillus subtilis*

Procedure:

1. Take out the three centrifuge tubes from the incubators
2. Transfer 200μL from each centrifuge tube into a separate well of a 96-well plate
3. Disinfect all equipment and move it into the BSC for aseptic operation
4. Transfer 200μL from the 0812 filtered hay infusion into a separate well of a 96-well plate
5. Place the 96-well plate into the OD microplate reader and record the result

Done by Sophia L, Aibgail

Date: 20260814
Title: Colony PCR (Q1, Q2, S)
Purpose: Amplify engineered DNA plasmid
Procedure:

1. Add the following (2 master mix)

pSTK-0C-SamyQ-BoPep4-v2t1
pSTK-0C-SamyQ-BoPep4-v2t2

	1 reaction	35 reactions
Forward primer (STK fw)	1μL	35 μL
Reverse primer (STK rv)	1μL	35 μL
Q-amp master mix	10μL	350 μL
ddH_2O	8μL	280 μL

pSTK-0C-Csn:Lea

	1 reaction	26 reactions
Forward primer (Csnfw)	1μL	26 μL
Reverse primer (LEA rv)	1μL	26 μL
Q-amp master mix	10μL	260 μL
ddH_2O	8μL	208 μL

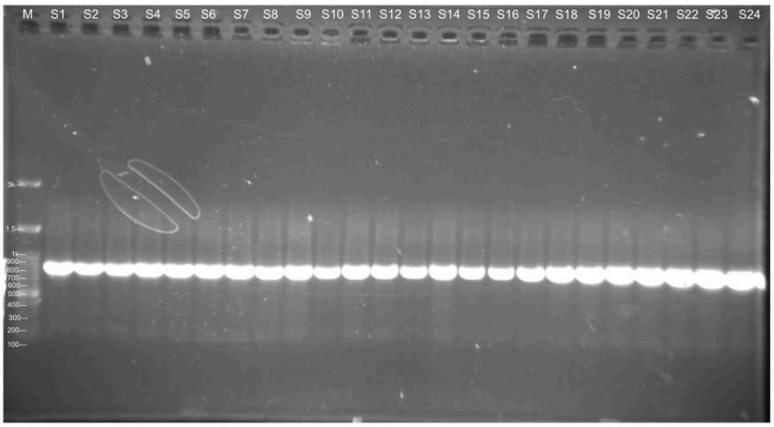
2. Circle 12 colonies from plate G and M, 16 colonies from L
3. Pick up colonies from each agar plate with tip and streak the labelled grid
4. Dispense the tips with colonies into the PCR strips and wait 5 minutes
5. Take out the tips and close the lid
6. Place into PCR machine for thermal cycle

95°C	2 min	1 cycle
95°C	30 sec	30 cycles
65°C	40 sec	
72°C	3 min	
72°C	1 min	1 cycle

Done by Sophie C.

Results:

0814 Colony PCR (Q1)



M: M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder

S1-S24

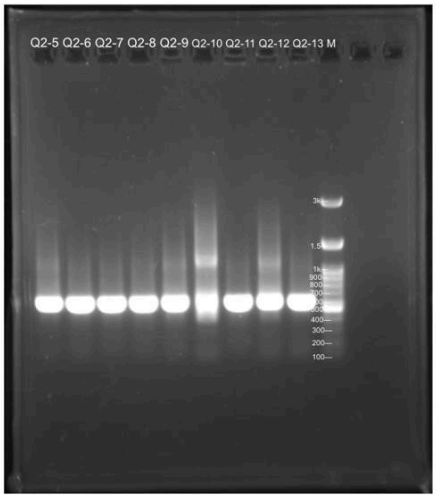
pSTK-0C-Csn-Lea

Csn F & STK-seq-rv

Bp: 785 bp

S2, S10, S16

0814 Colony PCR (Q1)



M: M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder

Q2(5)-Q2(13)

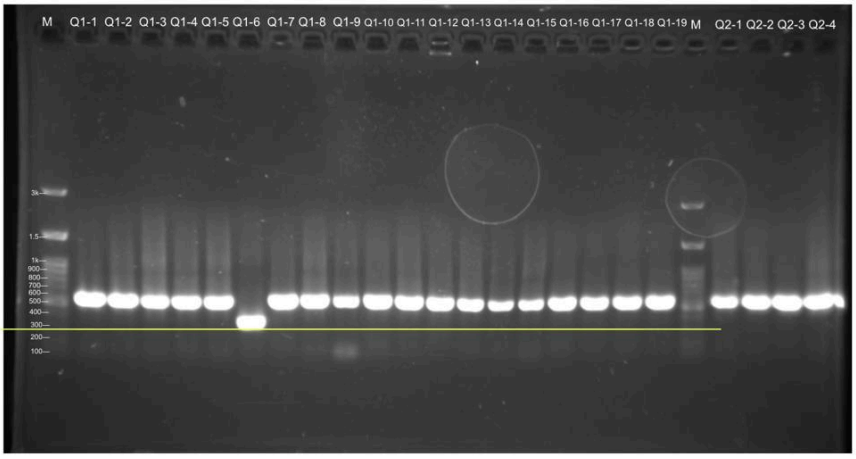
pSTK-0C-SamyQ-Bopep4 -v2t2

STK-seq-fw & STK-seq-rv

Bp: 536 bp

Q2-5, Q2-7, Q2-13

0814 Colony PCR (Q1)



M: M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder

Q1(1)-Q1(19)

pSTK-0C-SamyQ-Bopep4-v2t1

STK-seq-fw & STK-seq-rv

Bp: 536 bp

Q2(1)-Q2(4)

pSTK-0C-SamyQ-Bopep4-v2t2

STK-seq-fw & STK-seq-rv

Bp: 536 bp

Q1-4, Q1-10, Q1-19 -> liquid culture

Date: 20260814
Title: Make 1% agarose gel x3
Purpose: prepare gel for electrophoresis
Procedure:

1. Assemble tray, mold, comb
2. Weigh 1g agarose powder & measure 100mL 1XTAE buffer
3. Add all in conical flask
4. Wrap with plastic wrap (cover)& puncture holes
5. Microwave 30sec, agitate
6. Microwave 5sec. Repeat till no sediments
7. When no smoke, add 5μL safety dye
8. Agitate
9. Pour melted gel into mold
10. Let it solidify at RT

Done by Sophie C

Date: 20260814
Title: Fertilize plants (in soil)
Purpose: Fertilize plants
Procedure:

1. Prepare nutrient solution:
   -Add 0.5g fertilize powder to 500mL tap water in 500 mL flask (ratio: 1g fertilizer: 1000mL tap water)
   -Gently mix the solution
2. Add the solution to each plates

Done by Sophie H.

Date: 20260814
Title: Transformation of L and V
Purpose: Transfer
Procedure:

1. Set up water bath at 42℃
2. Wait until competent cell melted 50% , and add 5μL plasmid to each 100μL competent cell
3. Incubate on ice for 20min
4. Heat shock for 45sec
5. Incubate on ice for 3 min
6. Add 1mL of sOC medium to each tubes
7. Incubate for 1 hr in incubator (37℃)
8. Centrifuge 5000g for 5min
9. Place the centrifuged tube in BSC along with an empty microtube&TBC plates
10. Discard around 900mL supernatant
11. Repellent the rest
12. Spread the liquid on TBC plate
13. Place in incubator for 16hr
    Done by Naomi, Sophie H

Date: 20260814
Title: Make LBC plate
Purpose: prepare LBC plate
Procedure:

1. Add 250mL ddH_2O 9.25 LB agar powder, and 2mL glycerol to a 500mL flask
2. Add a stirring bar
3. Stir the mixture until all powder are dissolved
4. Cover the flask with aluminum foil and an autoclave indicator sticked on top
5. Autoclave for 20 minutes at 127°C.
6. Put it in BSC while wait for it to cool down (can also stir before putting in BSC)
7. Add 250μL of chloramphenicol and gently mix it
8. Distribute 250μL LBC medium in plate (20mL each)
9. Wait for it to cool down

Done by Sophic H, Alex Li, Naomi

Date: 20260814
Title: Make LB medium
Purpose: prepare LB medium
Procedure:

1. Add 11.g of TB powder, 250mL of ddH_2O and 2mL glycerol to a 500mL flask

2. Add a stirring bar to the flask
3. Stir until all powder are dissolved
4. Cover the flask with aluminum foil and an autoclave indicator sticked on top
5. Autoclave for 20 minutes at 127°C.
   Done by Naomi, Sophie H

Date: 20260814
Title: Fertilize plants
Purpose: Fertilize plants
Procedure:

1. Prepare nutrient solution
    - Add 0.5g fertilize powder to 500mL tap water(1g fertilizer:1000mL taps water)
    - Gently mix it
2. Add the solution to each plates
   Done by Sophie H

Date: 20260814
Title: Cotreatment(正瀚 protectant+100NaCl)
Purpose: test 正瀚 protectan
Procedure:
Aftertreatment (100NaCl)
pretreatment(100NaCl)

1. Use previous diluted 正瀚 protectant (1/500 dilution) into tube in the BSC
2. Add 1 mL of the diluted 正瀚 protectan into two cotreatment plates
    - Cotreatment (protectant+100NaCl)
    - Cotreatment (protectant+0NaCl)
3. Add 1mL of diluted 100mM NaCl into two cotreatment +aftertreatment
    - Cotreatment (protectant+100mM NaCl)
    - aftertreatment(100NaCl+protectant)
4. Evenly distribute the liquid by lightly shaking left to right
   Done by Abigail

Date: 20260814
Title: prepare 0.1 MS medium
Purpose: prepare medium for experiment
Procedure:

1. Prepare 700mL of 0.1 MS medium using the following protocol
2. Add the contents into a beaker
3. Use a magnetic stirring bar to stir until no sediments are left in the solution
4. Use pH probe to measure the pH and ensure the pH is between 5.7~5.8. If pH is too low slowly add 1M KOH to increase it
5. Wrap the beaker with aluminium foil and stick a piece of indicator tape on top

6. Auto clave for 20min at 127℃
   Done by Alex

Date: 20260814
Title: Add CH protectant in Box C2 (pretreatment) for heat stress testing
Purpose: test heat stress
Procedure:

1. Change the medium of Boxes C1,C2,C3 and C4. Add 300mL of 0.1MS medium when changing medium for each box
2. Take out the plants that aren't growing well or are to small
3. Evenly distribute the rest of arabidopsis into boxes C1,C2,C3,C4
4. Add 600μL of CH protectant into box C2
   Done by Alex

Date: 20260814
Title: Hay infusion
Purpose: test which nutrient additive best support *Bacillus subtilis* growth in lake water
Procedure:

1. Take out half of the hay from the 新竹 container
2. Use an electronic pipette controller with 25mL serological to draw all the liquid from the container and pour it into a 100mL beaker
3. Measure the total amount of liquid
4. Take out 5 bottles and label them as following
   - 10g coffer grounds
   - 10g sugarose
   - 10g hay
   - 10g hay / 5g sugarose / 10g coffee grounds
   - 10g hay / 10g sugarose / 5g coffee grounds

5. Take out the scale and measure the additive as labeled on the battle
6. Use the beaker and get 500mL of water
7. The water into the log sugerose bottle to see if water can cover the sugerose
8. Remove the water from the bottle, make sure not to pour the sugarcane out
9. Pour the liquid in the 1000mL beaker back into the 新竹 container
10. Add 10g of sugarose and log of coffer grounds into the 新竹 container and close the lid
    Done by Sophia L, Sophie H, Olivia Du, Mia

Date: 20260815
Title: Make LBC with glycerol
Purpose: prepare LBC plate with glycerol
Procedure:

1. Follow the recipe
2. Add a stirring bar and stir until no more sediments
3. Cover the flask with aluminium foil and stick indicator tip on top
4. Autoclave at 127℃ for 20min
5. Place new petri dishes and equipment in the BSC
6. Spray equipment with 70% ethanol
7. Add 500μL chloramphenicol to the mixture
8. Use a pipette gun to add 20mL of mixture into each petri dishes
9. Solidify at room temperature under the BSC
   Done by Abby K/T

Date: 20260815
Title: Colony PCR (L)
Purpose: Amplify engineered DNA plasmid
Procedure:
L:3P-EEBD-RBS0-csn:GanA-Gala-L3S2P21

1. Prepare the following master mix x35

Fw primer EEBD F	35μL
Rv primer CsnR	35μL
Q-amp master mix	350μL
ddH_2O	280μL

2. Distribute 20μL to each PCR tube
3. Pick up colony with pipette
4. Dispense pipette in master mix
5. Close the lid
6. Place in PCR machine for thermal cycle

95℃	5min	x1
95℃	20sec	35cycle
60℃	20sec	
72℃	30sec	
25℃	5min	x1

Date: 20260815
Title: Make 2% Agarose Gel x2
Purpose: prepare gel for gel electrophoresis of colony PCR
Procedure:

1. Assemble mold, tray, comb
2. Weigh 1.2g agarose powder and measure 60mL 1XTAE buffer
3. Add all in conical flask
4. Cover with plastic wrap&puncture holes
5. Microwave 30sec agitate
6. Repeat till no sediments
7. When no smoke, add 3μL safety dye
8. Pour melted gel into mold
9. Let solidify at RT
   Done by Sophie C

Date: 20260815
Title: Miniprep
Purpose: to extract plasmid from liquid culture
Procedure:
S2,S10,S16 : pSTK-OC-csn:LEA
Q1-19,Q2-13,Q2-5,Q1-4,Q2-7,Q2-10:pSTK-OC-SamyQ-Bopep4-V2t2

1. Pellet 1.8mL bacterial overnight culture by centrifuge 6800g for 3min
2. Discard the supernatant and repeat step 1 until collect all of the liquid culture
3. Add 250μL P1 buffer, and resuspend pelleted bacterial
4. Add 250μL P2 buffer, and mix thoroughly by inverting the tube 4-5times until the solution become clear(Do not allow the reaction more than 5 min)
5. Add 350μL N3 buffer and mix immediately and thoroughly by inverting 4-6 time
6. Centrifuge for 10 min at 17900g
7. Transfer 800μL supernatant into new column
8. Centrifuge 1 min for 17900g
9. Add 500μL PB buffer
10. Centrifuge 1 min for 17900g , and discard the flow-through
11. Add 750μL PE buffer
12. Centrifuge 1 min for 17900g , and discard the flow-through
13. Centrifuge empty column for 1min at 17900g , and discard the flow-through
14. Put into new column
15. Add 50μL EB buffer
16. Stand at RT for 1 min
17. Centrifuge for 1min at 17900g

	Concentration(ng/μL)	A260/230	A260/280
S10	48.00	2.000	1.920

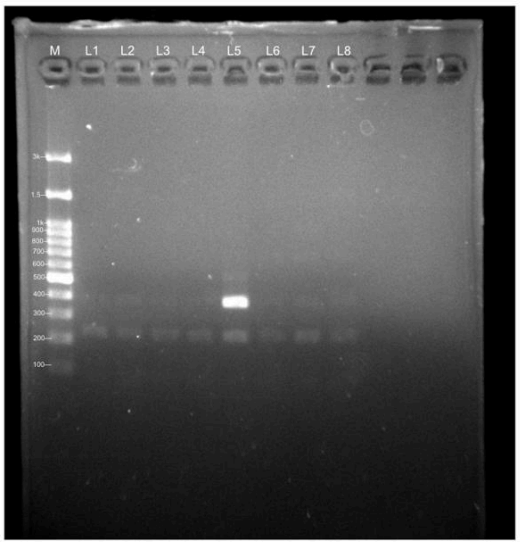
S2	56.00	1.931	1.931
S16	55.00	1.774	1.897
Q1-19	61.00	1.906	1.900
Q1-4	77.00	1.400	1.925
Q1-10	52.00	2.364	2.000
Q2-5	50.00	2.000	1.923
Q2-7	57.00	2.036	1.900
Q13	57.00	1.500	1.900

Done by Sarah, Sophie H

Date: 20260815
Title: Gel electrophoresis (L)
Purpose: Check the approximate length of colony PCR product
Procedure:
L:pSTK-Exp1-3p-EEBD-RBS0-can:GanA-Gala-L3S2P21

1. Load the samples in the following order:
   Gel1:
   M L1 L2 L3 L4 L5 L6 L7 L8
   5 5 5 5 5 5 5 5 5
   Gel2:
   M L9 L10 L11 L12 L13 L14 L15 L16 L17 L18 L19 L20 L21 L22 L23 L24 L25 L26 L27
   5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5
   L28 L29 L30 L31 L32
   5 5 5 5 5
2. Resolve the gel at 100V for 30min

0815 Colony PCR (L)



M: M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder

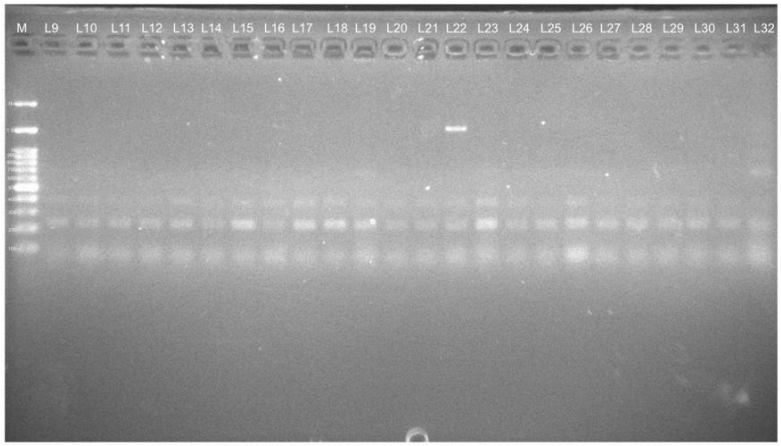
L1-L8

pSTK-EXP1-3P-EEBD-RBS0-Csn:GanA-Gala-L3S2P21

EEBD F & Csn R

Bp: 210 bp

0815 Colony PCR (L)



M: M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder

L9-L32

pSTK-EXP1-3P-EEBD-RBS0-Csn:GanA-Gala-L3S2P21

EEBD F & Csn R

Bp: 210 bp/371bp

Done by Sophic C

Date: 20260815
Title: Colony PCR
Purpose: Amplify engineered DNA plasmid
Procedure:
V1-V24:ABDI

1. Prepare the following master mix x28

Fw primer CcaR	28μL
Rv primer PcpcG2-veg	28μL
Q-amp master mix	280μL
ddH_2O	224μL

2. Distribute 20μL to each PCR tube
3. Pick up colonies with pipettes
4. Strike the grid three times
5. Dispense pipette tip in master mix
6. Close the lids and spin down
7. Place into PCR machine for the following thermal cycle

95℃	5min	x1
95℃	20sec	35cycle
60℃	20sec	
72℃	1min	
25℃	5min	x1

Done by AbbyK, Sophie C, Chloe

Date: 20260815
Title: Golden Gate Assembly
Purpose: assemble the parts
Procedure:
pSTK-OC-SamyQ-AccD-v2t1

part	amount
pSTK-001-pSTK-O-sfGFP 346ng/μL , 3090bp	1.45μL(⅛)
SamyA-AccD-v2t1(pTwist) 125ng/μL , 3401bp	1.78μL(⅛)
T4 DNA ligase buffer	2μL
Sterile water	12.78μL
T4 DNA ligase	1μL
BbsI-HF	1μL

pSTK-OC-SamyQ-AccD-v3t1

part	amount
pSTK-001-pSTK-O-sfGFP 346ng/μL , 3090bp	1.45μL(⅕)
SamyA-AccD-v3t1(pTwist) 142ng/μL , 3401bp	1.56μL(⅕)
T4 DNA ligase buffer	2μL
Sterile water	12.99μL
T4 DNA ligase	1μL
BbsI-HF	1μL

pSTK-OC-SamyQ-LEA-v2t1

part	amount
pSTK-001-pSTK-O-sfGFP 346ng/μL , 3090bp	1.45μL(⅕)
SamyA-AccD-v2t1(pTwist) 141ng/μL , 2840bp	1.3μL(⅕)
T4 DNA ligase buffer	2μL
Sterile water	13.25μL
T4 DNA ligase	1μL
BbsI-HF	1μL

pSTK-OC-SamyQ-LEA-v2t1

part	amount
pSTK-001-pSTK-O-sfGFP 346ng/μL , 3090bp	1.45μL(⅕)
SamyA-AccD-v2t1(pTwist) 111ng/μL , 2840bp	1.66μL(⅕)
T4 DNA ligase buffer	2μL
Sterile water	12.8μL
T4 DNA ligase	1μL

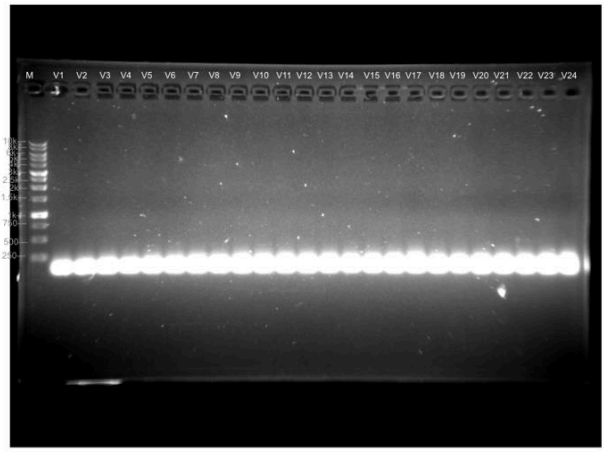
BbsI-HF	1μL

Done by Sophie C, Abby T/K

Date: 20260815
Title: Gel electrophoresis (V)
Purpose: Check the approximate length of colony PCR product
Procedure:
M:1kb DNA ladder
V:ABDI

1. Load the sample in the following layout
   M V1 V2 V3 V4 V5 V6 V7 V8 V9 V10 V11 V12 V13 V14 V15 V16 V17 V18 V19 V20 V22
   5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5
   V23 V24
   5 5
2. Resolve the DNA at 100V for 30 min

0815 Colony PCR (V - ABDI)



M: 1kb DNA ladder

V1-V24

A: PtagC-RBS-Ho1:pcyA-L3S2P21

B:Pliag-RBS0-ccaSm3-B0015

D: Pveg-RBS0-ccaR-B0015

I: PcpcG2-veg-EEBD-RBS0-csn:ACCD-B0015

ccaR F & pcpcG2-veg R

Bp: 1032 bp

Done by Abby K

Date: 20260815
Title: Make 1% agarose gel
Purpose: prepare gel for electrophoresis
Procedure:

1. Assemble mold, tray, comb
2. Weigh 0.2g of agarose powder and 20mL of 1XTAE buffer
3. Add into a flask
4. Cover with plastic wrap and poke a few holes
5. Microwave for 20 seconds, then agitate
6. Repeat for 5 seconds each until no sediments

7. When no smoke, add 2mL of safety dye
8. Pour melted gel into mold
9. Solidify at RT
   Done by Abby K , Sophie C

Date: 20260815

Title: hydroponic heat stress testing

Purpose: testing heat stress in hydroponic

Procedure:

1. Prepare 4 new boxes and add 300mL of old MS medium into each box. Label them as C1, C2,C3,C4
2. Add 600L CH protectant into C3 (co-treatment)
3. Take another box and add 300mL of water into it
4. Put a thermometer into the water
5. Put all boxes into the incubator(setting 37℃, actual temperature 33℃)
6. Wait until the thermometer reaches 33℃
7. Take out the new C1 box from incubator
8. Transfer the floating board with plants form old C1 box into the new C1 box that is at 33℃
9. Put the new C1 box back into the incubator
10. Repeat steps 7~9 until all boxes changed medium
    Done by Alex

Date: 20260815

Title: add zhenhen protectant to heat stress testing agar plates

Purpose: testing zhenhen protectant

Procedure:

pre-treatment 7:00pm

1. Dilute zhenhen protectant by 1/500 (40μL of peptide+19960μL of ddH_2O -> total:20mL)
2. Apply the sprayer to the centrifuge tube
3. Place the plate vertically and spray the plants with protectant with a distance of ~15cm in between (2 sprays)
   Done by Abby T
   Supervised by Gina

Date: 20260816
Title: Protein Extraction
Purpose: purify protein samples
Procedure:

1. Take 24 2mL centrifuge tubes and add 1800μL of overnight liquid culture(*Bacillus subtilis*)
2. Centrifuge 16000g for 30seconds
3. Take tubes out of microcentrifuge and pour the supernatant into a 50mL centrifuge tube
4. Repeat the steps 1-3 until the 50mL centrifuge tube is filled
5. Pour the supernatant into a syringe connected to 0.22um filter and filter the supernatant into a 50mL centrifuge tube
6. Repeat the previous steps until all culture is filtered
7. Centrifuge the Ni-NTA column at 175g for 2 min to remove the storage buffer
8. Add 6mL of equilibrate buffer into the column by dispensing the buffer along the walls of the tube
9. Add 237.5μL of imidazole into supernatant
10. Add all supernatant from 1 tube into the column
11. Add 6mL of wash buffer and remove the flow-through repeat this step 3 times
12. Centrifuge at 175g for 2 min
13. Collect the flow through from the last wash buffer step
14. Add 3mL of elute buffer
15. Centrifuge at 175g for 2min
16. Collect the flow-through as the purified protein

Date: 20260816
Title: Miniprep of H1 (0809 liquid culture)
Purpose: DNA plasmid purification
Procedure:
H1: 3P-EEBD-RBS0-Csn:LEA-B0015

1. Transfer 1.8ml of overnight liquid culture medium to a microcentrifuge tube. Centrifuge at 8000g for 3 minutes, and discard the flow-through. Repeat this step until all liquid cultures are used.
2. Resuspend pelleted bacterial cells in 250 µl Buffer P1
3. Add 250 µl Buffer P2 and mix thoroughly by inverting the tube 4-6 times
4. Add 350 µl Buffer N3 and mix immediately and thoroughly by inverting the tube 4-6 times.
5. Centrifuge for 10 min at 17,900 g
6. Transfer the supernatant from to the QIAprep spin column. Centrifuge for 60s and discard the flow-through
7. Wash the spin column by adding 0.5 ml Buffer PB. Centrifuge at 17900g for 60s, and discard the flow-through
8. Wash the spin column by adding 0.75 ml Buffer PE. Centrifuge at 17900g for 60s, and discard the flow-through
9. Centrifuge for 1 min to remove residual wash buffer

10. Add 50μL of EB buffer to the center of the column membrane, let stand for 1 min, and centrifuge for 1min.

Done by Abby T

Results:

Concentration (ng/μL)	A260/230	A260/280
178	2.177	1.914

Date: 20260816
Title: Level 2 Golden Gate Assembly
Purpose: Assemble level 1 genetic circuits
Procedure:

V: A+B+D+I (1:3)

Part	amount	dilution
Exp2 (163ng/μL,5104bp)	0.203	1.015μL (⅕) → 1μl of plasmid with 4μl of ddH_2O
A:PtagC-RBS-H01:pcyA-l3S2 P21 (109ng/μL,3 754bp)	0.690	1.38μL(½) → 1μl of plasmid with 1μl of ddH_2O
B:PliaG-RBS0-CcaSm3-B0015 (83ng/μL,3792bp)	0.890	1.78μL (½) → 1μl of plasmid with 1μl of ddH_2O
D:Pveg-RBS0-CcaR-B0015 (134ng/μL,3243bp)	0.471	1.413μL (⅓) → 1μl of plasmid with 2μl of ddH_2O
I:PcpcG2-veg-EEBD-RBS0-csn:AccD-B0015 (165ng/μL,3757bp)	0.444	1.332μL (⅓) → 1μl of plasmid with 2μl of ddH_2O
T4 DNA ligase buffer	2μL	2μL
Sterile water	13.3μL	13.3μL
T4 DNA ligase	1μL	1μL
BsmBI	1μL	1μL
Total	20μL	20μL

2. Run thermal cycle

55°C	5 min	45 cycles
16°C	5 min	
55°C	10 min	1 cycle
80°C	20 min	

Done by Abby T, Alex

Date: 20260816
Title: Spray CH bio peptide to co-treatment group, and place all groups into incubator at 37°C for thermo priming
Purpose: apply bio peptide and heat stress
Procedure:

1. Place the co-treatment plate vertically, spray evenly toward the entire plant from leaves to root with around 15cm of distance in between, 1 spray on the left side and 1 spray on the right side. Make sure the entire plant receives even treatments.
2. Place all groups on the rack and put the entire rack into the incubator, under the white light set, set at 37°C

Done by Abby T

Date: 20260816
Title: Ni-NTA Regenerate Buffer (for protein purification)
Purpose: prepare buffer for protein purification
Procedure:

1. Prepare 0.39g of MES and 0.584g of NaCl
2. Dissolve both in 100mL of ddH_2O in a flask
3. Adjust pH to 5.0 using NaOH (final :5.0 pH)
4. Transfer all to the jar label (N-NTA regenerate buffer)
   Formula:20mM MEs 0.1M NaCL 5.0pH (therme fisher)
   Done by Ryan Y

Date: 20260816
Title: Transformation
Purpose: transform and cultivate engineered bacteria on agar plate
Procedure:
pSTK-OC-SamyQ-AccD-V2t1
pSTK-OC-SamyQ-AccD-V2t1
pSTK-OC-SamyQ-LEA-V2t1
pSTK-OC-SamyQ-LEA-V2t1

1. Turn on water bath to 42℃ prepare ice bucket
2. Prepare 4 LBC plates

3. Thaw competent cells until 50% thawed and add 10μL plasmid
4. Incubate in ice for 20min
5. Heat shock at 42℃ for 45 seconds
6. Incubate on ice for 3 min
7. Add 1mL SOC medium to each tube
8. Incubate at 37℃ for 1 hour
9. Centrifuge at 5000g for 5 min
10. Take out 1 mL of supernatant
11. Resuspend cell, and spread entirely on LBC plates
    Done by Abby T/K

Date: 20260816

Title:Remove protectant from C3 and add protectant in C4

Purpose: testing protectant

Procedure:

1. Prepare 2 new boxes and add 300μL of 0.1X MS medium into both boxes. Label them as C3, C4
2. Add CH protectant into box C4(post-treatment)
3. Take another box and add 300mL of water into it
4. Put a thermometer into the water
5. Put both boxes into the incubator(setting 37℃, actual temp 33℃)
6. Wait until thermometer reaches 33℃
7. Take out new C3 box from incubator
8. Transfer the floating board with plants form old C3 box into new C3 box that is at 33℃
9. Put the new C3 box bak into incubator
10. Repeat steps 7~9 to change the medium of C4
    Done by Alex

Date: 20260816

Title:Spray CH bio peptide to co-treatment (7:00pm)+place all group into incubator at 37℃ for therm priming

Purpose: testing peptide

Procedure:

1. Spray diluted protectants to co-treatment
    - 2 sprays
    - 15cm distance
2. Place all groups of plates onto a rack , and put in incubator under white light
    Done by Abby T

Date: 20260817
Title: LB Medium
Purpose: Prepare medium for bacterial culture
Procedure:

1. Prepare the following

LB broth powder	6.25g
ddH_2O	250mL

2. Add all to a conical flask
3. Add a stirring bar
4. Stir on hot plate till no sediments
5. Remove stirring bar
6. Autoclave
7. Put it in the fridge
   Done by Sophie C

Date: 20260817
Title: Colony PCR
Purpose: to screen for successful cloning after transformation of QA, QB, QC, QD
Procedure:
QA:pSTK-OC-SamyQ-LEA v2t1
QB:pSTK-OC-SamyQ-AccD v3t1
QC:pSTK-OC-SamyQ-LEA v2t2
QD:pSTK-OC-SamyQ-AccD v2t1

1. Prepare the following master mix for
   QA+QC x26

Fw primer Biobrick F	26μL
Rv primer Lea R	26μL
Q0amp master mix	260μL
ddH_2O	208μL

QB+QDx26

Fw primer Biobrick F	26μL
Rv primer AccD R	26μL

Q0amp master mix	260μL
ddH_2O	208μL

2. Distribute 20μL to each PCR tube
3. Dip pipette into colony
4. Dispense the tip into PCR tube
5. Take out tip & close lid
6. Thermal cycle in PCR machine

95℃	5min	x1
95℃	20sec	35cycle
60℃	20sec	
72℃	1min20sec	
25℃	5min	x1

Done by Sophie C, Abby K

Date: 20260817
Title: Make 1% agarose gel x2
Purpose: for gel electrophoresis of colony PCR
Procedure:

1. Assemble comb, tray, mold
2. Weigh 0.8g agarose powder & 80mL 1XTAE buffer
3. Add all in conical flask
4. Cover with plastic wrap & puncture holes
5. Microwave for 30sec agitate
6. Microwave for 5 sec. Agitate. Repeat until no sediments
7. When no smoke, add 4μl safety dye
8. Pour melted gel into mold
9. Let solidify at RT
   Done by Sophie C

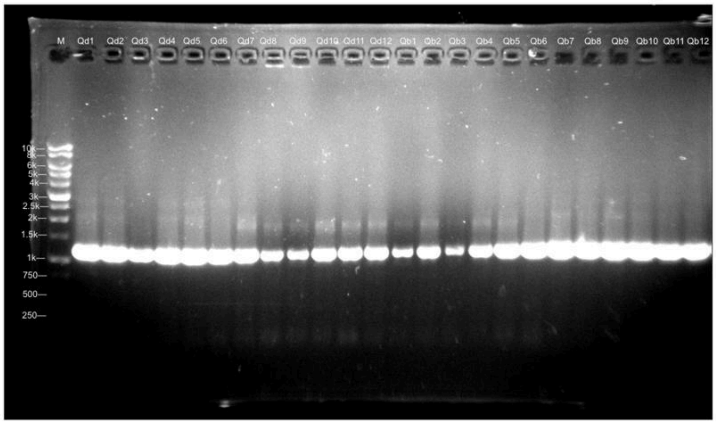
Date: 20260817
Title: Gel Electrophoresis of QD and QB
Purpose: Check the approximate length of colony PCR product
Procedure:
QD:pSTK-OC-SamyQ-AccD-v2t1
QB:pSTK-OC-SamyQ-AccD-v3t1

1. Load samples in the following order:
   M QD1 QD2 QD3 QD4 QD5 QD6 QD7 QD8 QD9 QD10 QD11 QD12 QB1 QB2 QB3
   5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5

QB4 QB5 QB6 QB7 QB8 QB9 QB10 QB11 QB12
5 5 5 5 5 5 5 5 5

2. Resolve at 100V for 30 min

8/17 colony PCR (ACCD)



M: 1kb DNA ladder

Qd1-12 (ACCD v2t1)

Qb1-12 (ACCD v3t1)

Biobrick F
ACCD Rv

Bp = 1331

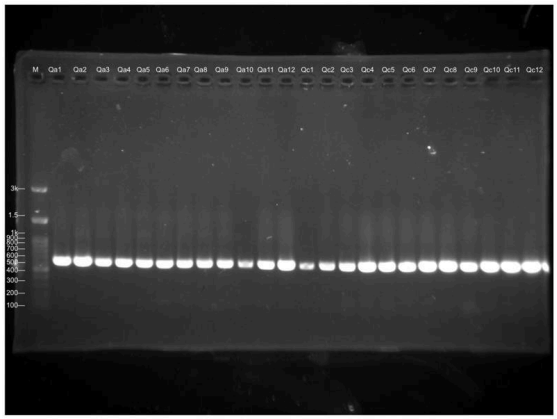
Done by Abby K

Date: 20260817
Title: Gel Electrophoresis of QA and QC
Purpose: Check the approximate length of colony PCR product
Procedure:
QA:pSTK-OC-SamyQ-Lea-v2t1
QC:pSTK-OC-SamyQ-Lea-v2t2

1. Load the sample in the following order
   M QA1 QA2 QA3 QA4 QA5 QA6 QA7 QA8 QA9 QA0 QA11 QA12 QC1 QC2 QC3
   5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5
   QC4 QC5 QC6 QC7 QC8 QC9 QC10 QC11 QC12
   5 5 5 5 5 5 5 5 5

2. Resolve at 100V for 30 min

8/17 colony PCR (LEA)



M: 3kb DNA ladder

Qa1-12 (LEA v2t1)

Qc1-12 (LEA v2t2)

Biobrick F
Lea Rv

Bp = 596

Done by Sophie C

Date: 20260817
Title: Transformation (V)
Purpose: Transform and cultivate engineered bacteria on agar plate
Procedure:

1. Set up water bath to 42℃
2. Thaw competent cell till 50% thawed, add 10μL plasmid
3. Incubate on ice for 20 min
4. Heat shock 42℃ for 45sec
5. Incubate on ice for 3 min
6. Add 1mL SOC medium
7. Incubate for 1 hr at 37℃
8. Centrifuge at 5000g for 5 min
9. Take out ~1000μL supernatant
10. Resuspend the pellet
11. Spread on LBC plate
12. Incubate at 37℃ for 16hr
    Done by Sophie C

Date: 20260817
Title: Add zhenhen protectant (pre-treatment) to experiment set #7 trial 3 (3:30pm)
Purpose: testing zhenhen protectant
Procedure:

1. Dilute the CH protectant by 1/500
2. Under the BSC , use 200μL pipette to remove the excess water
3. Add 3mL of CH protectant to the agar plate

4. Move it around the plate until the entire surface is covered
5. remove 2 mL of the CH protectant na discard it
6. Repeat steps 3-5 for the other two plates
7. Use breathable paper to tape up the edges of the plates
8. Place them back in the growth chamber
   Done by Abby K
   Supervised by Abby T

Date: 20260817

Title: Plant new seeds on soil pots

Purpose: plant seeds

Procedure:

1. Autoclave a bag of soil
2. Soak 4 flower pots in 1/10 bleach and wait for 20 min
3. Rinse the pot with water for 5 min
4. Fill the pot with soil and slightly press it down
5. Gently shake it vertically to soften the soil
6. Water the soil until the upper part of the soil beamer fully wet
7. Use a tip to get a seed from preservation solution
8. Dispense the seed on the soil in fixed position (4 seeds in a pot)
9. After all the pots are done, cover with petri dish and place them in the incubator
   Done by Abigail , Sophie C

Date: 20260817

Title: Spray CH bio peptide to after-treatment (7:00pm)

Purpose: testing peptide

Procedure:

1. Spray diluted protectants to after treatment
   - 2 sprays
   - 15 cm distance
2. Place all groups of plates onto a rack, and put in incubator (2**2**℃ for 17hr)
   Done by Sophie C

Date: 20260817

Title: plant new set of seeds (set F) for hydroponic system

Purpose: prepare new set of seed for hydroponic system

Procedure:

1. Prepare 47 eppendorf lids
2. Add 250µL of 0.8% agar into each lid
3. Wait until the agar solidifies

4. Clean tube rack with 10% bleach and rinse with ddH_2O until there are no smell of bleach
5. Add 2,2mL of 0,1X MS medium into 47 tube rack
6. Use an autoclaved toothpick to pick one seed and add the seed into the hole of the lid
7. Repeat step 6 until all lids have seeds
8. Close the lid of the tube rack
9. Wrap the tube rack with aluminum foil
10. Place the tube rack into the refrigerator at 4℃

    Done by Alex

Date: 20260818
Title: Electroporation competent cell preparation
Purpose: To prepare electroporation competent cell for electroporation
Procedure:

1. Overnight culture *Bacillus subtilis* WB800
2. Dilute the overnight culture of WB800 *Bacillus subtilis* in 16 fold in growth medium and grow at 37℃ to OD_{600} of 0.907
3. Cool the cells on ice with shake
4. Collect the cells in 2 mL microtube at 4℃ and 5000g for 5 min
5. Wash the cells with 1mL cold electroporation medium 2 times
6. Suspend the cells in 1/50 of the culture volume of the electroporation medium
7. The cells stored at -80℃
   Done by Gabriel

Date: 20260818
Title: Electroporation
Purpose: Introduce DNA sequence into competent cells
Procedure:

1. Following the protocol in 0811
2. Mix 2.4μL pveg-AccD(J) plasmid (410ng/μL) in the 100μL EPC from 0817
3. Electroporate the cell at 2.3kV 300st 35F
4. Recovery with recovery medium (1mL) for 3 hr
5. Centrifuge at 5000g for 5 min
6. Spread at LBE plate
   Done by Gabriel

Date: 20260818
Title: Colony PCR (QA, QB, QC, QD)
Purpose: Amplify the number of targeted gene
Procedure:
QA: pSTK-OC-SamyQ-LEA-v2t1
QB: pSTK-OC-SamyQ-AccD-v3t1
QC: pSTK-OC-SamyQ-LEA-v2t2
QD: pSTK-OC-SamyQ-AccD-v2t1

1. Prepare the following master mix x9

Fw primer Samy Q F	9μL
Rv primer sSTK-seq-rv	9μL
Q-amp master mix	90μL
ddH_2O	72μL

2. Distribute 20μL to each PCR tube
3. Pick up colonies with tip (QA1, QA3, QC3,QCs,QB,QB3,QD2,QD11)
4. Dispense tip in PCR tube. Take out tip
5. Close the lid
6. Thermal cycle in PCR machine

95℃	5min	x1
95℃	20sec	35cycle
60℃	20sec	
72℃	1min30sec	
72℃	5min	x1

Done by Sophie C

Date: 20260818
Title: Make 1% Gel x2
Purpose:prepare gel for electrophoresis
Procedure:

1. Assemble mold, tray, comb
2. Weigh 0.2g agarose gel, 20mL 1XTAE buffer (for 1)
3. Add all in conical flask
4. Cover with plastic wrap&puncture holes
5. Microwave for 30 sec agitate
6. Repeat until no sediments
7. When no smoke, add 1μL safety dye agitate
8. Pour melted gel into mold
9. Let it solidify at RT
   Done by Sophie C

Date: 20260818
Title: Gel electrophoresis
Purpose: To validate PCR product
Procedure:

1. Load samples in the following order
   M QA_1 QA_3 QC_3 QC_5 QB_1 QB_3 QD_2 QD_{11}
   5 5 5 5 5 5 5 5 5

   M: 3kB DNA ladder

   Qa1, 3 (LEA v2t1)
   Qc3, 5 (LEA v2t2)
   Bp: 770

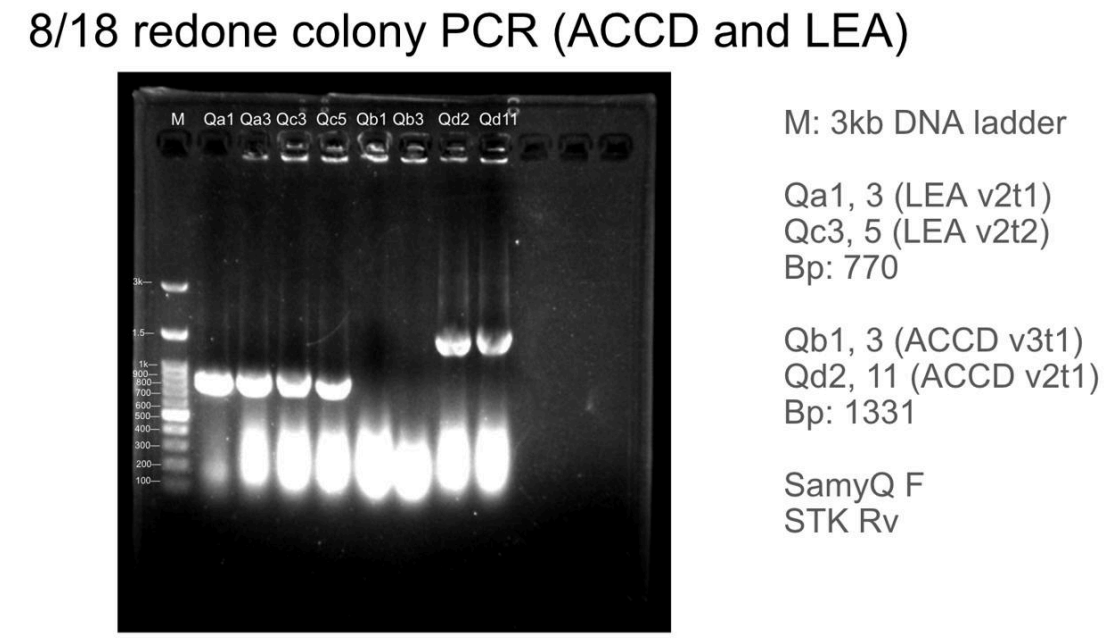
Qb1, 3 (ACCD v3t1)
Qd2, 11 (ACCD v2t1)
Bp: 1331

Primers: SamyQ F, STK Rv

2. Resolve gel at 100V for 30 mins

Done by Abby K

Results



Date: 20260818
Title: Colony PCR (L)
Purpose: to screen for successful cloning after transformation
Procedure:

L: 3P-EEBD-RBS0-Csn:GanA-Gala-L3S2P21

1. Prepare the following reactions in PCR tube

   L1:

Fw primer Biobrick prefix	1μL
Rv primer CsnR	1μL
Q-amp master mix	10μL

ddH_2O	8μL

L2:

Fw primer pSTK-EXP-fw	1μL
Rv primer CsnR	1μL
Q-amp master mix	10μL
ddH_2O	8μL

2. Pick up colony with tip
3. Dispense tip in pcr tube
4. Take out tip, close lid
5. PCR machine for thermal cycle

95℃	5min	x1
95℃	20sec	x35
60℃	20sec	
72℃	1min30sec	
25℃	5min	x1

Done by Sophie C

Date: 20260818
Title: Gel electrophoresis (L)
Purpose:
Procedure:
L: 3P-EEBD-RBS0-Csn:GanA-Gala-L3S2P21

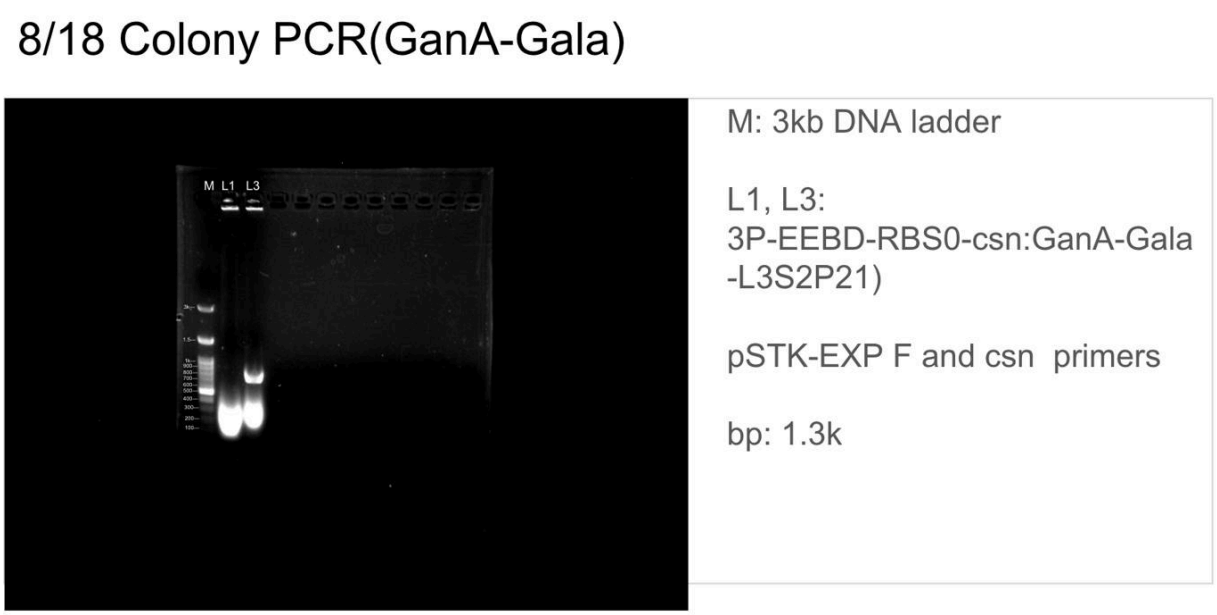
1. Load the samples in the following order:
   M L1 L2
   5 5 5

   M: 3kb DNA ladder
   L1, L3: 3P-EEBD-RBS0-csn:GanA-Gala-L3S2P21)
   pSTK-EXP F and Csnprimers
   bp: 1.3k

2. Resolve the gel at 100V for 30 minutes

Done by Sophie C

Results:



Date: 20260818
Title: Golden gate assembly (level 2)
Purpose: to assemble parts to level 2
Procedure:
Samples:

- X: ACDI
- V: ABDI

V:A+B+D+I

V	
Part	Amount
Vector Exp2 (163ng/μL, 5104bp)	1.017μL
A (109ng/μL, 3754bp)	1.119μL
B(83ng/μL, 3792bp)	1.484μL
D (134ng/μL, 3243bp)	0.786μL
I (165ng/μL, 3787bp)	0.75μL
T4 DNA ligase buffer	2μL

Sterile ddH_2O	10.84 µl
T4 DNA ligase	1µL
BsmB1	1µL

X: A+C+D+I (1:1)

Part	Original (pmol=0.05)
Vector: pSTK-EXP-2-sfGFP - Concentration: 163 - Bp: 5104	1.017 µL
(A)pSTK-1A-PtagC+RBS-Ho1:pcyA-L3S2P21 - Concentration: 109 ng/µl - Bp: 3754	1.119 µL
(C)pSTK-1B-PlepA-RBS0-ccaSm3-B0015 - Concentration: 98 ng/µl - Bp: 3828 bp	1.27 µL
(D)pSTK-1C-Pveg-RBS0-ccaR-B0015 - Concentration: 134ng/µl - Bp: 3243 bp	0.786 µL
(I)pSTK-1D-PcpcG2-veg-EEBD-RBS0-csn:ACCD-B0015 - Concentration: 165 ng/µl - Bp: 3857	0.75 µL
T4 DNA ligase buffer	2 µL
Sterile water	11.05 µL
T4 DNA ligase	1 µL
BsmB1	1 µL
Total	20 µL

Done by Sophie C.

Date: 20260818
Title: Add NaCl + CH peptide to contreatment (exp set #7 trial 3)
Purpose: contreatment test
Procedure:
NaCl: 7.5mM, 100mM
CH peptide: pretreat, contreat, aftertreat

1. Dilute protectant by 1/500, NaCk from 3M NaCl master mix
    - 75mM (total volume: 15mL 375µL (3M NaCl)+ 14.625mL (sterilized water)
    - 100mM (total volume : 15mL 500µL (3M NaCl)+14.5mL (sterilized water)
2. Under the BSC, use a 200µL pipette to remove the excess water
3. Add 3mL of CH protectant to the agar plates
4. Move it around the plate until the entire surface is covered
5. Wait for 1 min and remove 2mL of CH peptide and discard it
6. Mix 1mL of CH peptide (diluted) with 1mL of 75mM NaCl in a 15mL centrifuge tube by pipetting and label (CH peptide, 75mM NaCl)
7. Mix 1mL of CH peptide (diluted) with 1mL of 100mM NaCl in another 15mL centrifuge tube by pipetting and label (CH peptide, 100mM NaCl)
8. Pipette 2mL of protectant + NaCl master mix to each of the corresponding agar groups and more the plates until surface covered entirely
9. Pipette 1mL NaCl to 75mM-> (control. 75mM NaCl) plate / 100mM -> (control, 1–mM NaCl) plate and move the plate until surface covered entirely
10. Take pics with phone (wrapped in plastic wrap) to upload daily observation
11. Use breath able tape to tape out and seal edges of plates
12. Place them back in the growth chamber
    Done by Abigail

Date: 20260819
Title: Colony PCR (X, V, J)
Purpose: screen for successful cloning after transformation
Procedure:
X: ACDI (X1-X8)
V: ABDI (V1-V8)
J: Pveg-EEBD-RBS0-csn::ACCD-B0015 (J1-J8)

1. Prepare the following master mix (x18)
   X&V

Fw primer ccar	18μL
Rv primer csnR	18μL
Q-amp master mix	180μL
ddH_2O	144μL

   J:

Fw primer Csn	10μL
Rv primer ACCD	10μL
Q-amp master mix	100μL
ddH_2O	80μL

2. Distribute in PCR tubes (20μL each)

Done by Abby K, Abigail

Date: 20260819
Title: Gel electrophoresis (1% gel)
Purpose: to validate PCR product
Procedure:
1. Load the samples in the following order:

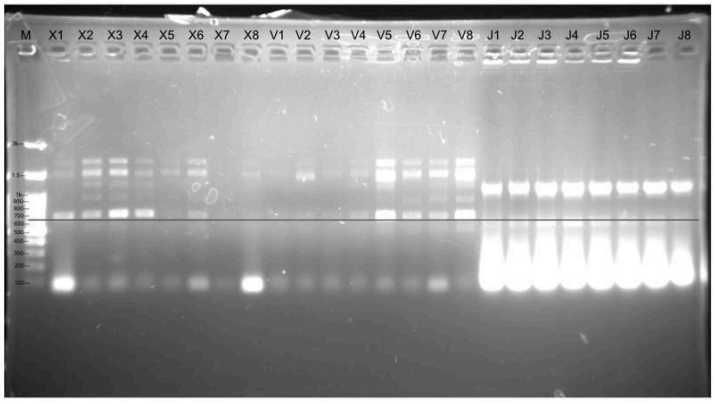
M	X1	X2	X3	X4	X5	X6	X7	X8	V1	V2	V3	V4	V5	V6	V7	V8	J1	J2	J3	J4	J5	J6
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5

J7	J8
5	5

2. Resolve the gel at 100V for 30min

Done by Sophie C

Results:



M: 3kb DNA ladder

X:ACDI
V:ABDI
ccar fw and csn rv
bp: 1424

J:
Pveg-EEBD-RBS0-Csn:ACCD-B0015
csn F ACCD R
bp: 1195

Date: 20260819
Title: Transformation (L)
Purpose: to validate PCR product
Procedure:
3P-EEBD-RBS0-Csn:GanA-Gala-B0015

1. Set up water bath to 42℃
2. Thaw competent cell till 50% thawed, add 10μL plasmid
3. Incubate on ice for 20 min
4. Heat shock 42℃ for 45sec
5. Incubate on ice for 3 min
6. Add 1mL SOC medium
7. Incubate for 1 hr at 37℃
8. Centrifuge at 5000g for 5 min
9. Take out ~1000μL supernatant
10. Resuspend the pellet
11. Spread on LBC plate
12. Incubate at 37℃ for 16hr

Done by Sophie C

Date: 20260819
Title: Prepare 420 mL 0.8% ½ MS agar
Purpose: add agar into lids for new set of seeds
Procedure:
The following protocol is for 420 mL ½ MS agar (0.8%)

1. ½ MS powder (w/o sucrose): 0.966 g
2. MES buffer: 0.21 g

3. Distilled water: 420 mL
4. Sucrose: 4.2 g
5. Agar: 3.36 g
6. Add ½ MS powder, MES buffer, sucrose, and distilled water into a flask
7. Shake the flask with hands until no sediments are in the solution
8. Use the pH probe to measure the pH of the solution, ensure the pH is between 5.7–5.8
9. If pH is too low, add 0.1 M KOH slowly to increase it
10. Add agar into the flask
11. Wrap the beaker with aluminium foil and stick a piece of indicator tape on top
12. Autoclave for 20 min at 127°C

Done by: Abby K

Date: 20260819

Title: Hay Infusion

Purpose: Preparing different ratios of nutrients additive

Procedure:

1. Take out half of the hay from the 新竹container
2. Use an electronic pipette controller with a 25mL serological pipette to draw all the liquid from the container and pour it into a 1000mL beaker
3. Measure the total amount of liquid
4. Take out 5 bottles and label them as follows:

1	10g coffee grounds
2	10g sugarcane
3	10g hay
4	10g hay 5g sugarcane 10g coffee grounds
5	10g hay 10g sugarcane 5g coffee grounds

5. Take out the scale and measure each as labeled on the bottle
6. Use the beaker and get 500ml of water
7. Pour the water into the 10g sugarcane bottle to see if the water covers the sugarcane
8. Remove the water from the bottle, make sure not to pour the sugarcane out
9. Pour the liquid in the 1000 mL beaker back into the 新竹 container
10. Add 10g of sugarcane and 10g of coffee grounds into the container and close the lid

Done by Sophia L, Sophie H, Olivia Du, Mia

Date: 20260819

Title: Add CH protectant in Box D2 (pre-treatment) for heat stress testing

Purpose: Add CH protectant in Box D2 (pre-treatment) for heat stress testing

Procedure:

1. Change the medium of boxes D1, D2, D3, and D4. Add 300mL of 0.1x MS medium when changing medium for each box
2. Take out the plants that aren't growing well or are too small
3. Evenly distribute the rest of arabidopsis into boxes D1, D2, D3, D4
4. Add 600µL of CH protectant into box D2

Done by Alex

Date: 20260819

Title: plant new seeds on agar x12 (on 0.8% agar)-> vernalize

Purpose: prepare new seeds

Procedure:

1. Open the BSC and turn on fan
2. Place the 0.8% agar plates on the 3D printed layout (seed template)
3. Take a clean bamboo stick (autoclaved) and dip in sterilized (autoclaved) water
4. Stick arabidopsis (col-o) seeds on the stick and tap lightly on the agar
5. Repeat until all plates are planted
6. Use breathable tape to seal each of the agar plates
7. Label the vernalization date, incubator date and seed type
8. Wrap the agar plates with aluminum foil
9. Vernalize in the 4℃ refrigerator for 4 days and avoid light

   Done by Abigail

Date: 20260819

Title: Add CH protectant in Box D2 (pre-treatment) for heat stress testing

Purpose: heat stress testing

Procedure:

1. Change the medium of boxes D1,D2,D3 and D4. Add 300mL of o.1X MS medium when changing medium for each box
2. Take out the plants that aren't growing well or are too small
3. Evenly distribute the rest of arabidopsis into boxes D1,D2,D#,D$
4. Add 600µL of CH protectant into box D2

   Done by Alex

Date: 20260820
Title: Miniprep
Purpose: to extract plasmid from liquid culture
Procedure:
A1: PtagC+RBS-Ho1:PcyA-L3S2P21
B2: Pveg-RBS0-CcaR-B0015
C6: PlepA-RBS0-CcaSm3-B0015
I3: PcpcG2-veg-EEBD-RBS0-Csn:ACCD-B0015

#7: L0-Bopep4-v2t2 (pSTK-0C)
#19: L0-Bopep4-v2t1 (pSTK-0C)

X2: ACDI
X3: ACDI
V2: ABDI
V8: ABDI

1. Pellet 1.8mL bacterial overnight culture by centrifuge 6800g for 3min
2. Discard the supernatant and repeat step 1 until collect all of the liquid culture
3. Add 250μL P1 buffer, and resuspend pelleted bacterial
4. Add 250μL P2 buffer, and mix thoroughly by inverting the tube 4-5times until the solution become clear(Do not allow the reaction more than 5 min)
5. Add 350μL N3 buffer and mix immediately and thoroughly by inverting 4-6 time
6. Centrifuge for 10 min at 17900g
7. Transfer 800μL supernatant into new column
8. Centrifuge 1 min for 17900g
9. Add 500μL PB buffer
10. Centrifuge 1 min for 17900g , and discard the flow-through
11. Add 750μL PE buffer
12. Centrifuge 1 min for 17900g , and discard the flow-through
13. Centrifuge empty column for 1min at 17900g , and discard the flow-through
14. Put into new column
15. Add 50μL EB buffer
16. Stand at RT for 1 min
17. Centrifuge for 1min at 17900g
18. Measure the concentration in Biodrop

Done by Sophie C., Abigail

Results

	Concentration (ng/μL)	A260/A280	A260/A230
A1	107	1.945	2.548

V8	212	1.876	2.000
B2	86	1.955	2.457
C6	134	1.887	1.675
X3	131	1.871	1.658
I3	108	1.862	1.385
#19	137	1.877	/
#7	101	1.836	1.063
X2	241	1.898	2.078
V2	217	1.937	2.192

Date: 20260820
Title: PCR of V, X
Purpose: to amplify specific segment of DNA
Procedure:

1. Add the following parts

V: ABDI

	1 reaction
Forward primer (EXP F)	1μL
CDS primer (pcyA R)	1μL
DNA (V)	1μL
Q-amp master mix	10μL
ddH_2O	7μL

Bp: 1712

	1 reaction
Forward primer (CcaR F)	1μL
CDS primer (PcpcG2 R)	1μL
DNA (V)	1μL
Q-amp master mix	10μL

ddH_2O	7μL

Bp: 1032

X: ACDI

	1 reaction
Forward primer (EXP F)	1μL
CDS primer (pcyA R)	1μL
DNA (X)	1μL
Q-amp master mix	10μL
ddH_2O	7μL

Bp: 1712

	1 reaction
Forward primer (CcaR F)	1μL
CDS primer (PcpcG2 R)	1μL
DNA (X)	1μL
Q-amp master mix	10μL
ddH_2O	7μL

Temperature: 65˚C

Done by Sophie C.

Date: 20260820
Title: Enzyme Digestion
Purpose: To confirm plasmid size for each samples before moving on to golden gate assembly
Procedure:
Samples

- #19: Bopep 4 v2t1 (pSTK-0C)
- #7: Bopep4 v2t1

1. Set up following reaction on ice
   Volume of DNA(μl)= $\frac{desired\ \ DNA\ mass\ (ng)}{DNA\ concentration\ (ng/\mu l\ )}$

*desired DNA mass = 500ng

A. Bopep4 v2t11
B. Bopep4 v2t2

Components	20 µL rxn
DNA - Concentration: 101 ng/µL	4.95µl
Cutsmart Buffer	2 µL
Restriction Enzyme	0.5µg
Nucleus free water	12.55 µL

Done by Sophie C., Abby K

Date: 20260820
Title: Gel electrophoresis
Purpose: To validate enzyme digestion and PCR product
Procedure:
From (pSTK-0C) enzyme digestion:

- #19 bopep4 v2t1
- #7 bopep4 v2t2

PCR:

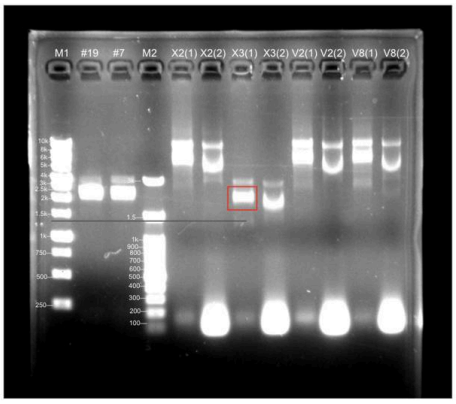
- #X2 ACDI
- #X3 ACDI
- #V2 ABDI
- #V8 ABDI

1. Load the samples in the following order
   M1 #19 #7 M2 #$X2_1$ #$X2_2$ #$X3_1$ #$X3_2$ #$V2_1$ #$V2_2$ #$V8_1$ #$V8_2$
   5 5 5 5 5 5 5 5 5 5 5 5
   M1: 1kB DNA ladder
   M2: 3kB DNA ladder

2. Resolve the gel at 100 V for 30 min

Done by Sophie C.

Results



M1: 1kb DNA ladder
M2: 3kb DNA ladder

#19: Bopep4 v2t1 (pSTK-0C) enzyme digestion
#7: Bopep4 v2t2 (pSTK-0C) enzyme digestion
bp: 183 bp

X:ACDI, V:ABDI
X2(1), X3(1), V2(1), V8(1)
EXP-seq-fw, pcyA R
bp: 1712

X2(2), X3(2), V2(2), V8(2)
CcaR F, PcpcG2-veg R
bp: 1032

Date: 20260820
Title: Prepare 420 mL 0.6% ½ MS agar
Purpose: Add agar into lids for new set of seeds
Procedure:
The following protocol is for 420 mL ½ MS agar (0.6%)

1. ½ MS powder (w/o sucrose): 0.966 g
2. MES buffer: 0.21 g
3. Distilled water: 420 mL
4. Sucrose: 4.2 g
5. Agar: 2.52 g
6. Add ½ MS powder, MES buffer, sucrose, and distilled water into a flask
7. Shake the flask with hands until no sediments are in the solution
8. Use the pH probe to measure the pH of the solution, ensure the pH is between 5.7–5.8
9. If pH is too low, add 0.1 M KOH slowly to increase it
10. Add agar into the flask
11. Wrap the beaker with aluminium foil and stick a piece of indicator tape on top
12. Autoclave for 20 min at 127°C

Done by: Abby K

Date: 20260820
Title: Level 1 Golden Gate Assembly of Bopep4
Purpose: To assemble level 0 parts to level 1
Procedure:

1. Prepare the following reactions

N: Pveg-EEBD-RBS0-SamyQ:Bopep4(v2t1)-B0015

• Vector (shuttle vector): pSTK-EXP-1-sfGFP	0.2780154321	1.4 ul (dilute 1/5)

• concentration: 162 ng/ µl • 6929 bp		
Insert promoter(Pveg) • concentration(ng/µL): 168 • bp: 2361	0.182 uL	1.82 uL (dilute 1/10)
Insert RBS(EEBD-RBS0) • concentration(ng/µL): 151 • bp: 2392	0.2059337748 µL	1.03 µL (Dilute 1/5)
Insert CDS(Samy Q:Bopep4(v2t1)) - concentration: 137 - 2263 bp	0.2147372263	1.07 (dilute 1/5)
Insert Terminator (B0015) • concentration(ng/µL): 107 • bp: 2213	0.268869 µL	1.34 µL (dilute 1/5)
T4 DNA ligase buffer	2 µL	2 µL
Sterile water		9.33 µL
T4 DNA ligase	1 µL	1 µL
BsaI-HFv2	1 µL	1 µL
Total	20 µL	20 µL

O: (STK) Module 4-PcpcG2-veg - EEBD- RBS0 - SamyQ:Bopep4(v2t2) - B0015

Vector (1D) • concentration(ng/µL): 132 • bp: 3120	0.15 µL	1.5 µL (dilute 1/1]\\ ]0)
Insert promoter(PcpcG2-veg) • concentration(ng/µL): 168 • bp: 235	0.01818452381 µL	0.90922619 µl (dilute 1/50)

Insert RBS(EEBD-RBS0) ● concentration(ng/μL): 151 ● bp: 2392	0.2059337748 μL	1.03 μL (Dilute 1/5)
Insert CDS(SamyQ:Bopep4 v2t2) ng/uL - concentration: 101 - 2263 bp	0.2912772277	1.17 μL (Dilute 1/4)
Insert Terminator (B0015) ● concentration(ng/μL): 107 ● bp: 2213	0.268869 μL	1.34 μL (dilute 1/5)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water		10.05 μL
T4 DNA ligase	1 μL	1 μL
BsaI-HFv2	1 μL	1 μL
Total	20 μL	20 μL

Done by Sophie C, Abby K

Date: 20260820

Title: hydroponic heat stress testing

Purpose: To test hydroponics plant growth when put under heat stress

Procedure:

1. Prepare 4 new boxes, add 300mL of 0.1X MS medium into each box. Label them as D!,D2,D3,D4
2. Add 600μL of CH protectant into D3(co-treatment)
3. Take another box and add 300mL water into it
4. Put a thermometer into water
5. Place all boxes into the heat box(setting 50℃, actual temp 35℃)
6. Wait until thermometer reaches 35℃
7. Take out new D1 box from heat box
8. Transfer the floating board from old D1 box to the new D1 box
9. Put new D1 box back into the hat box
10. Repeat steps 8~10 until all boxes changed medium
    Done by Alex

Date: 20260820

Title: Hay infusion

Purpose: observe the growth of *Bacillus subtilis* in 新竹 hay infusion and LB

Procedure:

1. Take the 0815 新竹 1000mL hay infusion
2. Take two centrifuge tubes and label them
3. Disinfect all equipment and move it into the BSC for aseptic operation
4. Use and electronic pipetted controller with a serological pipette to draw 15mL of hay infusion
5. Transfer the 15mL into a syringe fitted with a sterile filter
6. Filter the hay infusion and collect it into the 15mL tube
7. Use a new serological pipette to transfer 5mL of the filtered water into a centrifuge tube (labeled with “hay infusion”)
8. Use a pipette to add 200μL of *Bacillus subtilis* into the hay infusion centrifuge tube
9. Use the electronic pipette controller with a new serological pipette and add 5mL of LB into the LB centrifuge tube
10. Use a pipette to add 200μL of *Bacillus subtilis* into the LB centrifuge tube
11. Place the centrifuge tubes in the centrifuge overnight
12. Take out the centrifuge tube after a wight
13. Use the pipette to transfer 200μL of each filtered hay infusion and LB(with no *Bacillus subtilis*) into separate wells of a 96-well plate)
14. Transfer 200μL of hay infusion and LB (with *Bacillus subtilis*) into separate wells of a 96-wells plate
15. Place the 96-well plate into the OD microplate reader and record the result
16. Put the centrifuge back to the centrifuge
17. Repeat step 14 to step 16 every hour

Result

Hour	OD of LB	OD of Hay Infusion
18	0.489	0.091
19	0.486	0.095
20	0.486	0.097
21	0.503	0.107
22	0.499	0.124

Done by Sophia L, Olivia Du

Date: 20260821
Title: Miniprep
Purpose: to extract plasmid from liquid culture
Procedure:
L1, L2: pSTK-EXP1-P3P-EEBD-RBS0-Csn:GanA-GalA-L3S2P21 / 0820 TBA
#3: pSTK-0C-SamyQ:LEA v2t2 (D-C) / 0817 TBC
#1: pSTK-0C-SamyQ:LEA v2t1 (Q-A) / 0817 TBC
#2: pSTK-0C-SamyQ:ACCD v2t1 (Q-D) / 0820 TBC
#S16: pSTK-0C-csn:LEA / 0814 TBC

1. Pellet 1.8mL bacterial overnight culture by centrifuge 6800g for 3min
2. Discard the supernatant and repeat step 1 until collect all of the liquid culture
3. Add 250μL P1 buffer, and resuspend pelleted bacterial
4. Add 250μL P2 buffer, and mix thoroughly by inverting the tube 4-5times until the solution become clear(Do not allow the reaction more than 5 min)
5. Add 350μL N3 buffer and mix immediately and thoroughly by inverting 4-6 time
6. Centrifuge for 10 min at 17900g
7. Transfer 800μL supernatant into new column
8. Centrifuge 1 min for 17900g
9. Add 500μL PB buffer
10. Centrifuge 1 min for 17900g , and discard the flow-through
11. Add 750μL PE buffer
12. Centrifuge 1 min for 17900g , and discard the flow-through
13. Centrifuge empty column for 1min at 17900g , and discard the flow-through
14. Put into new column
15. Add 50μL EB buffer
16. Stand at RT for 1 min
17. Centrifuge for 1min at 17900g

Done by Sarah, Ryan W.
Supervised by Sophie C.

Results:

	Concentration	A260/A280	A260/A230
L1	34	2.000	1.789
L2	225	1.907	1.433
#3	74	1.947	2.000
#1	105	1.810	0.991
#2	103	1.839	1.084

#S16	90	1.765	0.909

Date: 20260821
Title: Transformation
Purpose: Transform and cultivate engineered bacteria on agar plate
Procedure:
Sample: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-Bopep4-v2t2-B0015

1. Set the water bath to 42°C and prepare ice
2. Thaw competent cells until they're 50% thawed and add 10μL of plasmid
3. Incubate on ice for 20min
4. Heat shock at 42°C for 45 seconds
5. Incubate on ice for 3 min
6. Add 1mL SOC medium
7. Incubate for 1 hr at 37°C
8. Centrifuge at 5000g for 5 min
9. Remove 1 mL of supernatant then vertex
10. Spread on their respective plates
11. Incubate at 37°C for 16hr

Done by Sophie C

Date: 20260821
Title: PCR
Purpose: to amplify DNA parts
Procedure:
L2, L1: pSTK-EXP1-P3P-EEBD-RBS0-Csn:GanA-GalA-L3S2P21

1. Prepare the following reaction:

	1rxn	3rxn
Forward Primer Exp1	1μL	3μL
Reverse Primer Csn	1μL	3μL
Qamp master mix	10μL	30μL
ddH_2O	7μL	21μL
DNA	1μL	1μL

2. Put it in PCR machine for thermal cycle:

95℃	5min	x1
95℃	20sec	x35
65℃	20sec	
72℃	1min 20sec	
25℃	5mmin	x1

Done by Sarah, Ryan W.

Date: 20260821
Title: Prepare 1.0M KOH
Purpose: Prepare KOH for balancing pH levels
Procedure:

1. Prepare the following in a centrifuge tube:
    a. 2.24g KOH
    b. 40mL ddH_2O
2. Add the KOH first. Make sure that no KOH remains on the weighing paper/boat
    a. KOH is hygroscopic meaning that it absorbs some water in air so it sticks to the surface
3. Add the ddH_2O slowly in an ice bucket (or ice water)

Done by Sophia Y.
Supervised by Chars

Date: 20260821
Title: Gel electrophoresis
Purpose: To validate PCR products
Procedure:
L2, L1: pSTK-EXP1-P3P-EEBD-RBS0-Csn:GanA-GalA-L3S2P21 (PCR)
L20, L10: pSTK-EXP1-P3P-EEBD-RBS0-Csn:GanA-GalA-L3S2P21
C:Water(1μL)+Qamp master mix (10μL)+primer (2μL) +ddH_2O (7μL)

1. Load the samples in the following order
   M L1 L2 C M L10 L20
   5 5 5 5 5 5 5
2. Resolve the gel at 100V for 30min

Done by Ryan W.

Results:

Date: 20260821
Title: Golden Gate Assembly (P, Q, R, S)
Purpose: assemble level 0 parts to level 1
Procedure:
P: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-ACCD-v2t1-B0015
Q: pSTK-1D-PcpcG2-veg-EEBD-RBS0-SamyQ-ACCD-v2t1-B0015
R: pSTK-EXP1-Pveg-EEBD-RBS0-Csn:LEA-B0015
S: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ:LEA-v2t1-B0015

1. Assemble the following

P: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-ACCD-v2t1-B0015

• Vector (shuttle vector): pSTK-EXP-1-sfGFP • concentration: 162 ng/ µl • 6929 bp	0.2780154321	1.4 µl (dilute 1/5)
Insert promoter(Pveg) • concentration(ng/µL): 168 • bp: 2361	0.182 µL	1.82 µL (dilute 1/10)
Insert RBS(EEBD-RBS0) • concentration(ng/µL): 151 • bp: 2392	0.2059337748 µL	1.03 µL (Dilute 1/5)
Insert CDS (SamyQ-ACCD-v2t1) - concentration: 103 - 3206 bp	0.4046407767	1.21 µl (1/3)
Insert Terminator (B0015) • concentration(ng/µL): 107 • bp: 2213	0.268869 µL	1.34 µL (dilute 1/5)
T4 DNA ligase buffer	2 µL	2 µL
Sterile water		9.2 µL
T4 DNA ligase	1 µL	1 µL
BsaI-HFv2	1 µL	1 µL
Total	20 µL	20 µL

Q: pSTK-1D-PcpcG2-veg-EEBD-RBS0-SamyQ-ACCD-v2t1-B0015

Vector (1D) ● concentration(ng/μL): 132 ● bp: 3120	0.15 μL	1.5 μL (dilute 1/10)
Insert promoter(PcpcG2-veg) ● concentration(ng/μL): 168 ● bp: 235	0.01818452381 μL	0.90922619 μl (dilute 1/50)
Insert RBS(EEBD-RBS0) ● concentration(ng/μL): 151 ● bp: 2392	0.2059337748 μL	1.03 μL (Dilute 1/5)
Insert CDS (SamyQ-ACCD-v2t1) ng/μL - concentration: 103 - 3206 bp		1.21 μl (1/3)
Insert Terminator (B0015) ● concentration(ng/μL): 107 ● bp: 2213	0.268869 μL	1.34 μL (dilute 1/5)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water		10.01 μl
T4 DNA ligase	1 μL	1 μL
BsaI-HFv2	1 μL	1 μL
Total	20 μL	20 μL

R: pSTK-EXP1-Pveg-EEBD-RBS0-Csn:LEA-B0015

● Vector (shuttle vector): pSTK-EXP-1-sfGFP ● concentration: 162 ng/ μl	0.2780154321	1.4 μl (dilute 1/5)

• 6929 bp		
Insert promoter(Pveg) • concentration(ng/μL): 168 • bp: 2361	0.182 μL	1.82 μL (dilute 1/10)
Insert RBS(EEBD-RBS0) • concentration(ng/μL): 151 • bp: 2392	0.2059337748 μL	1.03 μL (Dilute 1/5)
Insert CDS (Csn:LEA) - concentration: 105 ng/μl - 2660 bp	0.33 μl	0.99 μl (1/3)
Insert Terminator (B0015) • concentration(ng/μL): 107 • bp: 2213	0.268869 μL	1.34 μL (dilute 1/5)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water		9.42 μL
T4 DNA ligase	1 μL	1 μL
BsaI-HFv2	1 μL	1 μL
Total	20 μL	20 μL

S: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ:LEA-v2t1-B0015

• Vector (shuttle vector): pSTK-EXP-1-sfGFP • concentration: 162 ng/ μl • 6929 bp	0.2780154321	1.4 μL (dilute 1/5)
Insert promoter(Pveg) • concentration(ng/μL): 168 • bp: 2361	0.182 μL	1.82 μL (dilute 1/10)

Insert RBS(EEBD-RBS0) • concentration(ng/μL): 151 • bp: 2392	0.2059337748 μL	1.03 μL (Dilute 1/5)
Insert CDS (SamyQ:LEA-v2t1) - concentration: 90 ng/μL - 2645 bp	0.3820555556 μL	1.15 (1/3)
Insert Terminator (B0015) • concentration(ng/μL): 107 • bp: 2213	0.268869 μL	1.34 μL (dilute 1/5)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water		9.26 μL
T4 DNA ligase	1 μL	1 μL
BsaI-HFv2	1 μL	1 μL
Total	20 μL	20 μL

Thermal Cycles:

Level 1	Level 2	Time	Cycles
37℃	42˚C	5min	45 cycles
16℃	16˚C	5min	
37℃	55˚C	10min	1 cycle
80℃	80˚C	20min	1 cycle
RT			

Done by Chars

Date: 20260821
Title: Prepare 0.1 MS medium
Purpose: prepare medium for hydroponics testing

Procedure:

1. Prepare the following

MS powder without sucrose	0.69g
MES powder	1.02g
ddH_2O	1500mL

2. Place the materials in a 2000mL beaker
3. Use a stirring rod to stir until no sediments remain
4. Calibrate the pH probe
5. Record the initial pH
6. Add 1.0M KOH in small increments until the pH reaches 5.70~5.80
    - Initial pH:4.29, final pH:5.70

Done by Sophia Y.
Supervised by Chars

Date: 20260821
Title: Hay Infusion
Purpose: Prepare the medium for *Bacillus subtilis* to grow
Procedure:

1. Cut the hay into smaller pieces
2. Place the chopped hay into a flask until it reaches the 150 mL mark.
3. Place a container on the scale, and set it to zero
4. Transfer the hay from the flask to measure its weight (6.69g).
5. Return the weighed hay to the flask.
6. Measure 250 mL of distilled water using a beaker and pour it into the flask.
7. Cover the flask with aluminum foil and autoclave for 20 minutes at 127°C.
8.
9. Blend grapes with water in a blender, then pour the mixture into a cup.
10. Filter the mixture through gauze into another flask to collect the extract.
11. Repeat step 8 to step 9 using blueberries to obtain blueberry extract.

Done by Sophia L

Date: 20260821
Title: Remove D3's protectant and add protectant in D4
Purpose: testing protectant
Procedure:

1. Prepare 2 new boxes and add 300mL of 0.1X MS medium into both boxes label them as D3, and D4
2. Add 600µL CH protectant into box D4 (post-treatment)

3. Take another box and add 300mL of water into it
4. Put a thermometer in the water
5. Put both boxes into heat box(setting 50℃ actual temp 35℃**)**
6. Wait until thermometer reaches 35℃
7. Take out new D3 box from heat box
8. Transfer the floating board from old D3 box to new D3 box
9. Put new D3 box back into heat box
10. Repeat steps 7~9 to change the medium of D4
    Done by Alex

Date: 20260821

Title: Measure fresh weight of heat stress trial (agar)

Purpose: measuring heat stress pretest freshweight results

Procedure:

1. Use a tweezer, lift and pick up all arabidopsis plants to simultaneously form the agar plate in a single motion
2. Put plants on a tissue paper, lightly press to remove the agar sticking to the plants. (do not press the leaf area too hard)
3. Place all the arabidopsis in a weighting plate, weight the fresh weight (a plate at the time-control , pre-treat, co-treat, post-treat)
4. Pick all the plants up and put them inside a 15mL centrifuge tube, make sure all the plant and the roots goes inside the microtube
5. Label the fresh weight nad the group concentration on the microtube
   *problem observed:
   Ants climbing inside the incubator resulting in
   -> ants climbing into the heat stress pretest agar plates
   *possible solution:
   Tape should be stuck onto the plate tightly till there's no more gap between the tape and flu plate
   Done by Abigail
   Recorded by Audrey H

Date: 20260821

Title: planting seeds on 0.6% agar plates

Purpose: plant Arabidopsis on agar plates

Procedure:

1. Dip an autoclaved tooth pick in deionized water(autoclaved)
2. Reach into the eppendorf with seeds and pick up one at time
3. Plant 40 seeds on each plate according to the grid
4. Repeat with 13 plates
5. Wrap with aluminium foil and place in the refrigerator
   Done by Abby K, Abigail, Sophie C

Date: 20260822
Title: Colony PCR
Purpose: to screen for successful cloning after transformation
Procedure:
N: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-Bopep4-v2t1-B0015
X: ACDI

1. Prepare the following reactions
2. Distribute into PCR tubes
3. Dispense pipette into colonies, streak gild
4. Dispense in PCR tube
5. Close lid & spin down
6. Place in PCR machine for thermal cycle

95℃	5min	x1
95℃	20sec	x35
60℃	20sec	
72℃	1min 30sec	
25℃	5mmin	x1

Done by Sophie C, Abby K

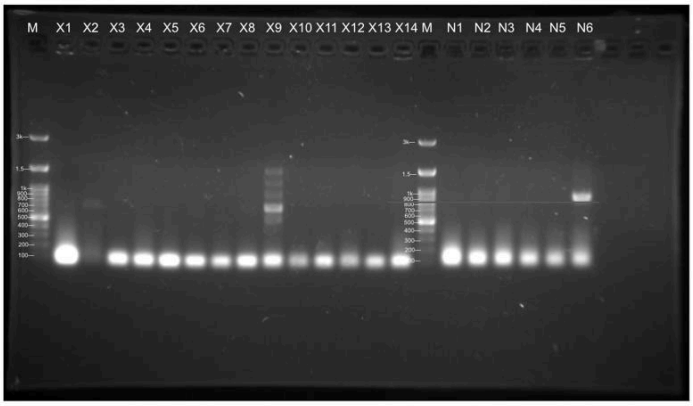
Date: 20260822
Title: Gel electrophoresis
Purpose: check PCR result
Procedure:
N: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-Bopep4-v2t1-B0015
X: ACDI

1. Load the sample in the following sample

M	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	X11	X12	X13	X14	M	N1	N2	N3	N4	N5	N6
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5

2. Resolve the gel at 100 V for 30 min

8/22 Colony PCR



M: 3kb DNA ladder

N: pSTK-EXP1-Pveg-EEBD-RBS0-Bopep4-v2t1-B0015
pSTK-EXP-seq fw, Bopep4 R
bp: 678

X: ACDI
CcaR F, Csn R
Bp: 1424

Done by Sophie C

Date: 20260822

Title: Measure chlorophyll of heat stress trial (agar)-chlorophyll extraction

Purpose: measure the chlorophyll for heat stress pretest

Procedure:

1. Take the microtubes from 0821 from measured freshweight at 0821
2. Add 4mL to each tube
3. Add 10 metal bits (403) into each microtubes
4. Wait for the chlorophyll to extract and use a vortex to crush the plants to spread up the process(vortex till the plants are no longer green)
5. Add the liquid to the 96 well, 200µL in each well. 3 well for each sample
6. Make sure to select the correct wavelength
    - 663
    - 645

	2	3	4	5	6
F	Control 1	pre–treatment	co-treat	after-treat	80% acetone
G	Control 1	pre-treatment	co-treat	after-treat	80% acetone
H	Control 1	pre-treatment	co-treat	after-treat	80% acetone

Done by Audrey H
Supervised by Gabreil

Date:20260822

Title: Transformation
Purpose: Transform and cultivate engineered bacteria on agar plate
Procedure:
X1:ACDI(1:1) / X2:ACDI(1:2) / X3:ACDI(1:3)
V1: ABDI(1:1)/ V2: ABDI(1:2)/ V3: ABDI(1:3)
R:pSTK-EXP1-Pvef-EEBD-RBS0-csn:LEA-B0015
S:pSTK-EXP1-Pveg-EEBD-RBS0-Samy Q-LEA-v2t1-B0015

1. Set up water bath at 42℃
2. Thaw competent cell fill 50% thawed, add 5μL plasmid to 100μL competent cell
3. Incubate on ice for 20min
4. Heat shack at 42℃ for 45 sec
5. Incubate on ice for 3min
6. Add 1mL SOC medium
7. Incubate at 37℃ for 1 hr
8. Centrifuge for 5 min at 5000g
9. Remove ~1mL supernatant
10. Resuspend pellet
11. Spread on respective plates (LBC,LBA)
12. Incubate at 37℃ for 16hr

Done by Sophie C, Abby K, Chloe

Date: 20260823

Title: Make 1% MS medium without vitamin and 1% agar

Purpose: prepare MS medium for experiment

Procedure:

1. Measure the following

   ½ MS medium without vitamin

MS powder without vitamin	3.525g
MES buffer	0.75g
Distilled water	1500mL

   1% agar

½ MS powder	1.15g
MES buffers	0.25g
sucrose	5g
agar	5g
Distilled H_2O	500mL

2. Add them into a flask and add a stirring bar
3. Stir on the hot plate until no sediment
4. Use pH probe to measure the pH. If too low, add 1M KOH until it reactes 5.7~5.8
5. Wrap both bottle, in aluminium foil and stick a piece of indicator tap on top

   Done by Sara, Alex

Date: 20260823

Title: Colony PCR

Purpose: Amplify DNA plasmid sequence

Procedure:

Q: pSTK-1D-PcpcG2-veg-EEBD-RBS0-SamyQ-AccD-V2t1-B0015

P:pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-AccD-V2t1-B0015

R:pSTK-EXP1-Pveg-EEBD-RBS0-can:LEA-B0015

S:pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-LEA-V2t1-B0015

1. Prepare the following reaction

   Qx8

PcpcG2-veg fw	9μL
AccD rv	9μL
Q-amp master mix	90μL
ddH_2O	72μL

Px8 Rx8 Sx8

pSTK-EXP-seq fw	27μL
pSTK-EXP-seq rv	27μL
Q-amp master mix	270μL
ddH_2O	216μL

2. Distribute into PCR tube
3. Dip pipette into colonies, streak grid
4. Dispense into PCR tube & wait for 3min
5. Close the lid & spindown
6. Place in machine for thermal cycle

95℃	5min	x1
95℃	20sec	x35
60℃	20sec	
72℃	2min	
72℃	5min	
25℃	0	x1

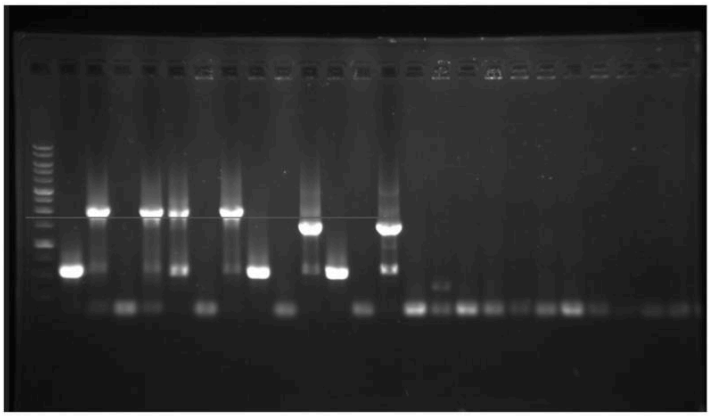
Done by Chloe

Date: 20260823
Title: gel electrophoresis(1% gel)
Purpose: check PCR result
Procedure:
Q: pSTK-1D-PcpcG2-veg-EEBD-RBS0-SamyQ-AccD-V2t1-B0015
P:pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-AccD-V2t1-B0015
R:pSTK-EXP1-Pveg-EEBD-RBS0-can:LEA-B0015
S:pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-LEA-V2t1-B0015

1. Load the samples in the following order
   Gel1:
   M P1 P2 P3 P4 P5 P6 P7 P8 R1 R2 R3 R4 R5 R6 R7 R8 S1 S2 S3 S4 S5
   5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5
   S6 S7 S8
   5 5 5
   Gel2:
   M Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8
   5 5 5 5 5 5 5 5 5
2. Resolve the gels at 100V for 30min

Done by Chloe, Abby K

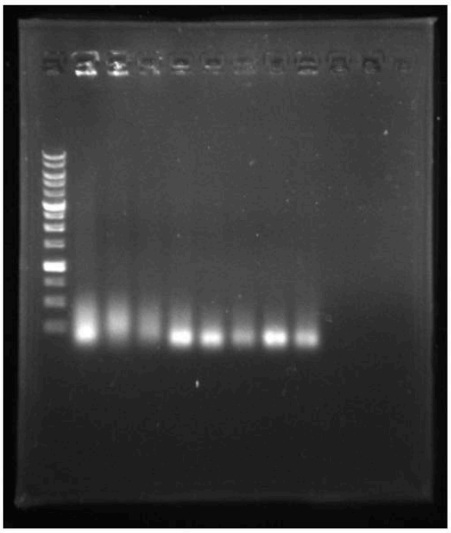
8/23 Colony PCR PRS



M: 3kb DNA ladder

P: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-ACCD-v2t1-B0015
bp:1984
R: pSTK-EXP1-Pveg-EEBD-RBS0-csn:LEA-B0015
bp:1438
S: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-LEA-v2t1-B0015
bp:1423
pSTK-EXP-seq fw, pSTK-EXP-seq

8/23 Colony PCR Q



M: 1kb DNA ladder

Q: pSTK-1D-PcpcG2-veg-EEBD-RBS0-SamyQ-ACCD-v2t1-B0015
PcpcG2-veg fw, ACCD R
bp: 1673

Date: 20260824
Title: miniprep
Purpose: isolate plasmid of level 2 liquid culture
Procedure:

1. Pellet 1.8mL of liquid culture by centrifuging at 6800g for 3min
2. Repeat until collected all medium
3. Add 250µL of P1 buffer and resuspend the pellet
4. Add 250µL of P2 buffer and mix by inverting the tube 4-6 times until the solution becomes clear-> do not allow lysis to proceed for more than 5 min
5. Add 350µL of N3 buffer and mix immediately by inverting the tube 4-6 times
6. Centrifuge for 10 min at 17900g
7. Transfer ~800µL of supernatant to column and collection tube
8. Centrifuge for 1 min at 17900g then discard the flow-through
9. Add 500µL PB buffer
10. Centrifuge for 1 min at 17900g then discard the flow-through
11. Add 750PE buffer
12. Centrifuge for 1 min at 17900g then discard flow-through
13. Centrifuge for 1 min at 17900g then discard flow-through
14. Place column on new 1.5mL microtube
15. Add 50µL EB buffer to the center of the column
16. Let stand at RT for 1 min
17. Centrifuge for 1 min at 17900g
18. biodrop

	concentration(hg/µL)	A260/280	A260/230
ACDI from 0820	70.00	1.917	1.725
ACDI from 0822	63.00	2.032	1.800

Done by Abby K

Date: 20260825
Title: Colony PCR
Purpose: to screen for successful cloning after transformation
Procedure:
X:ACDI

1. Prepare the following reaction
   11rxn

RV primer PcpcG2-veg	11μL
Fw primer CcaR-fw	11μL
Q-ap master mix	110μL
ddH_2O	88μL

2. Distribute the master mix to 10 PCR tubes
3. Pick up circled colonies and streak the grid
4. Dispense tips to their tubes for 5 minutes
5. After 5 minutes, remove the tips and spin down
6. Place the tubes in the PCR machine with the following thermal cycle setting

95℃	5min	x1
95℃	20sec	x35
60℃	20sec	
72℃	1min45sec	
72℃	5min	
25℃	0	x1

   Done by Naomi, Sarah

Date: 20260825
Title: Colony PCR
Purpose: to screen for successful cloning after transformation
Procedure:
Q: pSTK-1D-PcpcG2-veg-EEBD-RBS0-SamyQ-AccD-V2t1-B0015
S:pSTK-Exp1-Pveg-EEBD-RBS0-SamyQ-LEA-V2t1-B0015

1. Prepare the following reactions
   Qx13rxn

RV primer AccD R	13μL
Fw primer PcpcG2-veg F	13μL
Q-ap master mix	130μL

ddH_2O	104μL

2. Distribute master mixes into respective tubes
3. Pick up colonies and streak the grid
4. Dispense the tips into the corresponding tubes
5. Wait for 5 minutes before removing tips and spinning down
6. Place tubes in the PCR machine with the following thermal cycle

95℃	5min	x1
95℃	20sec	x35
60℃	20sec	
72℃	1min45sec	
72℃	5min	
25℃	0	x1

Done by Sophie C, Abby K

Date: 20260825
Title: LBK, LBC+glycerol, TBC, LBA+glycerol, LB, LBE+glycerol
Purpose: prepare plate
Procedure:

1. Formula as below:

LB

LB broth powder	6.25g
Agar powder	3.75g
ddH_2O	250mL

TB

TB broth powder	6.25g
Agar powder	3.75g
ddH_2O	250mL

2. Add one magnetic stirring bar to the flask
3. Place the flask containing the mixture on the hot plate, and stir until no sediments
4. Remove the stirring bar from the flask
5. Cover the flask with aluminum foil and apply on indicator tape to ensure the mixture is successfully autoclave
6. Place the flask in autoclave
7. Place the new petri dishes in BSC and turn on the UVC

8. Set up water bath at 55℃
9. Add 500μL ampicillin/ 25μL kanamycin / 1250μL erythromycin / 250μL chloramphenicol
10. Use serological pipette to add 20mL of melting agar into the petri dish
11. Let the agar plates dry for at least 1 hr in the BSC
    Done by Naomi, Sarah

Date: 20260825
Title: gel electrophoresis
Purpose: checking PCR result
Procedure:
X:ACDI
Q: pSTK-1D-PcpcG2-veg-EEBD-RBS0-SamyQ-AccD-V2t1-B0015
S:pSTK-Exp1-Pveg-EEBD-RBS0-SamyQ-LEA-V2t1-B0015

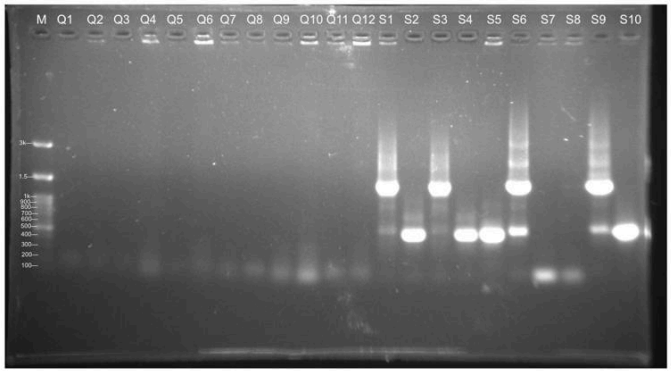
1. Load the gels in the following order

H	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	S9	S10	S11	S12	S1	S2	S3
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5

S4	S5	S6
5	5	5

2. Resolve the gel at 100V for 30min
   Done by SophieC, Sarah

8/25 Colony PCR (Q/S)



M: 3kb DNA ladder
Q:
pSTK-1D-PcpcG2-veg-EEBD-RBS0-SamyQ-ACCD-v2t1-B0015
PcpcG2-veg fw, ACCD R
bp: 1673
S:
pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-LEA-v2t1-B0015
bp:1423
pSTK-EXP-seq fw, pSTK-EXP-seq

Date: 20260825
Title: make acrymide gel
Purpose:prepare gel
Procedure:

1. Assemble casting fromes with two 1.5mm glass plates

2. Prepare solutions for the stacking (top) and resolving (bottom) gels

component	stacker	resolver
Resolver A	x	4.5mL
Resolver B	x	4.5mLx
Stacker A	1.5mL	x
Stacker B	1.5ml	x
total	3ml	9mL
temed	3μL	4.5μL
10% APS	15μL	45μL

3. Transfer resolving gel between glass plates
4. Add stacking gel to the top
5. Insert comb to form wells and allow it to set
   Done by Sophie C

Date: 20260825
Title: golden gate assembly of N,O
Purpose:assemble part
Procedure:
N:Pveg-EEBD-RBS0-SamyQ:Bopep4(V2t1)-B0015

1. Prepare the following reactions

pSTK-Exp-1-sfGFP	1.4μL(dill ⅕)
Pveg (168ng/μL , 2361bp)	1.82μL(dill 1/10)
EEBD-RBS0 (151ng/μL, 2992bp)	1.03μL(dill1/5)
B0015 (107ng/μL, 2213bp)	1.34μL(dill(⅕)
SamyQ-Bopep4 (v2t1) (137ng/μL, 2263bp)	1.07μL(dill1/5)
T4 DNA liq buffer	2μL
Bsa-HFv2	1μL

T4 DNA lig	1μL
ddH_2O	9.33μL

O:Pveg-EEBD-RBS0-SamyQ-Bopep4(v2t2)-B0015

pSTK-Exp-1-sfGFP	1.4μL(dill ⅕)
Pveg	1.82μL(dill 1/10)
EEBD-RBS0	1.03μL(dill1/5)
B0015	1.34μL(dill(⅕)
SamyQ-Bopep4 (v2t1) (101ng/μL, 2263bp)	1.17μL(dill1/5)
T4 DNA liq buffer	2μL
Bsa-HFv2	1μL
T4 DNA lig	1μL
ddH_2O	9.24μL

2. Put in PCR machine & apply the thermal cycle

37.0℃	5min	x45
16.0℃	5min	
37.0℃	10min	x1
80.0℃	20min	

Done by Chloe, Sophie C

Date: 20260825
Title: Transformation
Purpose:Spread bacterial liquid on agar plate
Procedure:
B:Pliag-RBS0-CcaSm3-B0015

1. Set up water bath to 42℃
2. Thaw competent cell until 50% thawed & add 2μL plasmid
3. Incubate on ice for 20min
4. Heat shock at 42℃ for 45sec
5. Recover on ice for 3 min
6. Spread on LBA plate in BSC

7. Incubate at 37℃ for 16h(overnight)
   Done by Chloe

Date: 20260825
Title: liquid culture
Purpose: Transfer colonies on agar to medium
Procedure:
A:pSTK-1A-PtagC+RBS-H01:pcyA-L3S2P21(#A1)
C:PcepA-RBS0-CcaSm3-B0015(#C6)
D:pSTK-1C-Pveg-RBS0-CcaR-B0015(#B2)
I:pSTK-1D-PcpcG2-veg-EEBD-RBS0-csn:AccD-B0015(#I3)
S:pSTK-Exp1-Pveg-EEBD-RBS0-SamyQ-LEA-V2t1-B0015(#S5, #S11)
P:pSTK-EXp1-Pveg-EEBD-RBS0-SamyQ-AccD-v2t1-B0015(#P2,#P7)
R: pSTK-EXp1-Pveg-EEBD-RBS0-csn:LEA-B0015 (#R2,#R5)
3mL TB + 3μL ampicillin per tube
Done by Chloe

Date: 20260826
Title: Miniprep (A, C, D, I, P, R)
Purpose: Collect concentrated bacterial liquid
Procedure:
A1: PtagC+RBS-Ho1:PcyA-L3S2P21
C6: PlepA-RBS0-CcaR-B0015
B2: Pveg-RBS0-CcaR-B0015
R2, R5: Pveg-EEBD-RBS0-Csn::EA:B0015
P2, P7: Pveg-EEBD-RBS0-SamyQ-ACCD
I3: PcpcG2-veg-EEBD-RBS0-Csn:ACCD-B0015

1. Pellet 1.8mL of liquid culture by centrifuging at 6800g for 1min
2. Repeat until all the medium were collected
3. Add 250μL of P1 buffer, resuspend the cell pellet by vortexting or pipetting
4. Add 350μL of P2 buffer, and mix thoroughly by inverting the tube 4~6 times
5. Centrifuge for 10min at 17900g
6. Transfer the supernatant to the spin column, centrifuge it for 60sec at 17,900g, then discard the flow through
7. Add 500μL PB buffer, and centrifuge for 1min at 17,900g
8. Discard flow through
9. Add 750μL PE buffer, and centrifuge for 1min at 17,900g
10. Discard flow through
11. Centrifuge empty column for 1min at 17,900g to remove residual wash buffer
12. Place the column to 1.5mL microcentrifuge tube
13. To elute DNA, add 50μL EB buffer to the center of the membrane
14. Let the tube stand for 1min, then centrifuge it for 1min at 17,900g

Biodrop Results:

Concentration	**A260/230**	**A260/280**
94ng/μL	1.918	1.918
86ng/μL	1.87	1.911
85ng/μL	1.889	1.932
253ng/μL	2.018	1.888
243ng/μL	2.229	1.898
220ng/μL	2.178	1.897
248ng/μL	2.234	1.893
32ng/μL	1.524	1.882

Done by Abby K, Abby T

Date: 20260726
Title: Colony PCR of B, Q
Purpose:Amplify the number of engineered circuit design
Procedure:
B: PliaG-RBS0-Ccasm3-B0015
Q: PcpcG2-veg-EEBD-RBS0-SamyQ:ACCD-v2t1-B0015 (bp: 2041)

1. Prepare the following reactions:

Q(18rxn)

	1 rxn	**18rxn**
pSTK-Seq fw	1μL	18μL
pSTK-Seq rv	1μL	18μL
Qamp master mix	10μL	180μL
ddH_2O	8μL	144μL

B(5rxn)

	1 rxn	**5rxn**
pSTK-Seq fw	1μL	5μL
pSTK-Seq rv	1μL	5μL
Qamp master mix	10μL	50μL
ddH_2O	8μL	40μL

2. Distribute the master mix into labeled PCR tubes (20μL each well)
3. Pick up colonies with a tip and streak on the grid
4. Dip the pipette tip into the respective master mix
5. Incubate at Room Temperature for 5min, then discard the pipette tips
6. Put the PCR tubes into PCR machine and run the thermal cycle as below:

95.0℃	5min	x1
95.0℃	20sec	
60.0℃	20sec	x35
72.0℃	5min	

72℃	5min	
25℃	0min	x1

Done by Abby T

Date:20260826
Title: Gel electrophoresis of Colony PCR (1% gel)
Purpose: Check the approximate length of PCR product
Procedure:

B: Pliag-RBS0-ccsAm3-B0015
Q: PcpcG2-veg-EEBD-RBS0-SamyQ-ACCD-v2t1-B0015

1. Load the gel in the following order:

M	B1	B2	B3	B4	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5

Q12	Q13	Q14	Q15	Q16
5	5	5	5	5

Result:



M: 1kb DNA ladder
Q:
pSTK-1D-PcpcG2-veg-EEBD-RBS0-Sam
yQ-ACCD-v2t1-B0015
pSTK_Seq fw, pSTK_Seq rv
bp: 2041
B: pSTK-1A-PliaG-RBS0-Ccasm3-B0015
pSTK_Seq fw, pSTK_Seq rv
bp:1991

→ liquid culture B, 8/27 Miniprep

Date:20260826
Title: Cotreatment + NaCl on soil
Purpose: Add salinity and protectant (co-treatment) on agar grown Arabidopsis
Procedure:

1. **Prepare 75mM NaCl and 100mM NaCl**

2. **Prepare diluted 正翰 protectant by the ratio of 1:500 (1μL of protectant and 499μL of autoclaved ddH_2O)**
3. **Add 2mL of respective NaCl solution and protectant in the ratio of 1:1 (1mL NacL solution and 1mL diluted protectant) into each respective agar plates with Arabidopsis**

Done by Abigail

Date: 20260826
Title: Transformation of N, O
Purpose: Spread bacterial liquid on agar plate
Procedure:
N: Pveg-EEBD-RBS0-SamyQ-Bopep4(v2t1)-B0015
O: Pveg-EEBD-RBS0-SamyQ-Bopep4(v2t2)-B0015

1. Set up water bath at 42℃ and ice tub
2. Wait until the competent cell is 50% thawed in the ice tub
3. Add 10μL plasmid
4. Incubate on ice for 20min
5. Heat shock for 45sec at 42℃ water bath
6. Incubate on ice for 3min
7. Add 1mL SOC medium in the microtube in BSC
8. Incubate at 37℃ for 1hr
9. Centrifuge for 5min at 5000g
10. Take out about 1mL of supernatant
11. Resuspend the pellet with pipette
12. Transfer all of the pellet on labeled LBA plate and spread it evenly
13. Place the spreaded agar plate upside down in incubator at 37℃ for 16hr

Done by Sophie C.

Date: 20260826
Title: Golden Gate Assembly of X *3
Purpose: Assemble parts
Procedure:
X: ACDI

part	**1:3**	**1:5**	**1:7**
pSTK-EXP-1-sfGFP Conc: 163ng/μL	1μL (dilute 1/5)	1μL(dilute 1/5)	1μL(dilute 1/5)

5164bp			
A 107ng/μL, 3754bp	0.68μL	1.14μL	1.6μL
C 134ng/μL, 3828bp	0.56μL	0.93μL	1.3μL
D 86ng/μL, 3243bp	0.74μL	1.23μL	1.7μL
I 108ng/μL 3757bp	0.68μL	1.13μL	1.58μL
T4 DNA ligase	0.5μL	0.5μL	0.5μL
T4 DNA ligase buffer	1μL	1μL	1μL
BsmbI	0.5μL	0.5μL	0.5μL
ddH_2O	4.34μL	2.57μL	0.82μL

Done by Sophie C
Supervised by Chloe

Date: 20260827
Title: Miniprep
Purpose: isolate plasmid form liquid culture
Procedure:
#2 PliaG-CcaSm3
#3 PliaG-CcaSm3

1. Pellet 1.8mL of liquid culture by centrifuging at 6800g for 1 min
2. Repeat until collected all medium
3. Add 250μL of P1 buffer and resuspend the pellet
4. Add 250μL of P2 buffer and mix by inverting the tube 4-6 times until the solution become clear
   ->do not allow lysis to proceed for more than 5min
5. Add 350μL of N3 buffer and mix immediately by inverting the tube 4-6 time
6. Centrifuge for 10min at 17900g
7. Transfer ~800μL of supernatant to column and collection tube
8. Centrifuge for 1 min at 17900g, discard the flow-through
9. Add 500μL PB buffer
10. Centrifuge for 1 min at 17900g, then discard the flow-through
11. Add 750μL PE buffer
12. Centrifuge for 1 min at 17900g, then discard the flow-through
13. Centrifuge for 1 min at 17900g, then discard the flow-through
14. Place column on new 1.5mL microtube
15. Add 50μL EB buffer to the center of the column
16. Let stand at RT for 1min
17. Centrifuge for 1 min at 17900g
18. biodrop

	Concentration(ng/μL)	A260/230	A260/280
#2	93	1.069	1.824
#3	88	1.692	1.913

Done by Sarah, Audrey H

Date: 20260827
Title: Electroporation of P, R
Purpose: Check the approximate length of colony PCR product
Procedure:

1. Prewarm the recovery medium at 37℃ and pre-cool a 0.2cm electroporation cube
2. Hix 403μL of plasmid P in 100μL cell in a 1.5mL microtube(P7:248ng/μL)
3. Mix 3.95μL of plasmid R in 100˜L cell in a 1.5mL microtube(R2:253ng/μL)
4. Set up electroporator and voltage resistance
5. Transfer mixture of cells&DNA to electroporation cuvette & tap suspension to bottom
6. Place in shake pad, close lid&wait until beep(1~2s)

7. Immediately remove from shock pad and add 1mL recovery medium
8. Transfer to liquid culture tube & incubate at 37℃ for 3hrs, shaking at 200rpm
9. Check&record parameters
10. Centrifuge at 5000g for 5min
11. Spread LB plate with erithmycin
    Done by Abby T/K
    Supervised by Gabriel

Date: 20260827
Title: Transformation
Purpose: Transform bacterial liquid on agar plate
Procedure:
X:ACDI(1:3)
X:ACDI(1:5)
X:ACDI(1:7)

1. Set up water bath at 42℃
2. Thaw competent cell till 50% thawed, add 5μL plasmid
3. Inicubate on ice for 20min
4. Heat shock for 45 sec
5. Incubate on ice for 3min
6. Add 1mL SOC medium
7. Incubate at 37℃ for 1 hr
8. Centrifuge for 5 min at 5000 g
9. Remove ~1mL supernatant
10. Resuspend the pellet
11. Spread on LBC
12. Incubate 37℃ overnight

Done by Sophie C

Date: 20260827
Title: Golden Gate Assembly
Purpose: assemble parts
Procedure:
Q: pSTK-1D-PcpcG2-veg-EEBD-RBS0-SamyQ-AccD-V2T1-B0015

1. Prepare the following

Vector 1D 132ng/μL,3120bp	1.5μL(1/10)
PcpcG2-veg 168ng/μL,235bp	0.91μL(1/50)

EEBD-RBS0 151ng/μL,2392bp	1.03μL(1/5)
pSTK-OC-SamyQ-AccD-V2T1 103ng/μL,2213bp	1.21μL(1/3)
B0015 107ng/μL,2213bp	1.34μL(1/5)
T4 DNA ligase buffer	2μL
Sterile water	10.01μL
T4 DNA ligase	1μL
BsaI-HFv2	1μL

2. Thermal cycle

37℃	5min	x45
16℃	5min	
37℃	10min	x1
80℃	20min	x1

Done by Sophie C

Date: 20260827
Title: protein extraction from cell and medium
Purpose:
Procedure:

1. Centrifuge B

Date: 20260828
Title: Colony PCR
Purpose: amplify engineered DNA plasmid
Procedure:
X:ACD1 0826's ACDE
XI:ACD1(1:7)
XII:ACDT(1:5)

1. Prepare the following reactions

X	1reaction	25reaction
CcaR	1μL	25μL
csn	1μL	25μL
Q-amp mastermix	10μL	250μL
ddH_2O	8μL	200μL

2. Distribute 10μL of each tube (24)
3. Pick colonies with a tip and streak on the grid
4. Dip the tip into the mix
5. Discard the tips at the 5mix
6. Run thermal cycle

95℃	5min	1cycle
95℃	20sec	35cycle
60℃	20sec	
72℃	1min	
72℃	5min	1cycle

Done by Olivia

Date: 20260828
Title: Gel electrophoresis
Purpose: checking colony PCR result
Procedure:
X:ACDI from 0826
XI:ACDI (1:7)
XII:ACDI(1:5)

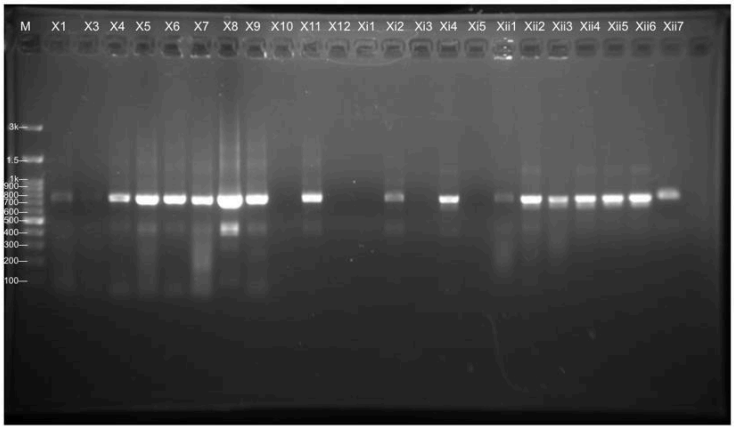
1. Load the gel in the following order

M	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	X11	X12	XI1	XI2	XI3	XI4	XI5	XII1	XII2
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5

XII3	XII4	XII5	XII6	XII7
5	5	5	5	5

2. Resolve the gel at 100V for 30mins

8/28 Colony PCR (X, X_I, X_II)



M: 3kb DNA ladder
X: ACDI from 8/28
- CCaR fw
- csn rv-seq
- bp: 760

X_I: ACDI (1:7)
- CCaR fw
- csn rv-seq
- bp: 760

X_II: ACDI (1:5)
- CCaR fw
- csn rv-seq
- bp: 760

X1, X4-9, X11, Xi2, Xi4, Xii1-7 → liquid culture

Done by Olivia

Date: 20260828
Title: SDS-PAGE
Purpose: separate proteins based strictly on their molecular weight (size) rather than their intrinsic charge or shape
Procedure:

1. Prepare 1.0% polyacrylamide gel
2. Loading the sample prepare form 260827

M	cell lysort	melium					medium	cell lysort	M
5	50	50	0	0	0	0	50	50	5

3. Resolve at 200V for 10min and 180V for 40min
4. Cut the gel from middle. Let part for Western Blot and Right part for Coomassie Blue
5. Transfer the left gel to PVDF for 90min
6. Blocking the PVDP in 20mL 5% BSA for 1hr
7. Incubate the PVDP in 1:1000 anti-His-tag mAb (CST) in 2% BSA overnight
8. Seain the right gel in coomassie blue for 45min
9. Destain the gel in reased destmin buffer for 30min
10. Destain the gel in fresh destain buffer overnight

Done by chars

Date: 20260829
Title: Gel electrophoresis
Purpose:checking colony PCR result
Procedure:

1. Load hte samples in the following format
   M:1kb DNA ladder
   D:Pveg-EEBD-RBS0-SamyQ:ACCD-V2T1-B0015 electroporation
   R:Pveg-EEBD-RBS0-csn:LEA-B0015 electroporation
   Gel1

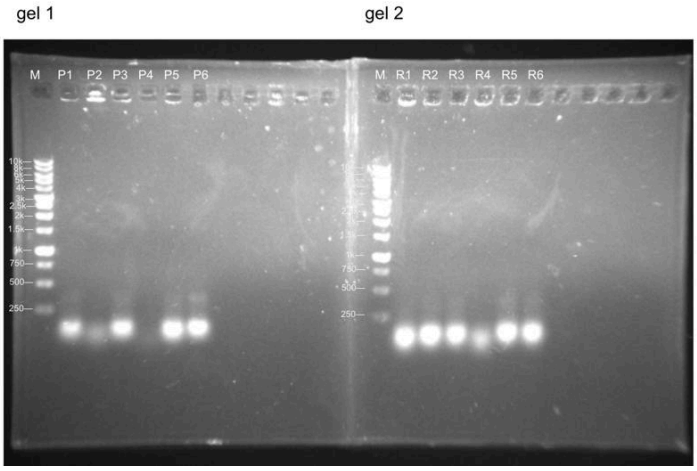
M	P1	P2	P3	P4	P5	P6
5	5	5	5	5	5	5

   Gel2

M	R1	R2	R3	R4	R5	R6
5	5	5	5	5	5	5

2. Resolve the DNA at 100V for 30min

8/27 Electroporation colony PCR P, R



M: 1kb DNA ladder
P: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-ACCD-v2t1-B0015
bp:1984
R: pSTK-EXP1-Pveg-EEBD-RBS0-csn:LEA-B0015
bp:1438
pSTK-EXP-seq fw, pSTK-EXP-seq

Done by Abby K

Date:20260829
Title: Western blot (J in WB800)
Purpose:
Procedure:

Date:20260829

Title: Colony PCR
Purpose:amplify engineered DNA plasmid
Procedure:
N:pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-Bopep4-V2T1-B0015

1. Prepare the following reaction x3 in PCR tubes

Fw primer pSTK-EXP-seq fw	1μL
Rv primer Bopep 4R	1μL
Q-amp master mix	10μL
ddH_2O	8μL

2. Pick up colonies with tip & streak grid
3. Dispense tip in mix
4. Take out tip and close lid
5. PCR machine for thermal cycle

95℃	5min	x1
95℃	20sec	x35
60℃	20sec	
72℃	45sec	
25℃	5min	x1

Done by Sophie C

Date:20260829
Title: Agar - ACCD trial 1 after treatment/CH trial4 co treatment w NaCl
Purpose:
Procedure:

Date:20260829
Title: Soil - 100mM NaCl
Purpose:

Procedure:

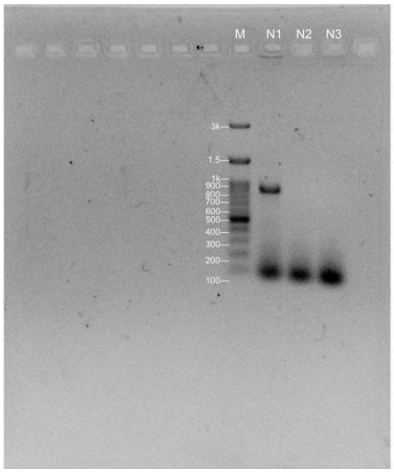
Date:20260829
Title: Gel electrophoresis
Purpose:checking colony PCR result
Procedure:
N:pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-Bopep4-V2T1-B0015

1. Load samples in following order

M	M1	M2	M3
5	5	5	5

2. Resolve gel at 100V for 30min

8/28 Colony PCR N



M: 3kb DNA ladder

N: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-Bopep4-v2t2-B0015
pSTK-EXP-seq fw & Bopep4 R
bp: 678

Done by SophieC

Date:20260829
Title: Miniprep
Purpose:
Procedure:
X6 ACDI
XII4 ACDI

1. Pellet 1-5mL bacterial overnight culture by centrifuge at 6800 g for 3min
2. Resuspend the pelleted bacterial cells in 250μL buffer P1, transfer to a micro-centrifuge tube

3. Add 250µL buffer P2 and mix thoroughly by inverting the tube 4-6 times until the solution becomes clear
4. Add 350µL of buffer N3, mix immediately and thoroughly by inverting the tube 4-6times. The solution will turn color less
5. Centrifuge for 10min at 17900 g
6. Apply ~800µL supernatant to QIA 2.0 spin column by pipetting
7. Centrifuge for 30sec at 17900 g & discard flow through
8. Wash the colun by adding 0.5mL buffer PB
9. Centrifuge for 30sec at 17900 g & discard flow through
10. Wash the column by adding 0.75mL buffer PE
11. Centrifuge for 30sec at 17900 g & discard flow through
12. Centrifuge for 1 min at 17900 g to remove residual wash buffer
13. Place column in clean 1.5mL micorcentrifuge tube
14. Add 50µL buffer EB to the center of the column to elute DNA
15. Let stand for 1min
16. Centrifuge for 1min at 17900 g
17. biodrop

	concentration(ng/µL)	A230/260	A260/280
#X6	102.0	2.488	1.902
#XII4	63	2.250	1,969

Done by Sophie C, Abby K

Date:20260829
Title: PCR
Purpose:
Procedure:
#X6 ACDI
#XII4 ACDI

1. Prepare the following reactions
2.

Date:20260829
Title: Golden gate assembly (L, O, M)
Purpose:
Procedure:

Date:20260829
Title: Golden gate assembly (V(1:5), V(1:7))
Purpose:
Procedure: